Baghbaan: research and integration plan

An all-in-one app for Kashmir's apple growers: what it should contain, and which tools, data, technologies and partners to build it with.

Companion to the *Baghbaan demo app walkthrough* (PDF) and the clickable prototype (claude.ai/artifact/T7wV1p9vo63Sqa659bEQCG; source in `prototype/`). Prepared November 2026.

How to read this report. Part 1 is the recommendation: what the app should contain and how each part works. Part 2 catalogues what we can integrate: data sources and APIs, technologies, datasets, apps to learn from, competitors and regulations. Part 3 covers architecture, business model, roadmap and risks. Appendices A–D hold the full desk research with a source link next to each fact. Facts marked **[V]** were read at the source, **[S]** come from search summaries and should be re-checked, and **UNVERIFIED** items are leads only. Crop-protection content follows SKUAST-K's 2026 guide, read from a copy of the booklet; check it against the printed booklet before anything reaches growers.

Summary

- **The opportunity is large and concentrated.** Jammu & Kashmir grew 21.10 lakh tonnes of apple on 1.73 lakh hectares in 2024-25, roughly 70–80% of India's apples. Seven to seven and a half lakh families and about 35 lakh people depend on horticulture, worth ₹10,000–12,000 crore a year (J&K Economic Survey 2025-26; Assembly reply, Sept 2026; Appendix B §1).
- **Growers lose money at five predictable points:** disease (2026 was a bad scab year, with B/C grades selling at ₹200–250 a box at Sopore against ₹1,000–1,300 for premium fruit), spurious inputs, highway closures (the 26 Aug–17 Sep 2025 NH-44 closure stranded 2,000+ trucks; traders put losses at ₹500–700 crore and freight jumped from ₹70–100 to ₹200–300 a box), weak price information, and labour and transport at harvest. There is no crop insurance for apple today (scheme on hold, Sept 2026) (Appendix B §2–5).
- **Recommendation: build Baghbaan around one question, "what is wrong with my orchard and what do I do?"** The home screen is the **Orchard Doctor**, a Yes / No and picture-based diagnosis that works offline. Around it sit **Buy** (licensed-dealer marketplace), **Learn** (short Kashmiri-audio lessons), **Mandi & news** (rates, where-to-sell, highway, rail, weather, advisories) and **Services** (trucks, labour, spray teams, equipment, experts, cold storage). **My orchard** quietly builds a spray diary and orchard record from what growers already do.
- **Design for the real conditions:** voice and pictures first; Kashmiri voice with Urdu-script text; offline-first with SMS and a missed-call voice line, because J&K has had more internet shutdowns than any other part of India; no signup form, and data asked only when it improves an answer.
- **Advice follows SKUAST-K's 2026 management guide.** It has 15 crop stages with doses per 100 L, and fungicide sprays are tied to rain forecasts. A verification round confirmed 270 of about 290 claims and corrected six: codling moth *is* present in the valley, Alternaria spots are not "frog-eye", and apple blotch leaf miner is the emerging pest growers mistake for Alternaria (Appendix D).
- **Integrate first:** SKUAST-K's spray schedule as the advice rule base; data.gov.in / Agmarknet mandi prices; IMD and a commercial weather API for spray windows and a Mills-type scab model; Bhashini and IndicTrans2 for Kashmiri; Kashmir-specific CC BY 4.0 leaf-image datasets for a later photo model; J&K Traffic Police advisories plus driver reports for roads; and partnerships with the Department of Horticulture's Kisan Khidmat Ghars for assisted onboarding.
- **Competition exists but leaves the core gap open.** Orchardly (claims 15,000+ growers), Farmit by Qul, Apple Doc and others offer advisory or IoT. None found offers Kashmiri voice, a neutral licensed-dealer marketplace, and labour and transport booking in one app (Appendix C §2).
- **Start with a two-district pilot** (Shopian and Baramulla/Sopore, the two largest dispatch mandis), earn trust with free diagnosis, rates and alerts, then add the marketplace and credit.

Part 1 · What the app should contain

1. The grower and the problem

Who uses it. Owners of small family orchards (often 5–40 kanal; 1 hectare ≈ 19.8 kanal), many over fifty, many more comfortable speaking Kashmiri than reading any script; their sons and daughters who handle the phone; younger agriculture graduates running new high-density blocks; and the people around them: dealers, transporters, labour leaders, commission agents and extension staff.

Where the money is lost.

PAIN POINT	WHAT HAPPENS TODAY	WHAT THE APP DOES
Disease and pests	Spray decisions come from the dealer or neighbours; sprays are mistimed relative to rain; the same fungicide group is repeated.	Orchard Doctor; weather-driven infection alerts; spray diary with group rotation; dose calculator.
Spurious or wrong inputs	Fake or mislabelled packs; products without an apple label claim; over-buying.	Licensed-dealer marketplace; brand comparison of the same molecule; genuine-pack QR check; buy-the-right-quantity.
Market information	Prices heard from the commission agent; no view across mandis or net of costs.	Daily rates by variety and grade; 30-day trends; "where should I sell?" net-of-freight calculator; price alerts.
Highway and logistics	NH-44 closures strand loaded trucks; freight spikes; no view of rail options.	Highway, Mughal Road and rail status with alerts; truck booking with shared loads; rail parcel option.
Labour and services	Phone calls to find teams; unclear rates; trust problems with unknown providers.	Verified, rated providers; fair-rate guides; booking and tracking; pay after work.
Knowledge	Package of Practices is a long PDF; extension staff are thinly spread.	Season-aware three-minute lessons with Kashmiri audio; library; calendar; expert escalation.
Credit	Informal credit from the dealer, tied to buying from him and settled at harvest.	Pay-after-harvest credit through a regulated lender, linked to the Kisan Credit Card.

2. The app at a glance

Five tabs plus a personal record. Every screen described here is in the clickable prototype.

TAB	PURPOSE	KEY SCREENS
Home · Orchard Doctor	The main page: today's alerts and the Yes / No diagnosis	Today (alert, crop stage, rates, highway, spray window, labour); Where is the problem?; questions; result; photo check; ask an expert
Buy	Genuine inputs from licensed dealers	Orchard kit; 12 categories; filtered lists; product (brands, dose, safety, reviews); order and payment; genuine-pack check
Learn	Practical knowledge, in Kashmiri	Continue; for this season; 12 topics; lesson with steps and quiz; pests & diseases library; orchard year
Mandi & news	Prices and the conditions that move them	Rates; where to sell; roads and rail; weather and disease risk; advisories, schemes and alerts
Services	Book help	Trucks and shared loads; labour and spray teams; equipment; expert visit; soil test; cold storage; beehives; nursery plants; buyers; booking tracker
My orchard (from the avatar)	The grower's own record	Blocks and varieties; spray diary; costs and income; harvest record; pest map; privacy and data

Other channels reach growers the app cannot: SMS keywords (SPRAY, RATE, ROAD), a WhatsApp number, a missed-call voice line in Kashmiri, Urdu and Hindi, assisted sign-up at Kisan Khidmat Ghars, and trained village helpers ("Bagh Mitras").

3. Home and the Orchard Doctor (the main page)

3.1 Home

The home screen answers "what should I do today?" for this grower's village and altitude:

- **The main question** in a green card: "What's troubling your orchard today?" with Start, a microphone (speak the problem in Kashmiri) and a camera (photo check).
- **Quick chips** for common entry points: spots on leaves, insects and mites, fruit problem, tree drying, what to spray now.
- **An alert with an action**, e.g. "Scab infection risk on Wednesday: 14 mm rain, leaves wet ~18 h at 11–16 °C, your last spray was 9 days ago → Spray before Wednesday's rain".
- **Crop stage for this altitude** (dormant → green tip → pink bud → bloom → petal fall → fruitlet → growth → harvest → leaf fall) and the next spray due.
- **Four tiles:** mandi rate and trend, NH-44 status, today's spray window, free spray teams nearby.
- **One quick question a day** (optional, one tap, reason shown): bees seen during bloom? pests seen this week? crop better or worse than last year?

The home content changes with the season: scab and bloom in spring, mites and Alternaria in summer, harvest, trucks and CA storage in autumn, 3–4% urea on yellowing leaves and pruning in late autumn and winter.

3.2 How a diagnosis works

1. **Where is the problem?** Leaves, fruit, flowers, branches and bark, roots and whole tree, or "not sure: take a photo".

2. **Yes / No / Not sure questions,** each with two pictures: what a Yes looks like and what a No looks like, plus a Listen button.
3. **A live "narrowing down" panel** showing the top candidates.
4. **One multiple-choice step** for severity (a few trees, about half, most), while the last spray is read from the spray diary rather than asked again.
5. **The result:** most likely cause with match strength and the reasons ("velvety olive-green spots; spots on young fruit"), urgency from the forecast, what to do now, products that fit (registered for apple and on the university schedule), how to prevent it next season, look-alikes to rule out, and buttons to ask an expert or save to the diary.

The engine. A small, auditable naive-Bayes model. Each problem has a prior for the season, altitude band and recent nearby reports; each question has the probability of a Yes given each problem, written by experts. After each answer the scores update, and the next question is the one with the largest expected information gain, so most diagnoses end in 2–4 questions. The engine stops at 90% confidence, and hands over to a human when answers conflict, when nothing fits, or for problems that must always go to an expert (root rot, sudden spread, fire-blight-like symptoms). In the demo, two Yes answers move scab from 33% to 91% to 99%.

Getting smarter. Expert confirmations and photo labels from real cases re-estimate the likelihoods each season. The question bank is versioned, and each version is signed off by the content partner.

Photo check (phase 2). An on-phone image model (LiteRT, MobileNet-class) trained on Kashmir field images (Appendix C §3–4) confirms, but does not replace, the questions. Models trained only on PlantVillage lab images drop sharply in accuracy on field photos, so field images must be collected with consent through the app.

Safety rules hard-coded in the engine: suggest only products with a CIB&RC label claim for apple and on the current SKUAST-K schedule; never an insecticide during bloom; never a banned or restricted molecule; show "read the label" with every product; escalate uncertain or high-consequence cases.

4. The problem library (diagnosis knowledge base, first version)

General symptom and management notes for the most important problems in Kashmir's apple belt. **Rows were checked against SKUAST-K's 2026 guide and monthly advisories, ICAR and peer-reviewed sources (Appendix D, section 4); doses and product names come only from the current SKUAST-K schedule.**

PROBLEM	PART	DISTINGUISHING SIGNS (BECOME YES / NO QUESTIONS)	SEASON / TRIGGER	LOOK-ALIKES	MANAGEMENT DIRECTION
Apple scab (*Venturia inaequalis*)	Leaves, fruit	Olive-green to black, velvety spots; dark, corky, cracking spots on fruit	Primary infection green tip to 2–3 weeks after petal fall; rain-splashed secondary infection all season (severe in July–Aug 2026)	Alternaria, Marssonina, spray burn	Spray before forecast rain from green tip; never repeat a product in consecutive sprays (Kashmir isolates show reduced sensitivity to difenoconazole and flusilazole); 3–4% urea when half the leaves yellow; open canopy
Alternaria leaf blotch (*Alternaria* spp.)	Leaves	Small (0.5–2 mm) light-brown round spots merging into blotches; defoliation	From mid-May; worst in warm, humid June–August	Apple blotch leaf miner, Marssonina, necrotic leaf blotch	Cover sprays before rain as per schedule; check for leaf miner before spraying fungicide
Marssonina blotch / premature leaf fall (*Diplocarpon coronariae*)	Leaves	Dark brown blotches on upper surface; leaves yellow with green islands and drop early	July–September after wet spells	Alternaria, mites	Timely summer sprays; sanitation; urea at leaf fall
Powdery mildew (*Podosphaera leucotricha*)	Shoots, flowers	White powder on young leaves, shoot tips, flower clusters; narrow curled leaves	Spring growth	Aphid curl	Prune out infected shoots in winter; sprays at pink bud / petal fall as per schedule
European red mite / two-spotted spider mite	Leaves	Dull, bronzed leaves; tiny moving dots under leaves	Hot, dry June–August	Alternaria, nutrient problems	Mineral oil before green tip (kills red-mite eggs only); spray only above SKUAST thresholds: 4–5 mites per leaf at petal fall, >5 fruitlet, >10 fruit development; rotate acaricides
San Jose scale	Bark, fruit	Grey crusty bumps on bark; red spots with grey centres on fruit	All year	Woolly aphid	Horticultural mineral oil at dormancy; prune heavily infested wood
Woolly apple aphid	Bark, roots	White cotton-like masses on wounds and cuts; galls on roots	All year	Scale	Brush colonies; listed insecticide; MM111 rootstock in new plantings (MM106 also resists it but is prone to collar rot on wet sites)
Aphids (green, rosy)	Leaves	Curled young leaves, sticky honeydew, ants	Spring	Mildew	Prune affected tips; spray only if 40% of young shoots are infested; never in bloom
Leaf-eating caterpillars (Indian gypsy moth, tent caterpillar)	Leaves	Eaten leaves, nests, egg masses	Spring	—	Destroy nests and egg masses; listed insecticide if spreading
Cankers (smoky blight, papery bark, others)	Bark	Sunken, cracked or peeling dead bark	All year	Borer damage	Scrape to healthy wood, paste wounds, burn debris; paint pruning cuts
Apple stem borer / root borer	Trunk, roots	Holes with sawdust-like frass; grubs on roots; declining tree	All year / summer	Canker	Clean and plug holes; destroy grubs; expert advice for soil treatment
Collar rot (*Phytophthora cactorum*)	Trunk base	Brown, soft, rotting bark at soil level; yellowing tree; wet soil	Wet spring and autumn	White root rot	Expose and scrape the collar, fungicide paste; drainage; keep manure off the trunk; *Trichoderma*

PROBLEM	PART	DISTINGUISHING SIGNS (BECOME YES / NO QUESTIONS)	SEASON / TRIGGER	LOOK-ALIKES	MANAGEMENT DIRECTION
White root rot (*Dematophora necatrix*)	Roots	Sudden wilting; white fungal threads on roots; patches of dying trees	Summer	Collar rot	Expert visit; drench affected and neighbouring trees; isolation trench
Bitter pit (calcium)	Fruit	Small sunken brown spots, mostly at the calyx end, worse in storage	Pre-harvest, storage	Hail, scale	Calcium sprays in fruit growth; avoid excess nitrogen and heavy pruning
Boron / zinc deficiency	Fruit, leaves	Cracking, corky core (boron); small bunched leaves, rosettes (zinc)	Growth	Scab cracking	Soil test; foliar micronutrients as per schedule
Sooty blotch & flyspeck	Fruit	Grey smudges and clusters of tiny black dots that rub off	Wet late summer	Scab	Summer sprays; airflow; fruit can be cleaned
Hail, frost	Fruit, flowers	Dents and bruises after storms; black flower centres after a cold night	Spring and summer storms; bloom	Disease	Nets; alerts; protective spray after hail; record damage
Poor fruit set	Flowers	Flowers drop without setting; few bees; cold rainy bloom	Bloom	Frost	Pollinizer varieties; beehives (SKUAST-K 2026 guide: 3–4 per hectare at 5–10% bloom; April 2026 advisory: 2 per acre); no insecticide in bloom
Codling moth	Fruit	Entry holes with frass; early fruit drop	Present in the Kashmir valley: DNA-confirmed in Baramulla after 2019, reported in six districts by 2023; about 3.5 generations a year	Other fruit borers	Pheromone-trap monitoring and SKUAST-K 'fruit borer' sprays; report new areas
Apple blotch leaf miner (*Leucoptera malifoliella*)	Leaves	Blotch-shaped mines with a larva or frass inside; often mistaken for Alternaria	May–September; 13–100% defoliation in south Kashmir (2023)	Alternaria	Destroy fallen leaves, scrape bark, traps; spray only the leaf-miner options in SKUAST-K advisories
Fire blight	Shoots, blossoms	Blackened blossoms and shoot tips bent like a shepherd's crook	Not known in India or J&K (EPPO: absent, unreliable old record)	Frost, canker	Escalate-only alert: report suspected cases to the department

Spray-schedule backbone. The Doctor, the home alerts and the shop all follow SKUAST-K's *Integrated Management Guide for Apple Orchards in Kashmir 2026* (No. AU/DEx/449-BL/01/26), issued through the Directorate of Horticulture. It has 15 stages: delayed dormancy (horticultural mineral oil 2 L per 100 L, before green tip), green tip and tight cluster (contact fungicide at 300 g per 100 L only if a full day of rain is forecast; skip if the next 7 days are dry), pink bud (scab spray; boron 1–1.5 g/L if needed; 3–4 beehives per hectare), flowering (no spray unless 10 days since the last fungicide and rain is due), petal fall and fruitlet (systemic fungicides before rain; mites only above thresholds; thinning at 10–12 mm), fruit development I–IV (fungicide only if rain is forecast; three calcium chloride sprays at 3–4 g/L), pre-harvest (storage-rot spray 15–20 days before picking CA fruit), harvest, post-harvest (3–4% urea when half the leaves are yellow) and dormancy (5% urea on fallen leaves if missed). Rules: never mix pesticides, keep 3–4 days between insecticide and fungicide sprays, never repeat a product in consecutive sprays, no spraying in full bloom. There is no separate high-density schedule. SKUAST-K's own 2026 documents disagree on a few points (Zineb + Hexaconazole dose, pre-harvest timing, when to stop spraying); the app must hold those rows back until SKUAST-K confirms. Full stage table with sources: Appendix D, section 1.

5. Buy: a licensed-dealer marketplace

Categories: fungicides; insecticides; mite control (acaricides); spray oils (HMO); fertilizers (urea, DAP, MOP, NPK, CAN, SSP); micronutrients (boron, zinc, calcium); organic and biological (vermicompost, FYM, neem, *Trichoderma*, biofertilizers); tools and sprayers; hail nets and supports; packing (cartons, trays); saplings and rootstocks (NHB-accredited nurseries); safety kits (PPE).

What makes it different from a general agri shop:

- **Kits matched to the orchard:** stage + area + sprayer drum + last spray group → the right products in the right quantity.
- **Filter by problem and stage,** not by brand. Each card shows the FRAC/IRAC group, the label claim and whether the group was used last time.
- **Same molecule, several brands:** price per pack across licensed dealers.
- **Dose calculator** in local units (kanal, trees, 200-litre drums).
- **Genuine-pack check:** scan the pack QR, match batch and expiry to the manufacturer's record (needs manufacturer data partnerships; start with the companies that already print QR codes). Failed checks go to the department and the dealer.
- **Safety:** PPE, waiting period before harvest, bee and water warnings on every product.
- **Fulfilment:** pickup at the village dealer (cheap and keeps dealers on side) or delivery.
- **Payment:** UPI, cash at pickup, and pay-after-harvest credit through a regulated lender.
- **Reviews only from verified buyers,** shown with district.

Compliance. Since the Insecticides (Second Amendment) Rules 2022 (rule 10E), a licensed seller may sell insecticides through an e-commerce platform, and the platform must verify the seller's state licence first; the Central Consumer Protection Authority enforced this against major marketplaces in May 2026. So run the shop as a marketplace for licensed dealers, store and show each licence, and list only products registered for apple. Malathion is no longer approved for apple and dimethoate lost its label claims on fruit eaten raw (Insecticides (Prohibition) Order 2023). Fertilizer sales go through authorised dealers under the Fertiliser (Control) Order; urea and DAP have fixed MRPs. Genuine-pack checks matter: 85 of 5,772 pesticide samples tested in J&K in 2025-26 were substandard, and a 2026 Shopian case over a mancozeb batch led to an FIR (Appendix D, section 5).

6. Learn: short lessons in Kashmiri

Format. Three-minute videos with Kashmiri audio (recorded by native speakers, not synthesized), picture step cards, a one-question check at the end, offline downloads, and lessons that surface by season. Digital Green's farmer-to-farmer video model has shown much higher adoption than conventional extension (Appendix A §1.2).

Curriculum (12 topics, about 120 lessons):

TOPIC	EXAMPLE LESSONS
Orchard calendar	Month by month for higher and lower belts
Soil & manure	Soil sampling; reading a Soil Health Card; manure and fertilizer by tree age
Planting & varieties	Rootstocks (seedling, M9, M26, MM106, MM111); varieties and pollinizers; planting a new block
High-density orchards	Trellis and drip; training feathered plants; fertigation; HDP subsidy
Pruning & training	Central leader and spindle; renewal pruning; wound care
Pests & diseases	One lesson per library entry; the 20-leaf mite check; scab and rain
Safe spraying	Calibration; mixing order; PPE; bees; waiting periods; disposing of packs
Water & drip	Irrigation in dry spells; mulching
Bees & pollination	Pollinizer ratios; renting hives; spraying around bloom
Harvest & storage	Maturity indices; grading and packing; CA storage
Selling smart	Per-box vs per-kg; reading mandi rates; commission and deductions; contracts with pre-harvest buyers
Schemes & insurance	HADP, HDP subsidy, KCC, Kisan Khidmat Ghars; insurance status

Content sources: SKUAST-K Package of Practices and extension material; ICAR-CITH Srinagar; KVKs; Department of Horticulture; NHB; Dr YSP UHF Nauni (Himachal) for HDP practice. Reuse needs permission; each lesson is signed off by a named expert.

7. Mandi & news

FEATURE	DATA SOURCE	NOTES
Daily rates by mandi, variety, grade	data.gov.in OGD API "Current Daily Price of Various Commodities from Various Markets (Mandi)", resource `9ef84268-d588-465a-a308-a864a43d0070` (free key; ₹/quintal); Agmarknet 2.0; e-NAM dashboard; NHB daily bulletin	Kashmir mandis report mainly August–March. J&K prices appear to be box prices converted at 16 kg; show ₹/kg and ₹/box and state the assumption. Add trained mandi reporters at Sopore, Shopian (Aglar), Parimpora, Narwal and Azadpur for same-day rates.
Where should I sell?	Rates + freight + commission + unloading	Net money per box for the grower's own load; show uncertainty while the truck is on the road.
Highway status	J&K Traffic Police advisories on X (@Traffic_hqrs), Traffic Control Unit phone lines, driver reports	No official API; parse advisories and confirm by staff; push changes by app, SMS and voice call.
Rail	Budgam–Delhi parcel train (since 13 Sep 2025; reported ~₹2.41/kg; 20,000+ t moved)	Show schedules and space; book through partners when possible.
Weather, spray window	IMD API (registration / IP whitelisting), commercial weather API (e.g. Open-Meteo paid plan), ERA5 for history	Spray window = wind, rain-free hours, temperature.
Disease risk	Scab infection model (Mills / modified Mills: leaf wetness hours × temperature); mite degree-days	Validate locally with SKUAST-K; later feed from in-orchard weather stations.
Advisories & schemes	Department of Horticulture, SKUAST-K, IMD Srinagar agromet bulletins	Every item can be listened to; dated and sourced.

Context the app should explain: imports (India–New Zealand agreement in force from 20 Oct 2026 with apple quotas from 2027; Iranian, Turkish and US apples in Delhi), CA storage capacity (2.92 lakh t against a stated need of ~6 lakh t), and the 2025 closure's lessons (rail and shared loads as backups).

8. Services: a trusted marketplace for help

What growers can book: trucks (open, shared part-loads, refrigerated), rail parcel space, labour teams (pruning, spraying, thinning, harvest, grading and packing), spray services with power sprayers, tractors and equipment, expert visits, soil-test sample pickup, CA cold storage, beehives, certified nursery plants, packing material, and buyers for produce.

Trust and safety (patterns from Indian platforms, Appendix A §4):

- ID and document checks (driving licence, RC, insurance for trucks); police-registration status for non-local labour, which J&K Police requires.
- Ratings from verified bookings only; "jobs done" counts.
- Fair-rate guides by district and season, with the government transport corporation's freight as a benchmark.
- PIN-closed jobs (the grower enters a code when work is done); pay after work; dispute desk.
- Shared loads: pool part-loads going to the same mandi on the same night.
- Live trip status tied to highway alerts.

Models to learn from: Bharat Intelligence (WhatsApp/voice booking of labour gangs), the government CHC farm-machinery app (PIN-closed jobs), Vahak and BlackBuck (load boards), ColdKonnect (cold-storage listings). Labour and transport aggregators fall under the Social Security Code's platform-worker

contribution (1–2% of turnover, capped at 5% of payouts).

9. My orchard and collecting information without being invasive

Rules.

1. **Value before questions.** Every screen works in guest mode. The app helps first, then asks.
2. **One question at a time, at the moment it improves an answer,** with "why we ask" and a Skip.
3. **Never ask twice.** Reuse what the grower already did: a purchase becomes a spray-diary entry; a diagnosis records the variety and block.
4. **Never ask** for Aadhaar, bank passwords, OTPs or land papers.
5. **Aggregate only.** Pest and crop reports leave the phone as counts by area.
6. **Reciprocity.** A shared report brings back a village alert.
7. **Shared phones.** A "helping someone" mode for family members, with the grower's own consent. Phone ownership is lower among women in rural J&K (58% vs 80% for men, 2025 survey), so design for shared devices.

The ladder (when → what we ask → what the grower gets): first open → language, village → local weather, stage, rates. First diagnosis → where, how many trees → a cause and plan. First purchase → orchard size, drum size → exact quantity. Each spray → product, date (from the order) → rotation, reminders, records. Daily, optional → one tap (bees, pests) → village alerts. Season end → crop better or worse → area estimate. Only if they want → phone, name → call-backs, bookings, credit.

Law. India's DPDP Act 2023 and DPDP Rules 2025: consent-manager provisions from about November 2026, main obligations from about May 2027. The app needs an itemised plain-language consent notice (in Kashmiri, Urdu, Hindi, English), withdrawal as easy as consent, breach reporting within 72 hours, log retention, a grievance channel, and verifiable parental consent for users under 18. Penalties reach ₹250 crore (Appendix C §7.1).

10. Designing for Kashmir's growers and networks

- **Pictures and voice first.** In studies, low-literacy users completed no tasks with text interfaces but most with voice or picture-based ones; illiterate farmers preferred audio-only advice (Appendix A §1.1).
- **Kashmiri voice, Urdu-script text.** Kashmiri speech recognition and synthesis are still limited (Bhashini lists Kashmiri; its Kashmiri TTS is Devanagari-script only), so pre-record prompts with native speakers and use Urdu script for reading.
- **Offline-first.** J&K has had several hundred internet shutdowns, including 550 days of no or partial mobile internet from August 2019. Diagnosis, lessons, schedules and last-known rates live on the phone; questions and orders queue; SMS keywords and a missed-call voice line are backups.
- **Simple screens.** Large targets, few levels, a replay button, Western digits for prices, long pauses before voice input stops.
- **Assisted channels.** Kisan Khidmat Ghars (2,000+ J&K farmer service centres), village helpers, dealers and FPOs as onboarding points. In GSMA's study of the IFFCO Kisan app, a signup profile form put off first-time users and most left within two months; avoid that.

Part 2 · Tools, technologies and data to integrate

11. Integration shortlist

PRIORITY	INTEGRATE	FOR	ACCESS / LICENCE
1	SKUAST-K spray schedule and Package of Practices	Doctor rules, alerts, shop filters, lessons	Partnership and reuse permission needed
2	data.gov.in / Agmarknet 2.0, e-NAM, NHB bulletin	Mandi rates	GODL-India (commercial use with attribution)
3	IMD API + commercial weather API + ERA5	Spray windows, scab model, alerts	IMD registration; Open-Meteo paid for commercial use; ERA5 CC BY 4.0
4	Bhashini (ASR, translation, TTS), AI4Bharat IndicTrans2 (Kashmiri in Arabic and Devanagari scripts)	Voice input, translation of content	Government platform; IndicTrans2 MIT
5	Kashmir apple leaf datasets (7,569-image Kashmir set; ICAR-CITH Srinagar set), AppleLeaf9, PlantVillage CC0 copy	Photo model	CC BY 4.0 / CC0; avoid non-commercial sets (MinneApple, Fuji-SfM)
6	LiteRT (TensorFlow Lite) on-device inference	Offline photo check	Open source
7	J&K Traffic Police advisories (@Traffic_hqrs), TCU lines	Road status	No API; parse and verify
8	Kisan Khidmat Ghar / Kisan Sathi portal	Assisted onboarding, scheme info	Partnership
9	SKUAST-K e-Plant Health Clinic	Expert escalation	Partnership
10	NHB National Nursery Portal	Verified sapling sellers	Public
11	UPI payments; NBFC / bank partner	Payments, pay-after-harvest	Licensed partners
12	WhatsApp Business API, SMS gateway, IVR provider	Channels beyond the app	Commercial

12. Technologies for better orchards (what to explain, offer or partner on)

TECHNOLOGY	WHAT IT DOES	HOW BAGHBAAN USES IT
High-density planting (M9/M26/MM106 rootstocks, trellis, drip)	Earlier and higher yields, better fruit; HD fruit fetched ≥₹1,600 per 10 kg box vs ≤₹1,200 per 16–20 kg traditional box in Aug 2025	Lessons, subsidy guide, certified nursery marketplace, HDP-specific schedules
Disease forecasting (scab infection periods, mite degree-days)	Spray when infection is likely instead of by calendar	Home alerts; later a network of in-orchard weather stations with leaf-wetness sensors
Anti-hail nets	Protect fruit from spring and summer hail	Shop category, installer booking, hail alerts
Better sprayers (air-blast, tower), calibration	Even coverage, less chemical, less drift	Lessons, rental, calibration service
Drones	Mapping, scouting; spraying in tall traditional canopies is limited	Pilot only, under Drone Rules 2021 and current pesticide-drone SOPs; check J&K flying zones
Satellite remote sensing (Sentinel-2, Landsat, Bhuvan)	Orchard mapping, stress and damage detection	Area-level crop and damage estimates for the department and lenders
Smart pheromone traps	Automatic pest counts	Pilot with partners for key pests
CA storage and grading lines	Longer selling season, better grades	Booking, price outlook, grading-team booking
Traceability (QR on boxes, spray records)	Buyers and exporters see the spray history	Spray diary export; buyer links in phase 3
LLM assistants	Answers in natural language	Only on top of the vetted knowledge base, with expert review of new answer types

13. Apps and competitors

Public and national apps to link or learn from: Kisan Suvidha, Meghdoot, Mausam, Damini (lightning), Agmarknet and e-NAM apps, Kisan e-Mitra, Kisan Call Centre (1800-180-1551), J&K's Kisan Khidmat Ghar / Kisan Sathi. **Private:** Plantix (photo diagnosis; covers scab and Marssonina), DeHaat, AgroStar, BigHaat, Gramophone, Farmer.CHAT (Digital Green), KissanAI. **Global apple tools:** RIMpro, WSU Decision Aid System, Cornell NEWA, Trapview, FruitScout, Aerobotics, Croptracker. Reviews of the national apps repeatedly mention login problems, stale data and English-only content (Appendix C §1).

Kashmir and Himachal competitors (details in Appendix C §2):

PLAYER	OFFER	GAP BAGHBAAN FILLS
Orchardly (Fruitbee Agritech, Srinagar)	Advisory, smart devices, services; claims 15,000+ growers and ₹19 crore revenue (FY26)	Neutral licensed-dealer marketplace; free offline diagnosis; logistics
Farmit by Qul	Urdu AI chat; SKUAST-based advice; blocks insecticides in bloom	Kashmiri voice; services marketplace; mandi decisions
Apple Doc (SKUAST-K incubated)	AI decision support for precision apple farming	Breadth beyond advisory
Kash Crop	GPS, weather, AI assistant	Scale, trust, marketplace
SKUAST-K e-Plant Health Clinic	Expert diagnosis	A partner, not a competitor

Positioning: the neutral, Kashmiri-first "one app for the orchard year", partnering with SKUAST-K and the department rather than competing with them.

14. Datasets

DATASET	USE	LICENCE
Kashmir apple leaf disease set (7,569 images)	Photo model training	CC BY 4.0
ICAR-CITH Srinagar apple disease set	Photo model training	CC BY 4.0
AppleLeaf9	Training (9 classes)	CC BY 4.0
PlantVillage (CC0 copy)	Pre-training only; poor field generalisation	CC0
Plant Pathology 2020/2021 (Kaggle)	Benchmarking	Check competition terms
MinneApple, Fuji-SfM	Yield/fruit detection research	Non-commercial: avoid in product
data.gov.in mandi prices	Rates, trends	GODL-India
IMD gridded data, ERA5	Weather history, model calibration	IMD terms; CC BY 4.0
Soil Health Card data, SoilGrids	Soil context	Government / CC BY
Sentinel-2, Landsat, Bhuvan	Remote sensing	Open
J&K Economic Survey	District statistics	Public

15. Regulations that shape the app

AREA	WHAT APPLIES	PRODUCT DECISION
Personal data	DPDP Act 2023, DPDP Rules 2025 (main duties ~May 2027)	Itemised consent in four languages; minimal data; export/delete; breach process; under-18 consent
Pesticide sales	Insecticides Act 1968; rule 10E (2022) allows online sale by licensed sellers after platform verification; label claims; Prohibition Order 2023	Marketplace for licensed dealers only; apple-registered products only
Fertilizer sales	Fertilizer Control Order; fixed MRPs for some products	Authorised dealers; show MRP
Saplings	Nursery registration / accreditation	Sell only from NHB-accredited or J&K-licensed nurseries
E-commerce	Consumer Protection (E-Commerce) Rules 2020	Seller details, grievance officer, returns policy
Labour and transport platforms	Social Security Code (platform workers); motor-vehicle rules; J&K Police labour registration	Contributions; document checks; registration status
Drones	Drone Rules 2021; pesticide-drone SOPs	Pilots only, with certified operators

Part 3 · Building it

16. Architecture

- **App:** Android first (Flutter or React Native), small install, offline database (SQLite), background sync, works on 2 GB RAM phones; a light PWA for sharing links.
- **Content:** a headless CMS with an expert review workflow: question bank, likelihood tables, lessons, products, advisories, each versioned and signed off.
- **Doctor engine:** shared rules library that runs identically on the phone and the server (the prototype's engine is ~60 lines of JavaScript).
- **Data services:** price ingestion (data.gov.in + reporters), weather ingestion (IMD + commercial API), road-status desk, risk models.
- **Channels:** WhatsApp Business API, SMS gateway, IVR in Kashmiri, Urdu and Hindi; an expert desk web app for advisors and the department.
- **Marketplace:** dealer and provider onboarding with document checks; orders, bookings, PIN-closed jobs; payments through a licensed payment aggregator; lender integration for credit.
- **Privacy:** consent records, data minimisation, encryption at rest and in transit, data in India, role-based access for experts.

17. Business model

STREAM	NOTES
Commission on input orders	From licensed dealers, not from growers; keep advice neutral by never ranking products by margin
Booking fees	Small fee on trucks, teams, storage
Credit partnerships	Origination fee from the lender for pay-after-harvest
Premium services	Weather stations, field visits, record exports for exporters
Institutional	Expert desk and anonymised area insights for the department, insurers (when apple insurance launches) and researchers

Lessons from others: BharatAgri shut down in November 2025 after high acquisition costs and low repeat orders; Otipy also failed. Keep acquisition cheap (assisted onboarding, dealers, FPOs), earn repeat use with free daily value (alerts, rates, roads), and do not hold inventory early.

18. Roadmap and measures

PHASE	SCOPE
1 · months 0–6: pilot in Shopian and Baramulla	Orchard Doctor (25 problems) reviewed with a university partner; mandi rates, highway and weather alerts; 40 Kashmiri-audio lessons; offline mode and SMS keywords; directory of verified transporters and labour leaders; 500 growers via village helpers and Kisan Khidmat Ghars
2 · months 6–12: marketplace and records	Licensed-dealer shop; spray diary, rotation and dose calculator; in-app bookings with tracking; photo check beta; Kashmiri voice line; expert desk
3 · months 12–24: scale	All apple districts; pay-after-harvest credit; weather-station network; buyer and exporter links; anonymised intelligence for the department

Measures: weekly active growers; diagnoses completed and confirmed right by experts; repeat use in the second season; share of purchases genuine-checked; sprays avoided or rotated; truck-days lost to closures avoided; repeat bookings; grower-reported net price per box.

19. Risks and how to handle them

RISK	MITIGATION
Wrong advice harms crops or people	University-signed content; conservative engine; label-first; expert escalation; audit trail
Dealers see the app as a threat	Make them the fulfilment network: pickup orders, credit through partners, visibility
Low trust in a new app	Free value first; local helpers; department and university partners visible in the app
Shutdowns and weak networks	Offline-first, SMS and IVR
Data misuse fears	Minimal data, aggregates, DPDP-grade consent, no selling of personal data
Price data gaps (seasonal reporting)	Mandi reporters; clearly dated rates
Regulatory change (pesticide e-commerce, platform workers)	Legal review before Phase 2; licensed partners
Kashmiri language technology gaps	Recorded audio; Urdu text; collect consented speech for future models

20. Answered since the first draft, and still open

Answered (sources in Appendix D):

- **Spray schedule:** SKUAST-K's 2026 guide is current; the app now follows it. It was read from a copy of the booklet, so check against the printed booklet before launch.
- **HDP subsidy:** the Revised Modified High Density Plantation Scheme gives a 50% subsidy until 31 March 2027. The minimum area is 1 kanal, and growers apply through the Kisan Sathi portal. Separately, anti-hail nets carry a subsidy of ₹27,000 per kanal.
- **Online sale of pesticides:** legal for licensed sellers whose licence the platform verifies (rule 10E, 2022).
- **Labour:** harvest labour in Shopian cost ₹1,200–1,500 a day in October 2026, and pickers earned ₹800–1,100 in 2025. The statutory minimum for unskilled work is ₹311.

- **Codling moth:** present in the Kashmir valley. **Fire blight:** not known in India or J&K.
- **CA storage:** about ₹35 per 17 kg crate per month at Lassipora.

Still open:

1. An official copy of the 2026 guide, and SKUAST-K's ruling on its internal conflicts.
2. Partnership and reuse terms for SKUAST-K content.
3. Whether the tentative fertilizer schedule for high-density orchards (July 2026) was adopted.
4. Current CIB&RC apple label status for chlorpyrifos, thiacloprid and carbendazim mixes.
5. Task wages for pruning, spraying and packing by district, and employers' duties for non-local labour.
6. Whether apple crop insurance will relaunch.
7. Grower testing: show the demo to 20 growers in Shopian and Sopore and record what they change.

Appendix A · UX, connectivity, data collection and services marketplace

Desk research notes, November 2026. Tags: [V] read at source (in Appendix D: confirmed by an independent verifier); [S] from a search summary or not re-confirmed; UNVERIFIED = lead only.

Prepared 12 Nov 2026. Scope: design guidance for a mobile app for apple orchard growers in Kashmir (users often older, some with low literacy; Kashmiri, Urdu, Hindi, English), with a services marketplace (trucks, labour gangs, spray services, equipment, experts, cold storage).

How to read the evidence tags

- **[V]**: I opened and read the cited page or PDF in this session.
- **[S]**: taken from a search-engine extract of the cited page. I did not re-open the page, so treat exact numbers with some care.
- **UNVERIFIED**: the sources conflict, the source is secondary or undated, or I could not confirm the fact.
- **"Proposal"**: my own design synthesis, not a sourced fact.

Limitation: the session's shared web-search budget ran out partway through. Some sub-topics are thinner than planned (see Section 6).

---

0. The 15 most actionable design rules (with evidence)

1. **Don't build a text-first app. Lead with pictures and voice, and offer text as an optional layer.** - In a mobile-banking test, no low-literacy subject could complete a transaction with a text interface. 72% completed it with spoken dialogue and 100% with a graphical UI. [V] https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/ToCHI2711_Medhi.pdf - In FarmChat, every illiterate participant preferred the audio-only interface. 16 of the literate or semi-literate participants preferred audio plus text. The authors recommend a hybrid: two big buttons plus an optional chat-history view. [V] https://ubicomplab.cs.washington.edu/pdfs/farmchat.pdf
2. **Keep navigation flat.** - Avoid menu hierarchies, scrolling, soft-key mappings and non-numeric text entry. - 56 of 90 low-literacy subjects could not navigate hierarchical menus. 48 of 90 did not understand scroll bars. [V] (Medhi et al., link above) - In FarmChat, digitally-illiterate users never scrolled. [V] (FarmChat PDF)
3. **Use voice, replay and pictures, with some specifics.** - Write numbers with English (Western) digits. In FarmChat, 11 of 23 literate or semi-literate users could not read Hindi numerals. - Show photos of the product package. Illiterate users valued these because they fear being sold the wrong pesticide. - Put a big "replay" button on every answer. - Allow 7–8 seconds of silence before speech input stops listening. [V] (FarmChat PDF)
4. **Push short, scheduled advisories and let farmers choose how often.** - When Gujarat farmers chose their own call frequency, pickup rose 7.6 percentage points (control 80.0%) and listening rose 9.4 pp (control 32.6%). [V] https://registry.precisiondev.org/registry_entry/frequency-a-b-test - A heads-up SMS 10 minutes before an advisory call raised pickup by 5.9 pp and listening by 4 pp in Odisha (about 32,000 users). [V] https://registry.precisiondev.org/registry_entry/impact-of-heads-up-sms-on-call-pickup-and-listening/
5. **Give value before asking for anything. Ask for each data item only when it is needed.** - In GSMA's case study of the IFFCO Kisan app, the drop-down profiling step was "a barrier for first time users". Nearly 80% of users churned within two months. GSMA recommended simplified profiling, auto-filling location, and personalising so users see the value of sharing. [V] https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2016/10/IFFCO-Kisan-Agricultural-App.pdf - 18% of shoppers abandon checkout when forced to create an account. [V] https://baymard.com/blog/reduce-cart-abandonment
6. **Ask for permissions and consent in context.** - Ask at the moment the user starts the feature that needs it. Explain why, allow "not now", and degrade gracefully. [V] https://developer.android.com/training/permissions/requesting - The DPDP Act requires consent to be "free, specific, informed, unconditional and unambiguous", and withdrawal must be about as easy as giving consent. [V] https://dpdpa.com/dpdpa2023/chapter-2/section6.html
7. **Build offline-first, with three channels.** The channels are the app (offline cache and outbox), voice/IVR (missed call then free call-back, which works on basic phones), and assisted service at Kisan Khidmat Ghars (KKGs) or village-level entrepreneurs (VLEs). - SFLC.in counts 447 shutdowns in J&K. [V] https://alpha.internetshutdowns.in/static-page/jammu-kashmir/ - In 2019, mobile and landline services were also shut, for a total of 550 days of no or partial connectivity. [V] https://pulse.internetsociety.org/en/shutdowns/internet-suspended-in-jammu-and-kashmir-india-5-august-2019/ - Of rural J&K households with internet, 99.0% connect over mobile networks and only 6.3% have fixed/Wi-Fi. [V] https://www.mospi.gov.in/sites/default/files/publication_reports/CMST_report_m.pdf
8. **Use trusted local intermediaries for onboarding.** Examples: KKG agri-entrepreneurs, CSC VLEs, mandi shopkeepers. Budget for their cost. - GSMA found agent-led models effective for onboarding and trust in low-literacy settings, though costly. [S] https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/programme/agritech/five-key-lessons-from-scaling-digital-agriculture-services/ - J&K already runs "over 2,000" entrepreneur-driven KKGs. [V] https://kashmirreader.com/2026/01/19/hadp-ushers-a-new-era-of-agricultural-transformation-across-jk/
9. **Design for shared phones and helpers.** - One FarmChat participant said: "My son doesn't let me use his phone...". 12 participants had only occasional access to a phone at home. [V] (FarmChat PDF) - The DPDP Rules let a person nominate someone to exercise their data rights. [V] https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc20251117695301.pdf - Proposal: an explicit "helper" role, with the grower's own voice or PIN consent.
10. **Don't assume a woman in the household has her own smartphone.**
    - Rural J&K, age 15+: 58.0% of women own a mobile phone vs 79.7% of men. 61.8% of women used the internet in the last 3 months vs 75.1% of men. [V] (CMS:T 2025 PDF above)
    - Among rural women who own a phone, 22.7% own only a basic phone. [V] (same)
11. **Make pest reports two-way.**
    - Use photo + voice note + optional location, checked by an expert before any public alert. The reporter gets advice back immediately.
    - India's National Pest Surveillance System (NPSS, launched 15 Aug 2024) uses farmer photos analysed by experts and AI. [S] https://knnindia.co.in/news/newsdetails/sectors/agri-food/government-launches-ai-powered-national-pest-surveillance-system-to-empower-farmers
12. **If you use incentives, keep them small and immediate, such as airtime.**
    - Airtime incentives raised IVR survey response in Bangladesh and Uganda. Doubling the amount made no significant difference. [V] https://pmc.ncbi.nlm.nih.gov/articles/PMC6747927
13. **Build trust and safety in the marketplace with these features:**
    - verified providers (truck RC/KYC; police verification for non-local labour gangs, required in J&K);
    - ratings;

- an upfront price checked against a public benchmark (e.g., J&K Road Transport Corporation (JKRTC) freight rates per box);
- a PIN to close each job (as in the government CHC app);
- documented payment terms.
- Evidence:
- Kashmir freight rose from about Rs 60 to Rs 200 per box in the Sept 2025 crisis. [V] https://kashmirlife.net/soaring-freight-charges-for-apples-continue-despite-govt-assurances-406072/
- A J&K Police advisory (reported 6 Aug 2026) requires newly arriving labourers to register at the nearest police station. [V] https://www.wionews.com/india-news/j-k-police-issues-new-advisory-for-non-local-labourers-after-recent-terror-attacks-15-lakh-reward-announced-for-let-terrorist-1786002596846
- The CHC/FARMS booking-closure PIN. [S] https://www.agrimachinery.nic.in/Files/FAQ_CHC_MobAPP.pdf

14. **Don't build the business on farmer subscriptions or thin input-commerce margins.**
    - BharatAgri shut down in Nov 2025, citing "High customer acquisition costs and low repeat orders". [V] https://www.peoplematters.in/news/business/bharatagri-shuts-shop-after-failing-to-secure-new-funding-round-47212
    - Arya.ag, which combines storage and finance, is profitable. [S] https://entrackr.com/fintrackr/aryaag-reports-rs-447-cr-revenue-in-fy25-profits-spike-70-9048421
15. **Plan for DPDP compliance by about May 2027.**
    - The Rules were notified 14 Nov 2025, with an 18-month phase-in. [V] (PIB link above)
    - Requirements:
    - a standalone, itemised consent notice;
    - an option to read it in English or any Eighth-Schedule language (Kashmiri, Urdu and Hindi qualify);
    - a link to withdraw consent;
    - parental consent for under-18s;
    - breach notification;
    - a response to data-rights requests within 90 days.
    - Sources: [V] https://dpdpa.com/dpdpa2023/chapter-2/section5.html ; [V] https://www.dpdpa.com/dpdparules/rule3.html

---

1. Designing for low-literacy, rural, multilingual farmers

1.1 Text vs graphics vs voice vs a human operator

- **Medhi et al. (ACM TOCHI 2011)** [V] https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/ToCHI2711_Medhi.pdf
- Ethnography with 90 low-literacy subjects in India, Kenya, the Philippines and South Africa, plus 2 quantitative studies with more than 70 subjects in India.
- "textual interfaces are unusable by first-time low-literacy users, and error prone for literate but novice users."
- Mobile banking: no low-literacy subject completed a transaction with the text UI. 72% completed it with spoken dialogue and 100% with a graphical UI. Spoken-dialogue users who succeeded were faster and needed less help.
- Health data entry: a live operator was about 10 times more accurate than text entry.
- Navigation findings: 56 of 90 could not navigate hierarchical menus (none of the 40 non-phone-users could). 48 of 90 did not understand scroll bars. 45 had trouble with soft keys.
- Design recommendations:
    - graphical cues;
    - voice annotation;
    - local-language text and audio;
    - minimal hierarchy;
    - no non-numeric text input;
    - no scrolling menus;
    - minimal soft-key mappings;
    - human mediators to introduce the system.
- **Text-free UI (Medhi, Sagar, Toyama, ITID 2007)** [S] https://ideas.repec.org/a/tpr/itintd/v4y2007i1p37-50.html
- Illiterate domestic workers in Bangalore strongly preferred text-free designs.
- Confirmed the need for semi-abstracted graphics and voice feedback, and a consistent help system.
- **FarmChat (Jain et al., ACM IMWUT 2018)** [V] https://ubicomplab.cs.washington.edu/pdfs/farmchat.pdf
- 34 potato farmers near Ranchi, with a knowledge base built from Kisan Call Centre logs.
- All illiterate participants preferred audio-only. 16 literate or semi-literate participants preferred audio plus text, mainly to re-read medicine names and doses. 7 preferred audio-only ("nothing to read, so no tension").
- 16 of 22 digitally-illiterate users preferred audio-only, and none of them scrolled.
- Physical and environmental factors mattered:
    - sunlight makes the screen unreadable in the field;
    - tractor noise drowns out audio;
    - one older user had shaky hands and could not hit small buttons.
- Images of medicine packages were highly valued.
- 11 of 23 literate or semi-literate users could not read Hindi numerals; the authors recommend English digits.
- Speech input needs a longer silence timeout (7–8 s).
- Farmers were "highly skeptical" of advice from friends, shopkeepers and agri-apps, so trust has to be earned.
- Recommended design: a hybrid of two large buttons plus an optional chat-history toggle, with picture-based multiple choice.
- Agri-entrepreneurs preferred audio plus text for themselves but recommended audio-only for farmers in their village. This is an argument for a separate "assistant mode" UI for intermediaries.

1.2 Video-based extension (Digital Green)

- **2009 trial** [V] https://www.microsoft.com/en-us/research/?p=156320
- 13 months, 16 villages, 1,470 households.
- Digital Green's participatory, locally made videos increased adoption of some practices **seven-fold** over Training-and-Visit extension.
- **10 times more effective per dollar** on a cost-per-adoption basis.
- **J-PAL RCT in Bihar (2014–2016)** [V] https://www.povertyactionlab.org/evaluation/evaluation-digital-greens-agricultural-extension-program-india
- Women farmers in 420 villages.
- Video did not raise measured System of Rice Intensification (SRI) adoption, possibly because the survey did not capture partial adoption.
- Bayesian estimates show yields up 12–18% and profits up 9–24%. The video approach was cost-effective: additional profit was about 16 times the extra cost.
- **IFPRI RCT in Ethiopia** [S] https://essp.ifpri.info/2022/01/03/accelerating-technical-change-through-icts-evidence-from-a-video-mediated-extension-experiment-in-ethiopia/
- Reached 24% more farmers and produced up to 44% higher uptake.
- Cost about $4 per additional adoption. [S] https://bigdata.cgiar.org/digital-intervention/video-enabled-extension/
- **Implication for Kashmir:** short, locally filmed Kashmiri-language videos of real growers (scab spray timing, high-density orchard training, grading and packing) are well supported. They can be screened at KKGs or shared over WhatsApp, and should be cached for offline viewing.

1.3 IVR and voice services

- **Avaaj Otalo (Gujarat; CHI 2010)** [S] https://hci.stanford.edu/publications/paper.php?id=144
- Voice forum used by 51 farmers over a 7-month pilot. The most popular feature was the Q&A forum where farmers ask questions and browse others' questions and answers.
- **Cole & Fernando (Economic Journal 2021)** [S] https://www.atai-research.org/wp-content/uploads/2020/06/ueaa084.pdf
- RCT with 1,200 cotton farmers. More than half called the service in the first 7 months.
- Treated farmers used more effective pesticides and spent less on ineffective or hazardous ones.
- Trust in phone-based advice rose from near 0 to more than 6 on a 10-point scale.
- Yield gains (cumin +28%, cotton +8.6% in a sub-group) are from the working paper. UNVERIFIED whether the published version keeps them.
- Peer-effects follow-up: [S] https://voxdev.org/topic/agriculture/mobile-phone-based-extension-peer-interactions-and-farmer-information-exchange
- **Awaaz.De (Avaaj Otalo's successor)**
- Its Khedut Saathi service let subscribers forward episodes to up to 5 friends (viral voice sharing). [S] https://www.slideshare.net/awaazde/khedut-saathi-viral-voice-messaging-for-small-farmers-in-india
- **Gram Vaani / Mobile Vaani** [V] https://solve.mit.edu/solutions/35346
- The user gives a **missed call** and automatically receives a **free call-back**; navigation is by keypad.
- Users can record messages; 3–4% of calls produce a recording. Machine learning plus human moderators filter content.
- Scale: more than 100,000 monthly unique users before COVID-19 and more than 10,000 calls per day (self-reported).
- Bihar evaluation: only about 23% of interviewed women had heard of Mobile Vaani. [S] https://bmjopen.bmj.com/content/12/10/e052336.full
- A related Uttar Pradesh study found that how many messages women heard depended on phone ownership, ease of phone access and women's empowerment. [S] https://files.comminit.com/content/development-mhealth-behavior-change-communication-strategy-case-study-rural-uttar-pradesh
- Lesson: voice channels need offline outreach, especially to reach women.
- **IFFCO Kisan "Green SIM"** [V] (GSMA PDF)
- 3–4 daily push voice messages.
- Average pickup rate 38–52% across 1.8 million active users; average listening about 42 seconds.
- **Implication:** an IVR line with missed-call call-back and short (under 1 minute) push voice messages in Kashmiri. It reaches basic-phone owners and keeps working when mobile data is cut, as long as voice is up.

1.4 WhatsApp and AI chat services in Indian agriculture

- **Digital Green FarmerChat** (WhatsApp, Telegram and web; generative AI over Digital Green's video transcripts and call-centre logs) [V] https://www.isdm.org.in/ekam/development-management-hub/ask-and-they-receive
- More than 225,000 users, more than 160,000 of them in India; more than 20 languages.
- **Voice queries are 5 times text queries.**
- 30% of queries are about pests and diseases.
- Features added from farmer feedback: text-to-speech, audio answers, follow-up questions, a weather widget.
- Early rollout went through extension agents first, which added a human review step. [S] https://openai.com/index/digital-green
- **Oilseeds Kisaan Mitra** (ICAR-IIOR, Hyderabad): a WhatsApp AI advisory where farmers save a number and send questions. [S] https://agrospectrumindia.com/news/7/34198/oilseeds-kisaan-mitra-indias-first-nationwide-whatsapp-ai-advisory-for-oilseed-farmers.html
- **Kisan e-Mitra** (PM-KISAN scheme chatbot, launched 21 Sept 2023) [V] https://apolitical.co/en/navigator/case-studies/kisan-e-mitra-indias-voice-enabled-ai-support-for-farmers
- Voice or typed questions in 11 languages, aiming for all 22.
- More than 20,000 queries per day (April 2025).
- Uses Bhashini for language. It can answer personal status questions because it is linked to structured beneficiary records.
- **Farm Vaidya** (Telugu voice AI agents; IFPRI was technical adviser) [S] https://www.ifpri.org/blog/generative-ai-powered-voice-technology-in-agricultural-advisory-services-lessons-from-india/
- Farmers' local names for crops and pests are missing from standard datasets, so the team collected real farm speech to build speech-to-text.
- Agents must adapt to the crop season.
- **Bharat Intelligence** (Nashik farm-labour platform) [V] https://thebetterindia.com/innovation/ai-farm-labour-platform-bharat-intelligence-labour-os-nashik-12108095
- Booking happens on WhatsApp and by voice call. "There's no app to download."

- **Implication:** a WhatsApp number that accepts voice notes and photos and replies with voice notes is a low-friction second channel. Treat it as a front door into the same back end as the app. AI answers for scab or pest advice should be reviewed by experts until they are proven accurate.

1.5 Assisted ("agent") models

- **Kisan Khidmat Ghars (J&K's Holistic Agriculture Development Programme, HADP)**
- Launch event (Greater Kashmir, 5 Aug 2024): [V] https://greaterkashmir.com/national/kisan-khidmat-ghars-will-serve-as-one-stop-service-hub-for-farmers-lg
  - 500 KKGs inaugurated in Aug 2024, with 1,500 more planned.
  - Services: input booking, soil health, plant diagnostics, market intelligence, custom hiring, artificial insemination.
  - A KKG app was launched offering credit support, price forecasting, scheme applications, **booking expert consultations**, a dealer locator and skilling courses.
  - The centres are run by "agri-entrepreneurs".
- Status by Jan 2026 (Kashmir Reader): [V] https://kashmirreader.com/2026/01/19/hadp-ushers-a-new-era-of-agricultural-transformation-across-jk/
  - "Over 2,000" entrepreneur-driven, IT-enabled single-window centres.
  - Farmer onboarding runs through the **Kisan Sathi** portal: more than 3.7 lakh farmers registered, more than 5.9 lakh applications. Credit linkage is integrated.
- In Rajouri, KKGs offer micro-ATMs and POS machines. [S] https://www.devdiscourse.com/article/headlines/3775688-j-k-140-new-kisan-khidmat-ghars-in-rajouri-district-to-bring-government-services-to-farmers-doorsteps
- **This is the obvious assisted channel to partner with in J&K.**
- **Common Service Centres (CSCs)**
- Village-level entrepreneurs (VLEs) offer scheme access, prices and weather. [S] https://epubs.icar.org.in/index.php/JAEM/article/view/168941
- A 2025 MANAGE study found VLEs lack agri-specific training and that farmers know little about CSC services. [S] (same)
- Sources disagree on the CSC count. One says more than 600,000 centres by Mar 2025: [S] https://www.constructionworld.in/policy-updates-and-economic-news/csc-celebrates-16-years-of-rural-digital-empowerment/76197 . Another says 4,07,122 in Gram Panchayats as of 30 Jun 2026. Both UNVERIFIED.
- **EM3 AgriServices** (pay-per-use farm machinery) [V] https://techcrunch.com/2017/08/15/em3
- Call centres plus village representatives. "the majority of its business is offline" because "The majority of farmers are not on smartphones."
- **Digital Labour Chowk** [S] https://thebetterindia.com/startup/chandrashekhar-mandal-digital-labour-chowk-job-portal-daily-wage-workers-10469048
- Signed up mobile-shop and grocery owners near labour chowks (informal day-labour hiring points) as "business associates" who register workers.
- **Intermediated use** (Sambasivan et al., CHI 2010) [S] https://microsoft.com/en-us/research/?p=159423
- Low-income users often use technology through a digitally skilled intermediary, such as a mother reached through her daughter's phone.
- Design for this explicitly rather than treating it as misuse.

1.6 What reduces drop-off (evidence summary)

LEVER	EVIDENCE	SOURCE
Fewer profiling questions at signup; auto-filled location	IFFCO Kisan: drop-down profiling a barrier; about 80% churned within 2 months (benchmark 61–80% drop-off over 3 months)	[V] https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2016/10/IFFCO-Kisan-Agricultural-App.pdf
Push "bite-sized" notifications (weather alerts) instead of pull-only	Same GSMA report recommends personalised push to bring idle users back	[V] same
Let users pick message frequency	+7.6 pp pickup, +9.4 pp listening (Gujarat, n=3,788)	[V] https://registry.precisiondev.org/registry_entry/frequency-a-b-test
Heads-up SMS 10 minutes before a call	+5.9 pp pickup, +4 pp listening; larger effect for non-smartphone users	[V] https://registry.precisiondev.org/registry_entry/impact-of-heads-up-sms-on-call-pickup-and-listening/
Fewer, better-framed recommendations	Pickup +4 pp over a 65% control mean (Odisha)	[S] https://registry.precisiondev.org/registry_entry/fewer-and-revised-recommendations/
Simplicity over feature bloat; "friendly helper" guidance	GSMA AgriTech accelerator lessons	[S] https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/programme/agritech/five-key-lessons-from-scaling-digital-agriculture-services/
Peer content and a Q&A forum	Avaaj Otalo's most-used feature was the farmer Q&A forum	[S] https://hci.stanford.edu/publications/paper.php?id=144
Repeat purchase, not just installs	BharatAgri (more than 1 million users) closed because of high acquisition cost and low repeat orders	[V] https://www.peoplematters.in/news/business/bharatagri-shuts-shop-after-failing-to-secure-new-funding-round-47212

1.7 Kashmiri-language specifics

Speakers - Census 2011 counted 6,797,587 Kashmiri speakers nationally, 6,680,837 of them in J&K. The figures include related dialects. [S] https://en.wikipedia.org/wiki/States_of_India_by_Kashmiri_speakers ; [S] https://kashmirtimes.com/are-there-26-lakh-hindi-speakers-and-only-20000-urdu-speakers-in-jk/

Official status - Before 2019 only Urdu and English were official languages. - The **J&K Official Languages Act 2020** made Kashmiri, Dogri, Urdu, Hindi and English official. [S] https://prsindia.org/billtrack/prs-products/prs-bill-summary-3595 ; [S] https://www.dailyexcelsior.com/news/archive/hindi-kashmiri-dogri-introduced-as-official-languages-in-addition-to-urdu-english

Who reads Kashmiri vs Urdu - Literate Kashmiris have mostly been schooled in Urdu. - A 2017 order made Kashmiri a compulsory subject for Classes 9–10, starting 2018–19. Only 5 of 27 sanctioned Kashmiri lecturer posts are filled. [V] https://www.outlookindia.com/national/teacher-shortage-hits-kashmiri-language-education-in-schools (21 Feb 2026) - "Many Gen Z Kashmiris cannot read or write the Koshur script because it was not formally taught in many schools until recently". This is a qualitative claim. [V] https://kashmirlife.net/is-kashmiri-becoming-a-ghost-in-its-own-homeland-426308/ - A 2013 survey (Bhat, "Power of a written language...") reportedly found more than 95% of respondents can read and write Urdu but only 32.7% can do so in Kashmiri. UNVERIFIED: this comes from a search extract; I could not open the paper. https://www.academia.edu/5313616/POWER_OF_A_WRITTEN_LANGUAGE_A_CASE_OF_URDU_LITERACY_AND_DISCOURSE_IN_KASHMIR - A claim attributed to Kak (2001), that 97% of educated Kashmiris cannot read Kashmiri script, is also UNVERIFIED. - Both languages use Perso-Arabic script, so Urdu readers can partly decode Kashmiri

text. [S] (same paper extract) - **Implication:** use **Urdu (Nastaliq) as the default reading script**, with Kashmiri as **spoken audio** and Kashmiri text optional. Offer Hindi (Devanagari) and English.

Keyboards - Gboard lists five Kashmiri options: Arabic script; Arabic script with transliteration; Devanagari script; Devanagari script with transliteration; and Latin script. [V] https://support.google.com/gboard/answer/6380730 - Whether Gboard voice typing supports Kashmiri: UNVERIFIED. - Keyman offers a third-party Perso-Arabic Kashmiri keyboard. [S] https://help.keyman.com/keyboard/rac_kashmiri

Speech recognition (ASR) - AI4Bharat's **IndicConformer 600M multilingual** lists Kashmiri (ks) among its 22 languages. It is MIT-licensed but gated, and **no Kashmiri word-error rate is published**. [V] https://huggingface.co/ai4bharat/indic-conformer-600m-multilingual - The IndicVoices speech dataset includes Kashmiri. [S] https://huggingface.co/datasets/ai4bharat/IndicVoices/blob/main/README.md

Text-to-speech (TTS) - AI4Bharat **Indic Parler-TTS** supports Kashmiri only **unofficially** ("Competitive results... like Kashmiri (55.30)"). [V] https://huggingface.co/ai4bharat/indic-parler-tts - IndicVoices-R, an open TTS dataset, includes Kashmiri. [S] https://proceedings.neurips.cc/paper_files/paper/2024/file/7dfcaf4512bbf2a807a783b90afb6c09-Paper-Datasets_and_Benchmarks_Track.pdf - Whether Bhashini offers Kashmiri ASR or TTS: UNVERIFIED.

Machine translation - Microsoft Translator added Kashmiri on 5 Oct 2023. Speech support is not stated. [V] https://news.microsoft.com/source/asia/2023/10/05/microsoft-translator-expands-to-20-indian-languages-empowering-linguistic-diversity/ - Google Cloud Translation does **not** list Kashmiri. [V] https://docs.cloud.google.com/translate/docs/languages - Kashmiri is not among Google Translate's 110 languages added in June 2024. [V] https://support.google.com/translate/answer/15139004?hl=en

Implication (proposal) - For a fixed UI vocabulary (menus, confirmations, advisories), use **pre-recorded human Kashmiri audio**. - Treat Kashmiri speech recognition as experimental. Always pair voice input with picture buttons, and fall back to Urdu or Hindi speech recognition. - Collect consented local apple, pest and disease vocabulary, as Farm Vaidya did. - Have local experts review all machine translation.

---

2. Connectivity realities in Kashmir

2.1 Shutdown history

SFLC.in tracker (J&K page) [V] https://alpha.internetshutdowns.in/static-page/jammu-kashmir/ - **447 internet shutdowns in J&K.** The page does not state an as-of date. - Per year: 2012: 3 · 2013: 5 · 2014: 5 · 2015: 5 · 2016: 10 · 2017: 32 · 2018: 65 · 2019: 55 · **2020: 116** · 2021: 79 · 2022: 43 · 2023: 14 · 2024: 2 · 2025: 11 · 2026: 2.

The 2019–21 shutdown - From 5 Aug 2019: "Internet, mobile, and landline services were shut down." - Restoration: landlines 18 Aug 2019; postpaid mobile 14 Oct 2019; broadband for essential purposes 15 Jan 2020; 4G on 5 Feb 2021. - In all, 550 days of partial or no connectivity, including 213 days with no internet. [V] https://pulse.internetsociety.org/en/shutdowns/internet-suspended-in-jammu-and-kashmir-india-5-august-2019/ - From Jan 2020, 2G only, on verified SIMs, with whitelisted websites. [V] (SFLC page)

India-wide counts (Access Now) - 2024: 84 shutdowns (116 in 2023). [V] https://sabrangindia.in/india-saw-84-internet-shutdowns-in-2024-access-now-report/ - 2025: 65 shutdowns. [V] https://www.accessnow.org/press-release/one-region-half-the-worlds-shutdowns-internet-shutdowns-in-the-asia-pacific-in-2025/ - Press coverage says J&K had 12 shutdowns in 2024. [S] https://thewire.in/rights/india-top-democratic-country-internet-shutdowns-2024-report . This conflicts with SFLC's figure of 2 for 2024 (different methods), so the 2024 J&K count is UNVERIFIED.

Design meaning - Shutdowns cluster around security events and can last months. - In the worst case (2019), voice and SMS also went down. Offline capability plus assisted, paper or QR fallbacks therefore matter, not just IVR.

2.2 Phones and internet in J&K (NSS 80th round, CMS: Telecom, Jan–Mar 2025)

Source: [V] https://www.mospi.gov.in/sites/default/files/publication_reports/CMST_report_m.pdf (state tables in Appendix A; figures for age 15+)

INDICATOR (J&K, AGE 15+)	RURAL MEN	RURAL WOMEN	RURAL ALL	URBAN ALL	J&K ALL
Used a mobile phone in last 3 months (Table 1)	90.3%	80.6%	85.8%	87.8%	86.3%
Owns a mobile phone (Table 2)	79.7%	**58.0%**	69.7%	82.5%	72.7%
Owners with a basic phone only, no smartphone (Table 2.1)	17.5%	22.7%	19.5%	14.7%	18.2%
Used internet in last 3 months (Table 3)	75.1%	61.8%	68.9%	74.4%	70.3%
Self-reported: sent a message with an attachment (photo, document) (Table 6)	77.2%	64.7%	71.4%	79.5%	73.3%
Self-reported: able to bank online (Table 12)	63.5%	45.5%	55.2%	68.0%	58.2%

Rural J&K **households** (Tables 13, 14, 14.1): - Smartphone-only households: 53.2%. Basic-phone-only: 7.3%. Both: 39.2%. So **about 92% have at least one smartphone** and **about 47% still have a basic phone**. - Landline: 2.1%. - 92.5% have internet at home. Of those, **99.0% use mobile networks** and only 6.3% fixed or Wi-Fi. - Of the 7.5% without internet, 48.0% say they "do not know how to use it". - The ICT-skill figures are self-reported and were not tested. [V] (report section 2.2)

Older figures - NFHS-5 (J&K surveyed in Phase I, 2019–20): 75.2% of women in J&K had a mobile phone they used themselves; 73.3% in rural areas. [S] https://www.thekashmirmonitor.net/nfhs-survey-57-3-of-women-own-houses-in-jk-84-have-bank-accounts-75-use-mobile-phones/ - TRAI: about 1.27 crore wireless subscribers in the J&K service area (Sept 2025). [S] https://m.greaterkashmir.com/article/reliance-jio-added-40000-mobile-subscribers-in-jk-service-area-in-september-trai/449178 - No figure for rural J&K over age 60 was found. UNVERIFIED for elderly growers.

2.3 Implications: offline-first, with SMS and IVR fallbacks (proposal, grounded in 2.1–2.2)

- **Offline-first app.**
- Cache advisories, spray calendars, videos and price history.
- Queue bookings and pest reports in an outbox that syncs when the network returns.
- Show "sent / waiting for network" states with icons.
- Store a booking token and PIN on the device so a job can close offline and reconcile later.
- **Voice/IVR line with missed-call call-back.**
- Useful when data is down but voice works, and for the roughly 1 in 5 rural phone owners with basic phones only.
- Precedent: the Mobile Vaani design. [V] https://solve.mit.edu/solutions/35346

- **SMS for confirmations** (booking IDs, PINs, prices) and **heads-up SMS before advisory calls**. [V] https://registry.precisiondev.org/registry_entry/impact-of-heads-up-sms-on-call-pickup-and-listening/
- **Assisted channel at KKGs.** The centre's entrepreneur can book on a grower's behalf and print or QR-code the booking. This survives total blackouts once local services resume. [V] https://kashmirreader.com/2026/01/19/hadp-ushers-a-new-era-of-agricultural-transformation-across-jk/
- **Keep the app light.** Most rural users rely on mobile data, so a small download and compressed media matter.

3. Non-invasive data collection

3.1 Progressive profiling and value before asking

- **IFFCO Kisan (GSMA 2016)** [V] https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2016/10/IFFCO-Kisan-Agricultural-App.pdf
- "user profiling can be a major roadblock to adoption and usage."
- "the drop down profiling can be a barrier for first time users."
- State and district had to be re-entered for weather and market modules.
- "Personalising the app based on user profiles would show users the value of sharing their personal information."
- **Progressive disclosure (Nielsen, NN/g, 2006)** [V] https://www.nngroup.com/articles/progressive-disclosure/
- "Initially, show users only a few of the most important options."
- Keep to at most 2 disclosure levels.
- **Forced accounts:** 18% of users abandon checkout when forced to create an account. [V] https://baymard.com/blog/reduce-cart-abandonment
- **Ask preferences early:** asking farmers their preferences at the start of the season gave the highest engagement. [V] https://registry.precisiondev.org/registry_entry/frequency-a-b-test

Proposed "data ladder" (proposal). Each rung unlocks a visible benefit, explains "why we ask" in audio, and can be skipped or deleted.

WHEN	ASK	BENEFIT SHOWN IMMEDIATELY
First open (no account)	Language (tap a flag or face plus audio); village or tehsil (tap a map or say the name; GPS optional with in-context permission)	Local weather, scab or frost alerts, mandi prices
First question or booking	Mobile number by OTP or missed call; name (may be recorded by voice)	Booking confirmations, call-backs
Opening the spray calendar	Varieties (picture cards); orchard age band; approximate tree count or area in local units (kanal/marla; conversion UNVERIFIED)	Personalised spray schedule, product dose calculator
Booking labour or trucks	Drop-pin of orchard gate or landmark; expected boxes	Quotes and arrival times
Only when the user asks for credit or schemes	Kisan Credit Card (KCC), bank, land records, via the KKG assisted mode	Scheme or credit application

3.2 Just-in-time permissions and consent

- **Android guidance** [V] https://developer.android.com/training/permissions/requesting
- "Ask for a permission in context, when the user starts to interact with the feature that requires it."
- "Don't block the user."
- Degrade gracefully if denied.
- "persistent nagging to reconsider is not respectful of their choice."
- Denying twice counts as a permanent denial, so ask only at the moment of need.
- Use "Only this time" for location, camera and microphone.
- **Material Design:** request critical, obvious permissions up front; request others in context. [S] https://m1.material.io/patterns/permissions.html
- **Proposal:** ask for camera when the user taps "report pest", location when they tap "orchard pin", and microphone when they tap "speak". Each request comes with a 5-second Kashmiri audio explanation and an "I'll do it later" button.

3.3 Community and crowd-sourced pest reporting examples

- **National Pest Surveillance System (NPSS), India**
- Launched 15 Aug 2024. Farmers photograph pests or damage, and experts and scientists analyse them, supported by AI. Aims to reduce farmers' dependence on pesticide retailers.
- [S] https://knnindia.co.in/news/newsdetails/sectors/agri-food/government-launches-ai-powered-national-pest-surveillance-system-to-empower-farmers ; [S] https://visionias.in/current-affairs/monthly-magazine/2024-09-12/economy/national-pest-surveillance-system-npss
- Crop coverage and user numbers are UNVERIFIED; sources conflict.
- **FAO FAMEWS (fall armyworm)**
- A crowd-sourced field-scouting app. Data is validated by national focal points and mapped globally.
- One FAO release reports 60,000 reports from 64 countries. A 2024 FAO brochure cites more than 10,000 users in more than 60 countries.
- [S] https://www.fao.org/newsroom/detail/fall-armyworm-scaling-up-response-across-the-globe/en (UNVERIFIED that the 60,000 figure comes from this particular release) ; [S] https://www.fao.org/3/cd0185en/cd0185en.pdf
- **CROPSAP (Maharashtra)**
- Scouts report from fixed and random plots twice a week. Experts analyse the data, and farmers get SMS advisories twice a week.
- Started after a 2008–09 soybean pest outbreak.
- [S] https://krishi.icar.gov.in/jspui/bitstream/123456789/22404/1/Tbu_Crp_Information%20and%20Communication%20Technology%20based%20Pest%20Surveillance.pdf
- Subscriber counts (15–50 lakh) are UNVERIFIED; sources conflict.
- **SKUAST-K apple scab forecasting (Kashmir)**
- About 20 automatic weather stations, 2 per district, for a disease forecasting and decision-support system. An app for growers was planned.
- Launch status UNVERIFIED. [S] https://risingkashmir.com/skuast-k-installs-20-automatic-weather-stations-in-kashmir

- In 2020, scab reportedly affected about 40% of the previous year's crop. [S] https://www.downtoearth.org.in/agriculture/fungal-scab-attack-can-spell-doom-for-kashmir-s-apple-growers-this-year-74631
- **Mobile Vaani:** user voice recordings are moderated before broadcast. [V] https://solve.mit.edu/solutions/35346
- **Proposal: "report and get help"** 1. The grower sends a photo and/or voice note. 2. They immediately get an answer: an automated first answer plus an expert follow-up. 3. Aggregated, location-blurred reports feed village-level alerts ("scab reported in 3 orchards near you"). 4. Never show who reported. Precise GPS is used only with consent.

3.4 Incentive designs

- **Airtime for IVR surveys** [V] https://pmc.ncbi.nlm.nih.gov/articles/PMC6747927
- Bangladesh: response rate 19.2% with no incentive vs 23.9% with 50 Taka (US$0.60) vs 24.8% with 100 Taka.
- Uganda: 35.2% vs 39.3% vs 40.3%.
- No significant difference between the two amounts.
- **Non-cash incentives (proposal):** earn priority booking slots, free expert call-backs, or a "verified grower" badge for completing your orchard profile or submitting validated pest reports.
- **Implication:** small, immediate, certain rewards beat large ones. Avoid making service access conditional on giving data, which DPDP also restricts (consent must be "unconditional").

3.5 DPDP Act 2023 and DPDP Rules 2025 (brief)

Rules and timeline - Notified **14 Nov 2025**, with an **18-month phased compliance** period. [V] https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc20251117695301.pdf - Most obligations (Rules 3 and 5–16, including notice) apply about **13 May 2027**. Consent-manager registration applies after 12 months. - [S] https://www.mondaq.com/india/data-protection/1713500/regulatory-update-notification-of-the-digital-personal-data-protection-rules-2025 ; [S] https://www.storyboard18.com/amp/digital/breaking-govt-notifies-final-dpdp-rules-sets-staggered-rollout-with-key-obligations-for-data-fiduciaries-84201.htm . The exact day is UNVERIFIED (13 or 14 May 2027).

Consent (s.6(1)) [V] https://dpdpa.com/dpdpa2023/chapter-2/section6.html - Must be "free, specific, informed, unconditional and unambiguous with a clear affirmative action". - Limited to the data necessary for the stated purpose. - Withdrawal must be about as easy as giving consent (s.6(4)).

Notice (s.5) [V] https://dpdpa.com/dpdpa2023/chapter-2/section5.html - Must state the personal data and the purpose, how to exercise rights, and how to complain to the Data Protection Board. - Must offer the notice "in English or any language specified in the Eighth Schedule". The Eighth Schedule includes Kashmiri, Urdu and Hindi; AI4Bharat lists Kashmiri (ks) among the "22 officially recognized languages of India". [V] https://huggingface.co/ai4bharat/indic-conformer-600m-multilingual

Rule 3 [V] https://www.dpdpa.com/dpdparules/rule3.html - The notice must be understandable **independently** of other information. - It must give an **itemised** description of the data and the purposes. - It must include a website or app link and other means to withdraw consent with comparable ease, exercise rights, and complain to the Board.

Legitimate use without consent (s.7(a)) [V] https://dpdpa.com/dpdpa2023/chapter-2/section7.html - A business may process data that a person voluntarily provided for a specified purpose, unless she indicated she does not consent. The Act's example is a pharmacy texting a payment receipt. - This plausibly covers using a phone number to confirm a booking the grower requested. That is my interpretation and needs legal review.

Children (s.9) [V] https://dpdpa.com/dpdpa2023/chapter-2/section9.html - Under-18s need **verifiable parental consent**. - No tracking, behavioural monitoring or targeted advertising directed at children. - This matters because younger family members often operate the phone for an elderly grower.

Rights and duties under the Rules [V] (PIB link above) - Rights to access, correct, update and erase, and to nominate another person to act on one's behalf. - Data-rights requests must be answered within **90 days**. - Breaches must be notified to affected people in plain language "without delay". - Organisations must display a contact person for data queries. - Consent managers must be India-based. - Penalties: up to **Rs 250 crore** for failing to maintain reasonable security safeguards; up to Rs 200 crore for breach-notification or child-related violations; up to Rs 50 crore for other violations.

Proposal: an audio consent notice in Kashmiri (with an Urdu/Hindi/English text version), itemising each data item next to its picture. Add a "Delete my data / Stop" option reachable within two taps, or via IVR or a KKG.

3.6 Gender

- **Phone access gap in rural J&K (age 15+):** phone ownership is 58.0% for women vs 79.7% for men. Internet use is 61.8% vs 75.1%. Self-reported online banking ability is 45.5% vs 63.5%. [V] (CMS:T 2025)
- **Women's roles in orchards**
- Women help with picking, sorting and packing during harvest rushes. [S] https://www.deccanherald.com/amp/story/india/jammu-and-kashmir/kashmirs-apple-growers-race-against-weather-extend-working-hours-to-save-harvest-4183948
- SheThePeople (Aug 2019) cites a 2014 report, *Women participation in Apple Cultivation in Rural Kashmir*:
  - women are 25% of the Kashmir apple industry workforce;
  - "Nearly all the apple picking is done by women only";
  - 68.5% of women in Shopian, 62% in Kupwara and 59% in Budgam take part in moving packed boxes to loading points.
  - [V] for the article https://www.shethepeople.tv/she-means-business/25-of-workforce-in-kashmirs-apple-industry-is-female/ ; the underlying 2014 report is UNVERIFIED (not seen, and its figures are internally inconsistent).
- Women's role in farm decision-making is described as limited in Kashmir agriculture generally (not apple-specific). [S] https://www.jkpi.org/women-in-kashmirs-agricultural-economy-challenges-and-contributions/
- Voice-programme reach among women depends on phone ownership, access and empowerment. [S] https://files.comminit.com/content/development-mhealth-behavior-change-communication-strategy-case-study-rural-uttar-pradesh
- **Proposal**
- Don't make a personal phone a precondition. Support household phones, basic-phone IVR, and women-led KKG sessions.
- Include content on post-harvest tasks (grading, packing, sorting) and women-led labour crews. About 50% of Bharat Intelligence's Nashik workers are women, often in husband-and-wife teams. [V] https://thebetterindia.com/innovation/ai-farm-labour-platform-bharat-intelligence-labour-os-nashik-12108095
- Let women hold their own profile and own consent, separate from the male household head.

3.7 Elderly growers relying on younger family members

- **FarmChat evidence** [V] https://ubicomplab.cs.washington.edu/pdfs/farmchat.pdf
- A participant: "My son doesn't let me use his phone as he thinks I might break it."
- An older user with shaky hands preferred big buttons.
- Agri-entrepreneurs used the app on others' behalf.
- **Intermediated use** is a normal pattern in low-income communities. [S] https://microsoft.com/en-us/research/?p=159423
- **The DPDP Rules let a person nominate someone to exercise their rights.** [V] (PIB link above)
- **Proposal: "family helper" mode**
- The grower stays the account owner; consent is recorded in the grower's own voice or confirmed by an IVR call to their number.
- The helper can act on their behalf, and every action is logged and read back by voice to the grower.
- Helpers under 18 trigger the child rules.
- Large text and big touch targets; voice read-back of every booking.

---

4. Services marketplace models (India) and how they handle trust and safety

4.1 Agricultural labour booking

- **Bharat Intelligence (Nashik, grapes; operating since 2025)** [V] https://thebetterindia.com/innovation/ai-farm-labour-platform-bharat-intelligence-labour-os-nashik-12108095
- Booking: by WhatsApp or voice call; a whole season can be booked in advance.
- Crews: workers organised into teams, guaranteed work across the season (up to 300 days a year).
- Verification: skill and reliability are mapped through village data and local influencers (sarpanches, CSC operators). A formal process is not described.
- Pricing and pay: farmers pay about Rs 25,000 per acre. Workers are paid by output, not daily wage.
- Reported results:
  - 3,000 workers deployed;
  - 86% repeat-farmer rate;
  - about 50% of workers are women;
  - monthly earnings up from Rs 6–7k to about Rs 20k.
- **This is the closest analogue for Kashmir pruning, spraying, picking and packing gangs.**
- **Digital Labour Chowk** (daily-wage, mainly construction) [S] https://thebetterindia.com/startup/chandrashekhar-mandal-digital-labour-chowk-job-portal-daily-wage-workers-10469048 ; [S] https://digitallabourchowk.com/?p=1034
- Verified worker accounts; app in 15 languages.
- Shopkeepers near labour chowks act as registration agents.
- Says it charges no commission. Claims more than 1 lakh workers helped (self-reported).
- **Kerala "Kaarshika Labour Bank" app** (planned, with ratings and reviews): current status UNVERIFIED. [S] https://opendigest.in/farm-worker-is-just-a-click-away-kerala-to-launch-app-for-farm-labour-services/
- **Haryana "ShramMitra"** (state app replacing labour chowks): [S] https://currentaffairs.adda247.com/shrammitra-app-launched-in-haryana-to-replace-traditional-labour-chowk-system/
- **FarmbyAPP and FarmWork** advertise "verified labour crews". These are marketing claims only, UNVERIFIED. [S] https://www.farmbyapp.com/ ; https://farmworkindia.com/

4.2 Truck and load booking

- **Vahak** (load board / marketplace) [V] https://www.vahak.in/mob/online-load-booking
- Says it is "100% free and has no commissions or hidden fees."
- Users can check transporters' "ratings, connections, recommendations".
- Its terms say "the parties are liable to each other for the payments, safety & security of the consignments". So the platform does not guarantee payment on that page.
- Its app listing claims checks of RC and permit, GST and business ID, and driver KYC. [S] https://appgoblin.info/apps/search.find.lorries.invite.transporter.add.lorries.deliveries.codenicely.search.vahak
- Press mentions "escrow payments". [S] https://www.maritimegateway.com/?p=39588 . Escrow is UNVERIFIED and not on Vahak's own page.
- Revenue reportedly comes from add-ons (GPS, insurance, financial services, premium services). [S] https://www.itln.in/logistics/vahak-plans-premium-services-offer-cargo-insurance-1348213
- **BlackBuck (Zinka Logistics; listed company)** [S] https://inc42.com/buzz/blackbuck-posts-inr-280-cr-profit-in-q4-on-tax-credit ; [S] https://www.indianstartuptimes.com/?p=27336
- Most revenue comes from tolling (FASTag) and telematics for truck owners, plus fuel payments and vehicle finance.
- The load marketplace moved from classifieds to transactional booking ("SuperLoads") and is still small.
- FY25 revenue about Rs 427–462 crore depending on source (UNVERIFIED exact).
- Lesson: the money is in recurring services to truck owners, not in brokering loads.
- **TruckSuvidha**
- Load and truck board. Verifies customers' mobile number, PAN card and service-tax number. [S] https://startuptalky.com/trucksuvidha-success-story/
- Claims 40,000+ transporters (self-reported). [S] https://startinup.up.gov.in/crm/Welcome/startup_user_details/Mjc4MQ==
- **e-NAM** (the national online agri-market platform) lists institutional logistics providers integrated with it, BlackBuck among them. [S] https://enam.gov.in/eNAM-Logistics-Providers
- **Porter:** not researched; search budget exhausted. UNVERIFIED.

4.3 Farm machinery rental

- **Government CHC Farm Machinery app ("FARMS")** [S] https://www.agrimachinery.nic.in/Files/FAQ_CHC_MobAPP.pdf

- Custom Hiring Centres (CHCs) get an in-app notification and SMS for each booking.
- The farmer gives a **6-digit PIN** received at booking to close the job.
- A claim of more than 40,000 CHCs and 1.2 lakh machines is UNVERIFIED (secondary source). [S] https://www.oliveboard.in/blog/chc-farm-machinery-app/
- **EM3 AgriServices** [V] https://techcrunch.com/2017/08/15/em3
- Pay-per-use, priced per hour or per acre.
- Call centres plus village representatives; 10 service centres and about 8,000 farms (2017).
- Business mostly offline. Recent status UNVERIFIED.
- **Mahindra Trringo**
- Launched 2016 as a franchise-based tractor and equipment rental, booked **by phone call**. [S] https://www.rushlane.com/mahindra-uber-for-tractors-launches-in-karnataka-12204416.html
- More than 100,000 hours of work by early 2018. [S] https://www.mahindra.com/news-room/press-release/en/mahindra-trringo-touches-lives-of-india-s-farmers-by-crossing-1-lac-hour-of-work-in-the-field
- Merged into M&M on 6 Jun 2024, with net worth of only Rs 1.82 crore, a signal of limited scale. [V] https://www.autocarpro.in/news/mahindra-heavy-engines-mahindra-two-wheelers-trringocom-complete-merger-with-mm-120906
- **KKGs in J&K** already list "custom hiring services". [V] https://greaterkashmir.com/national/kisan-khidmat-ghars-will-serve-as-one-stop-service-hub-for-farmers-lg

4.4 Expert tele-consultation

- **Kisan Call Centre (1800-180-1551)**
- Toll-free from any phone, 6 am–10 pm, 7 days a week. Answers in 22 languages. [S] https://mkisan.gov.in/Home/KCCFeature
- Whether **Kashmiri** is supported is UNVERIFIED.
- FarmChat cites a report that about 40% of 1.11 million KCC calls went unanswered (year not captured). [V] (FarmChat PDF, ref. 3)
- **IFFCO Kisan "Ask Expert":** over 95% of queries answered on average. [V] (GSMA PDF)
- **KKG app:** "booking expert consultations". [V] https://greaterkashmir.com/national/kisan-khidmat-ghars-will-serve-as-one-stop-service-hub-for-farmers-lg
- **NPSS:** photo upload leading to expert advice. [S] (link in 3.3)
- **FarmerChat:** AI first, with extension-worker review. [V]/[S] (links in 1.4)
- **Proposal:** a tiered service. 1. Instant AI or library answer (Kashmiri audio). 2. Scheduled call-back from a SKUAST-K/KVK or horticulture expert within X hours. 3. Paid or free orchard visit booked via a KKG.
- Show the expert's name, photo and institution to build trust. FarmChat found trust requires "competency and unbiased viewpoints". [V]

4.5 Cold-storage booking

- **ColdKonnect (Hyderabad)** [V] https://thedailyguardian.com/india/cold-konnect-a-three-generation-bet-on-indias-farmers-769044/
- Two apps: farmers and traders search and request storage or reefer trucks; storage owners and fleets get sensors and GPS.
- Listings are verified before going live.
- Partners pay a yearly subscription with "no commission on bookings or enquiries".
- Scale is not disclosed; the article is a press release.
- **Coldtivate / Virtual Cold Chain Assistant:** pay-per-use cooling launched in India in Jan 2021. [S] https://www.climatepolicyinitiative.org/publication/cooling-as-a-service-with-virtual-cold-chain-assistant
- **Integrated Cold Chain Availability Platform (ICCAP):** a national database linking cold-chain assets. [S] https://en.vikaspedia.in/viewcontent/agriculture/post-harvest-technologies/integrated-cold-chain-availability-platform
- **Kashmir controlled-atmosphere (CA) capacity (Feb 2026)** [V] https://www.greaterkashmir.com/article/jk-to-expand-ca-storage-capacity-for-horticulture-cm-omar-abdullah/476631
- 68 CA stores with **2.92 lakh MT**, against a requirement of **6.00 lakh MT**.
- Plans: +15,000 MT in 2025–26 and +38,000 MT in 2026–27; about 40 new stores worth Rs 1,400 crore.
- The article says nothing about how CA space is booked or what it costs.
- Most capacity is in South Kashmir (55 stores, 2.57 lakh MT); north Kashmir has 5 stores (0.19 lakh MT). [S] https://kashmirlife.net/south-kashmir-apple-output-nears-9-5-lakh-mt-as-ca-storage-reaches-2-57-lakh-mt-425691/
- Apple production was more than 21 lakh MT in 2024–25. [S] https://kashmirlife.net/apple-production-crosses-21-lakh-mts-in-jammu-kashmir-government-targets-expansion-of-cold-storage-capacity-411454/
- **Proposal:** CA space is scarce and concentrated. Start with a "request a slot / join waitlist" flow linked to CA operators, with transparent per-box rates (rates UNVERIFIED; not found), rather than instant booking.

4.6 Trust and safety patterns seen across platforms

MECHANISM	WHO USES IT	SOURCE
Provider KYC and document checks (RC, permit, GST, PAN)	Vahak (claimed), TruckSuvidha	[S] links in 4.2
Ratings and recommendations	Vahak; Kerala labour bank (planned)	[V] https://www.vahak.in/mob/online-load-booking ; [S] https://opendigest.in/farm-worker-is-just-a-click-away-kerala-to-launch-app-for-farm-labour-services/
PIN or OTP to close a job	Government CHC/FARMS app	[S] https://www.agrimachinery.nic.in/Files/FAQ_CHC_MobAPP.pdf
Local influencer vetting of workers (sarpanch, CSC operator)	Bharat Intelligence	[V] https://thebetterindia.com/innovation/ai-farm-labour-platform-bharat-intelligence-labour-os-nashik-12108095
Shopkeeper or agent registration desks	Digital Labour Chowk; EM3 village representatives	[S]/[V] links above
Verified listings before going live; subscription instead of commission	ColdKonnect	[V] link in 4.5

MECHANISM	WHO USES IT	SOURCE
Payment liability left to the parties (no guarantee)	Vahak's stated terms	[V]
Output-based pay for crews	Bharat Intelligence	[V]

Not found in the sources: - **Cancellation policies.** Porter, Vahak and the CHC app were not documented here; UNVERIFIED. - **Proposal:** publish a simple, symmetric rule. Example: a free cancellation window; after it, a small fee to whichever side cancels. Weather-triggered cancellations (rain on harvest day) should be free for both sides. State the rule by voice when booking.

4.7 Kashmir-specific transport and labour context

Freight shocks - In Sept 2025, highway disruption and a truck shortage pushed rates from about **Rs 60 to Rs 200 per box** (Shopian grower), and above Rs 200 via the Mughal Road. [V] https://kashmirlife.net/soaring-freight-charges-for-apples-continue-despite-govt-assurances-406072/ - Growers demanded uniform freight charges. JKRTC trucks were sent to the Shopian and Sopore mandis. [V] (same) - Reported JKRTC concessional rates per box: Delhi Rs 130, Haryana Rs 120, Jalandhar Rs 90, Jammu Rs 70. Private trucks charged about Rs 200–230. [S] https://risingkashmir.com/index.php/fruit-on-the-brink-govt-deploys-rtc-fleet-to-rescue-apple-transport ; [S] https://www.etvbharat.com/en/!state/exorbitant-transportation-charges-put-kashmir-apple-growers-into-more-distress-enn25092103015 - The Shopian mega mandi shut amid the transport crunch. [S] https://m.greaterkashmir.com/article/shopians-mega-fruit-mandi-shuts-amid-transport-crunch-soaring-freight-charges/433617

Unions and strikes - In the January 2024 nationwide truckers' strike, the **Kashmir Valley Fruit Growers-cum-Dealers Union** (chairman Bashir Ahmad Basheer) warned of losses and asked for negotiations. [V] https://kashmirlife.net/trucker-strike-triggers-apple-concerns-in-kashmir-336977/ - **No union-published rate card or formal booking system was found** (UNVERIFIED either way). - **Implication:** a marketplace showing JKRTC benchmark rates next to private quotes, with verified trucks, addresses a real pain point. Expect pushback from incumbent transporters.

Non-local labour - Orchardists, especially in south Kashmir, hire non-local workers for picking and packing. They come mostly from Bihar and Uttar Pradesh. [S] https://www.deccanherald.com/india/apples-left-hanging-as-kashmir-growers-halt-harvest-amid-highway-blockade-3730124 - At one large cold store, 150 of 210 workers are migrants. [S] https://themigrationstory.com/post/backbone-of-operations-how-migrants-keep-south-asias-largest-apple-cold-storage-running/ - Estimates of the non-local workforce range from about 4 lakh to 5–10 lakh. UNVERIFIED. [S] https://thekashmirimages.com/2025/02/24/kashmirs-labor-scenario-is-kashmir-shaping-a-mini-bihar - Reported daily wages: Rs 800–900, up from Rs 450–500. Undated, so UNVERIFIED. [S] https://www.nationalheraldindia.com/india/panic-among-kashmirs-apple-orchardists-due-to-adverse-weather-forecast-high-cost-because-of-workers-exodus - Supply collapses after attacks on migrants: the 2019 exodus, and later incidents. [S] https://www.deccanherald.com/national/north-and-central/kashmir-exodus-of-migrant-labourers-affect-harvesting-763613.html

Police registration - After attacks in South Kashmir, a J&K Police advisory (article dated 6 Aug 2026) says newly arriving labourers must **"register at the nearest police station"**. Identity and place of stay undergo "strict security verification before permission is granted". Labourers are advised to relocate from isolated sites (e.g., brick kilns) and to stay indoors after dark. [V] https://www.wionews.com/india-news/j-k-police-issues-new-advisory-for-non-local-labourers-after-recent-terror-attacks-15-lakh-reward-announced-for-let-terrorist-1786002596846 - District orders (Samba, Kathua) require contractors and property owners to disclose or verify outside workers. Dates unclear. [S] https://m.greaterkashmir.com/article/disclosure-of-outsiders-working-residing-in-j-ks-samba-mandatory-for-property-owners-contractors/462844 ; [S] https://www.uniindia.com/north-india/security-jk-labourer-verification/206692 - In the past, the IGP Kashmir called one viral order on non-locals "fake". Verify any order against official channels. [S] https://thekashmirwalla.com/?p=77493 - A J&K Single Window procedure exists for principal employers of inter-state migrant workers. I could not open it, so thresholds and applicability to orchards are UNVERIFIED. https://singlewindow.jk.gov.in/assets/services/sop_file/84/sop_principal_emp_migrant_labour.pdf - **Implication (proposal):** - The labour module should record each gang member's police-verification status, with an upload or "verified at police station X on date" field. Store only what is needed, with consent. - Make the gang leader (thekedar, or labour contractor) the accountable party. - Add a safety layer: share a safety-alert channel and avoid publishing workers' locations. - Treat all of this as sensitive data under DPDP.

Marketing structure - Most produce moves through commission agents. One J&K study: 61% of production goes producer → commission agent → wholesaler → retailer. - Pre-harvest contractor channels carry about 17%. [S] https://ums.iust.ac.in/UploadedDocuments/EmployeeDocuments/ResearchPapers/1325_NF8MBD.pdf - Channel choice depends on the grower's source of finance. [S] https://www.pbr.co.in/2020/May7.aspx - Azadpur (Delhi) is the main destination. [S] (same IUST paper) - **Implication:** commission agents are both the incumbent lenders and the transport organisers. The app should complement them, not try to disintermediate them on day one.

5. Business models of Indian agri platforms

5.1 Snapshot (FY25 unless noted)

COMPANY	MODEL	KEY NUMBERS	SOURCE
DeHaat	Full-stack: inputs, output trading, advisory; franchised DeHaat Centres run by micro-entrepreneurs	Operating revenue Rs 3,010 cr; **~80% from agri-output (spices, staples), ~20% inputs**; net loss Rs 207 cr (or Rs 369–370 cr profit including non-cash fair-value gains); EBITDA margin −5.78%; reports EBITDA-positive Q1 FY26	[V] https://entrackr.com/fintrackr/dehaat-cuts-losses-by-15-to-rs-207-cr-in-fy25-10504383 ; [S] https://inc42.com/buzz/dehaat-turns-profitable-in-q1-fy26-eyes-full-year-profitability/ ; [S] https://inc42.com/features/inside-dehaats-inr-3000-cr-scale-and-its-long-road-to-profitability
AgroStar	Agri-input commerce, app plus agronomy advice	Revenue about Rs 850 cr; net loss about Rs 143.5 cr (down 56%); about 97% of revenue is product sales, services about Rs 13 cr	[S] https://entrackr.com/fintrackr/agrostar-crosses-rs-850-cr-revenue-in-fy25-cuts-losses-11171664 ; [S] https://lapaasvoice.com/startup/agrostar/business-model/
BigHaat	Inputs plus produce trading	Revenue about Rs 1,100 cr; claims EBITDA positive; net loss about Rs 25–29 cr; **85% of revenue from produce sales**; gross margin about 15%	[S] https://entrackr.com/exclusive/exclusive-bighaat-crosses-rs-1100-cr-revenue-in-fy25-turns-ebitda-profitable-9755082 ; [S] https://www.indianstartuptimes.com/?p=27459
Gramophone (Indore)	Inputs plus advisory; credit via Jai Kisan	FY24 revenue Rs 98.2 cr (down 69% after closing its output vertical); FY25 about Rs 67–71 cr; Unnati acquisition (Jan 2026) UNVERIFIED	[S] https://inc42.com/buzz/gramophones-fy24-revenue-slumps-69-to-inr-98-cr ; [S] https://www.indianstartuptimes.com/?p=27461
Ninjacart	B2B fresh-produce marketplace and fulfilment (margin between buy and sell prices)	Revenue Rs 1,634 cr (down 19% after dropping low-margin FMCG trading); loss Rs 256 cr; targets profit in FY27	[S] https://www.theweek.in/wire-updates/business/2025/12/26/dcm46-biz-agri-ninjacart.html

COMPANY	MODEL	KEY NUMBERS	SOURCE
Arya.ag	Warehousing, finance (NBFC Aryadhan, partner banks) and commerce	Net revenue about Rs 447 cr; PAT about Rs 32 cr (+70%); financing facilitated Rs 14,181 cr (Rs 2,000 cr via Aryadhan and partners); revenue mix about 50–55% storage, 25–30% finance; "near-zero NPAs" (self-reported)	[S] https://entrackr.com/fintrackr/aryaag-reports-rs-447-cr-revenue-in-fy25-profits-spike-70-9048421 ; [S] https://yourstory.com/2025/05/aryaag-pat-jumps-70-rs-32-cr-in-fy25-facilitates-rs-14181-cr-agri-loans ; [S] https://finance.yahoo.com/news/even-global-crop-prices-fall-070000157.html
BlackBuck (logistics comparator)	Truck-owner services: tolling, telematics, fuel, finance; small load marketplace	FY25 revenue about Rs 427–462 cr	[S] links in 4.2

Pattern: the big "farmer platforms" earn most of their revenue from trading (output commodities or inputs) at thin margins. Advisory itself earns almost nothing (AgroStar services about 1.5% of revenue). The clearly profitable model, Arya.ag, earns fees on physical infrastructure (storage) plus secured lending.

5.2 Input credit and "pay after harvest"

- **Gramophone with Jai Kisan (2022)** [V] https://agfundernews.com/gramophone-evolves-from-marketplace-to-full-stack-platform-with-new-lending-solutions-for-farmers
- Underwriting used farm size, crops, irrigation and transaction history collected during advisory.
- Eligible farmers got an input purchase limit. They **paid interest and fees upfront and repaid the principal after harvest.**
- **Jai Kisan**
- Acquired an NBFC licence via Kushal Finnovation. [S] https://inc42.com/buzz/jai-kisan-secures-nbfc-licence-after-picking-up-stake-in-kushal-finnovation
- Co-lends with banks, including public-sector banks. [S] https://inc42.com/buzz/rural-fintech-startup-jai-kisan-nets-inr-26-5-cr-from-existing-investors/
- **farMart**
- Credit linked to input purchases at partner retailers, repaid in a lump sum at harvest. [S] https://bfaglobal.com/wp-content/uploads/2020/06/About-farMart.pdf
- Its cashless model worked locally but could not scale without collection staff, and it shifted to merchants. [V] https://inc42.com/?p=235037
- **DeHaat** planned input credit based on members' transaction history. Old source; current status UNVERIFIED. [S] https://agfundernews.com/?p=12335
- **Arya.ag** lends against warehouse receipts. It partnered with NBFC MAS Financial for small farmers and farmer producer organisations (FPOs). [S] https://agritimes.co.in/investment-finance/arya-ag-tie-up-with-mas-financial-services-to-offer-agri-finance-to-small-farmers-and-fpos
- **J&K:** the Kisan Sathi portal routes credit applications to banks digitally. [V] https://kashmirreader.com/2026/01/19/hadp-ushers-a-new-era-of-agricultural-transformation-across-jk/
- **Why agri-fintech struggled (Inc42, Dec 2020)** [V] https://inc42.com/?p=235037
- Thin data: banks want 4–5 years of history; land records are outdated.
- Cash repayments are hard to scale.
- An NBFC licence takes more than a year, so startups operate as business correspondents and lenders ask for 20–40% guarantees.
- Local intermediaries lend in about 5 days vs about 20 days for government loans.
- Informal borrowing rises mid-season, when unplanned costs hit.
- **Implication for Kashmir (proposal)**
- Input credit should run only through a regulated partner (a bank or NBFC, possibly J&K Bank or KCC-linked).
- Repayment should be tied to apple sale proceeds, for example through commission-agent settlement or CA-storage receipts, the Arya.ag analogue.
- This needs collections and legal design before launch.

5.3 Failure lessons

- **BharatAgri** (advisory plus input e-commerce, more than 1 million users) shut down in Nov 2025 after failing to raise funds: "High customer acquisition costs and low repeat orders." [V] https://www.peoplematters.in/news/business/bharatagri-shuts-shop-after-failing-to-secure-new-funding-round-47212
- **Otipy** (farm-to-consumer) shut on 17 May 2025 amid a cash crunch, with about 300 employees affected. A reported Rs 50 cr in unpaid dues and a CAC of Rs 150 vs LTV of Rs 80 are single-commentator figures, UNVERIFIED. [S] https://inc42.com/buzz/otipy-shuts-operations-amid-cash-crunch ; [S] https://yourstory.com/2025/05/farm-to-consumer-startup-otipy-shuts-down-employees-customers-seek-clarity-layoff
- **Gramophone** revenue fell 69% after it closed its output vertical. [S] https://inc42.com/buzz/gramophones-fy24-revenue-slumps-69-to-inr-98-cr
- **Ninjacart** cut low-margin (1–2%) FMCG trading. [S] https://www.theweek.in/wire-updates/business/2025/12/26/dcm46-biz-agri-ninjacart.html
- **Trringo** was absorbed into its parent with a net worth of Rs 1.82 cr. [V] (link in 4.3)
- **Funding collapse:** agritech funding fell 78% to $178M in 2023, rose about 30% in 2024, then fell 58% year on year in H1 2025 to $96M (Tracxn-type data; other trackers differ). [S] https://entrackr.com/report/indias-agritech-at-inflection-point-funding-revival-profitable-models-and-consolidation-11036024
- **Industry analysis:** the early direct-to-farmer model assumed scale would make up for low margins, but the opposite held. Durable businesses address productivity, logistics and finance. [S] https://yourstory.com/2026/04/what-happened-when-scale-met-the-farm-reality
- **Offline-heavy operations are the norm** (EM3) [V], and agent models are costly (GSMA) [S].

Implications for the Kashmir app (proposal) 1. Earn from **transactions that already involve money and pain**: freight, labour gangs, CA space, spraying services. Use a small, transparent booking fee or a provider subscription (the ColdKonnect model). 2. Keep advisory free; it drives acquisition and trust. 3. Use the KKG network for distribution rather than paid customer acquisition. 4. Add credit only with a regulated partner. 5. Measure repeat bookings per season, not downloads.

6. What I could not verify, or did not research (search budget ran out)

Language and literacy - Literacy in Kashmiri vs Urdu: the 32.7% vs >95% survey figure and the Kak (2001) "97%" claim are UNVERIFIED. - Whether Gboard voice typing, Bhashini, Google or Microsoft speech services support **Kashmiri speech**: UNVERIFIED. No published Kashmiri word-error rate for IndicConformer. - Whether the Kisan Call Centre answers in **Kashmiri**: UNVERIFIED.

Connectivity - 2019–20 SMS restoration dates and the prepaid-voice timeline: not confirmed (ISOC gives landline, postpaid, broadband and 4G dates only). - J&K shutdown count for 2024: conflicting (SFLC says 2; press coverage of Access Now says 12). - Rural smartphone access for J&K adults over 60: no state-level age breakdown found.

Gender - Women's roles in Kashmir apple farming: only journalism and a 2014 report cited second-hand. No peer-reviewed apple-specific study was found.

Kashmir marketplace context - CA-storage **rent per box** and how CA space is booked in Kashmir: not found. - Truck-union rate cards in Kashmir and how growers currently book trucks: not found. - Current daily wages for picking and packing gangs (2025–26): only undated or modelled figures. - Applicability of migrant-worker registration (J&K Single Window SOP, labour codes) to individual orchard owners: UNVERIFIED.

Platforms - **Porter**, **Kisan Rath**, **Plantix** community features, **Jugalbandi** (WhatsApp), **Samunnati**, Celcius cold-chain, Gram Vaani's current scale, and cancellation policies of booking apps: not researched. - CHC/FARMS app usage numbers: only a secondary claim. - Several financial figures (BlackBuck, BigHaat, Gramophone FY25, Arya.ag PAT growth of 70% vs 79%) vary across trackers.

Appendix B · Market, logistics, weather, schemes and labour

Desk research notes, November 2026. Tags: [V] read at source (in Appendix D: confirmed by an independent verifier); [S] from a search summary or not re-confirmed; UNVERIFIED = lead only.

Research for an app for apple growers in Kashmir. Compiled 12 November 2026.

How to read this file - Every fact has its source link beside it. Every statistic gives its year or date. - **UNVERIFIED** means one of three things: I could not confirm it against a primary or reputable source in this session, the sources disagree, or it comes from background knowledge I did not re-check. - "(search summary)" means the fact came from a search-engine snippet of that page. I could not open the page itself. - Rupee figures are as published. "LMT" means lakh metric tonnes (1 LMT = 100,000 t). "q" means quintal (100 kg). - **Limits of this research:** the shared web-search budget ran out partway through. Labour wages, the number of CA store units, the November 2018 and 2019 snow events and some helpline numbers could not be checked (see section 8). I did not call the data.gov.in API, because using a key found in search results was blocked. API details come from documentation and third-party write-ups.

0. Quick reference: the best data sources to build on

NEED	BEST SOURCE	ACCESS	NOTES
Daily mandi prices (machine-readable)	data.gov.in OGD API, dataset "Current Daily Price of Various Commodities from Various Markets (Mandi)", resource ID `9ef84268-d588-465a-a308-a864a43d0070` (dataset page; usage notes)	Free API key from a data.gov.in account	Prices are in Rs per quintal. Kashmir markets report only during the season, roughly August to March (SKUAST-K preprint, 2026). Test the key and the filters before building on it.
Authoritative price portal (manual or scrape)	Agmarknet 2.0, upgraded November 2025 (agmarknet.gov.in), with the Agmarknet 2.0 mobile app (PIB via Global Agriculture, 17 Dec 2025)	Web and app. No public API documentation found.	4,367 mandis linked as of December 2025 (same source).
e-NAM trade prices	enam.gov.in live dashboard, which is public (PIB doc, April 2026, search summary)	Dashboard only. No documented public API.	17 J&K mandis are on e-NAM (J&K Economic Survey 2025-26).
National Horticulture Board bulletin	NHB "Daily Price-Arrival Market Bulletin". Centres include Srinagar, Jammu and Delhi (NHB report page)	Web form	Whether the data is current is UNVERIFIED.
J&K Horticulture (Planning & Marketing) rates	dhpm.jk.gov.in has "Wholesale Rates – Within State / Outside State" menus	Web	On 12 Nov 2026 both pages held no data, only placeholder entries (within, outside). Not usable as a feed.
Road status (NH-44, Mughal Road, SSG)	J&K Traffic Police on X, @Traffic_hqrs: timed updates and next-day advisories. Traffic Control Unit (TCU) phone lines (section 3.3).	X / phone	No official API found. Daily advisories are republished by Kashmir Life and others.
Weather (official)	IMD API platform api.imd.gov.in, API reference. Endpoints for district warnings, district nowcast, city forecast, AWS data and district rainfall forecast.	Register on the portal. Organisations contact IMD's nodal officer (IMD API doc).	The index also lists Highway Nowcast/Warning and Agromet Advisory APIs, but their details are not on the page.
Weather (easy fallback)	Open-Meteo	Free tier for non-commercial use only: 600 calls/min, 5,000/hour, 10,000/day, 300,000/month. Paid plans for commercial apps. Data under CC BY 4.0.	An app with ads or subscriptions needs a paid plan (terms).
Agromet advisories	IMD Agromet Advisory pages (Gramin Krishi Mausam Sewa). Meghdoot app. IMD Srinagar "Agromet bulletin" PDF (IMD Srinagar, IMD home).	Web/PDF/app	No district agromet API found (the API index item has no details).
Official statistics	J&K Economic Survey 2025-26 (PDF, February 2026): district-wise apple area and production, and mandi-wise dispatches	PDF	The best single official source for 2024-25 figures.

1. Sector statistics

1.1 Area and production (J&K)

METRIC	VALUE	YEAR	SOURCE
Area under apple	**1,73,073 ha**, 50.01% of the UT's horticulture area of 3,46,098 ha	2024-25	J&K Economic Survey 2025-26, Stat. Appendix Tables 4.2.2 and 4.2.3.1. The same 173,073 ha figure appears in Mongabay, 13 Oct 2026, citing the Digest of Statistics 2024-25.
Area under apple, 20 years earlier	1,07,930 ha	2004-05	Mongabay, 13 Oct 2026
Apple production (official)	**2,110.02 thousand MT (21.10 LMT)**	2024-25	Economic Survey 2025-26, Table 4.2.4
Split by division	Kashmir 20.79 LMT; Jammu 30,393 MT	2024-25	Assembly reply (Kashmir Life, 30 Oct 2025). This matches the district table: the 10 Kashmir districts sum to 2,079.63 thousand MT (my arithmetic).
Earlier government series	20.40 LMT (2022-23); 20.64 LMT (2023-24); 20.56 LMT (2024-25, provisional, March 2025)	2022–25	Kashmir Life, 6 Mar 2025. The 2024-25 figure was later revised to 21.10 LMT.
IASRI series (as cited)	J&K 20.45 lakh t vs Himachal 5.83 lakh t	2024-25	The Tribune explainer, 7 Nov 2026
2025 crop, in-season estimate	22.15 LMT (government estimate)	2025	Government statement in the Assembly (Kashmir Vision, 29 Oct 2025). A final 2025-26 figure was not found (UNVERIFIED).

METRIC	VALUE	YEAR	SOURCE
2026 crop	No official estimate found. Reports cite hail and scab losses; one report estimates a 20–30% cut in output.	2026	Kashmir Vision, 22 Oct 2026 (UNVERIFIED estimate). USDA, by contrast, says J&K weather in MY 2025/26 was more favourable than in 2024/25 (USDA FAS IN2026-0056, 3 Nov 2026).
Average productivity	~12.2 t/ha	2024-25	My calculation from the Economic Survey area and production figures
Total horticulture (fresh and dry fruit)	26.92 LMT on 3.46 lakh ha (2024-25); 27.35 LMT on 3.47 lakh ha (2025-26, up to Nov 2025)	2024–25	Economic Survey 2025-26, Table 4.2(ii)

1.2 District-wise apple area and production, 2024-25 (official)

Source for all rows: J&K Economic Survey 2025-26, Statistical Appendix Tables 4.2.3.1 and 4.2.4. The share and yield columns are my arithmetic.

RANK	DISTRICT	APPLE AREA (HA)	PRODUCTION ('000 MT)	SHARE OF UT OUTPUT	YIELD (T/HA)
1	Baramulla (incl. Sopore belt)	25,928.10	479.25	22.7%	18.5
2	Anantnag	19,149.57	285.00	13.5%	14.9
3	Shopian	21,811.50	269.66	12.8%	12.4
4	Kupwara	19,593.67	233.05	11.0%	11.9
5	Kulgam	18,255.00	218.40	10.4%	12.0
6	Pulwama	15,928.00	212.85	10.1%	13.4
7	Budgam	15,174.38	162.82	7.7%	10.7
8	Ganderbal	7,808.00	122.16	5.8%	15.6
9	Bandipora	5,854.00	75.71	3.6%	12.9
10	Srinagar	1,779.75	20.73	1.0%	11.6
–	Jammu-division districts (Ramban 7.59, Kishtwar 7.02, Doda 5.75, Rajouri 4.18, Kathua 1.78, Poonch 1.63, Reasi 1.24, Udhampur 1.21)	~21,791 combined	~30.4	~1.4%	low
	UT total	**173,073.10**	**2,110.02**	100%	12.2

An older Greater Kashmir piece (2 Jan 2025) gives the same Baramulla, Shopian and Anantnag numbers (GK).

1.3 J&K's share of India's apple output

- USDA FAS says J&K is "approximately 70 percent" of India's production, Himachal nearly 20% and Uttarakhand 2.5–6% (report dated 3 Nov 2026). USDA FAS IN2026-0056
- USDA's India totals: **2.55 MMT** (MY 2024/25), **2.505 MMT** (MY 2025/26, revised down because of Himachal weather losses) and a forecast of **2.59 MMT** (MY 2026/27). Same source.
- My arithmetic: J&K's official 2.110 MMT against USDA's 2.55 MMT for 2024-25 gives about **83%**. Using the IASRI figures in the Tribune, J&K is about 78% of the J&K plus Himachal total. The methods differ, so quote it as "about 70–80%" and give the source.
- Other shares in circulation: 75% (Times Headline, 2025, search summary) and 78% (ETV Bharat, 24 Apr 2026).
- Himachal comparison for 2026: about 1.37 crore boxes dispatched by the end of harvest, against 3.49 crore the season before (Tribune, 7 Nov 2026).

1.4 Livelihoods and value

METRIC	VALUE	DATE	SOURCE
People dependent on horticulture	~35 lakh (3.5 million), directly or indirectly	ES 2025-26 (Feb 2026); Assembly, Sept 2026	Economic Survey 2025-26; Rising Kashmir, 25 Sep 2026
Families	~7 lakh (ES 2025-26); **7.5 lakh** (Assembly reply, 24 Sep 2026)	2026	Same two sources
Annual value	"Roughly Rs 10,000 crore in revenue" (ES 2025-26); "~Rs 12,000 crore annually, ~10% of UT GDP" (Assembly, Sept 2026). Media estimates range from Rs 15,000 crore (GK, Dec 2025) to Rs 20,000 crore (GK, Jul 2025), both UNVERIFIED.	2025–26	As listed
Labour intensity	"One hectare of orchard generates about 400 man-days of work per year"	Sept 2026	Rising Kashmir, 25 Sep 2026
Apple trees	~45 million trees	2026	ETV Bharat, 12 Nov 2026 (UNVERIFIED)
Fruit dispatched outside J&K	Fresh fruit 17.13 LMT (2021-22), 18.31 (2022-23), 13.59 (2023-24), 13.35 (2024-25), 8.60 (2025-26 to Nov 2025). Fresh-fruit revenue Rs 4,171.66 crore (2024-25).	2021–25	Economic Survey 2025-26, Table 4.2(ii)

1.5 High-density (HD) plantations, status

- High- plus medium-density plantations cover **20,034.44 ha** cumulatively up to 2025 (1,119.21 ha HD and 18,915.23 ha medium density). 29.13 lakh HD plants were distributed in 2024-25, against 6.41 lakh in 2021-22. ES 2025-26
- 1,103.79 ha of HD apple in the Kashmir Division since the scheme began, about 4,428 farmers. About 90% of that area is Gala and Red Delicious strains. Assembly reply, Greater Kashmir, 27 Sep 2026
- A private drive by Qul Fruits aimed to plant about 1.5 million HD trees by April 2026 (Kashmir Reader, 14 Mar 2026, search summary).

2. Markets and prices

2.1 Major Kashmir and J&K fruit mandis: fruit dispatched outside J&K, 2024-25

Source: J&K Economic Survey 2025-26, Stat. Appendix Table 4.2.7 "Mandi-wise Fruit Exported outside J&K – 2024-25". Also reported by FreshPlaza, 10 Feb 2026.

MANDI	DISTRICT	FRUIT SENT OUTSIDE J&K ('000 MT)	TURNOVER (RS CRORE)	SHARE OF MANDI DISPATCHES (MY CALC)	NAME ON AGMARKNET (WHERE SEEN)
Sopore	Baramulla	289.22	1,387.50	45.4%	"Sopore (Nowpora) (F&V)"
Aglar, Shopian	Shopian	121.19	628.34	19.0%	"APMC/MAC Shopian"
Jablipora	Anantnag	64.86	282.90	10.2%	not seen in the Nov 2026 snapshot
Handwara	Kupwara	36.16	113.00	5.7%	"Handwara Kupwara (F&V)"
Parimpora, Srinagar	Srinagar	32.29	165.06	5.1%	reported in Jan 2026 (aggregator)
Prichoo	Pulwama	22.44	116.99	3.5%	UNVERIFIED
Narwal, Jammu	Jammu	17.70	113.18	2.8%	"MAC F&V Market Narwal Jammu"
Kulgam	Kulgam	16.81	67.62	2.6%	reported late 2025 (aggregator)
Pachhar	Pulwama	13.56	70.56	2.1%	UNVERIFIED
Kanispora	Baramulla	10.39	47.94	1.6%	–
Zaloosa, Charar-e-Sharief	Budgam	5.65	26.12	0.9%	"Zaloosa-Chararishrief (F&V)"
Zazna	Ganderbal	4.67	19.97	0.7%	"Market Administrative Committee Zazna"
Kupwara	Kupwara	2.63	132.38 (as printed)	0.4%	–
Total		**637.57**	**3,171.56**		

Notes: - Sopore is commonly called "Asia's second-largest fruit mandi", after Azadpur (Rising Kashmir, 3 Nov 2026; Wikipedia). A claimed turnover of about Rs 7,000 crore in 2025, cited on Wikipedia from Greater Kashmir, is UNVERIFIED and does not match the official Rs 1,387.50 crore above. The official figure covers outside-J&K dispatches only. - Sopore volumes in early November 2026: about 200–250 trucks a day and 2.5–3.5 lakh boxes a day, at about 3,000–3,500 boxes per truck (trader estimates, Rising Kashmir, 3 Nov 2026; Daily Excelsior, 3 Nov 2026). - J&K has **19 wholesale mandis, 17 on e-NAM**. Cumulative e-NAM trade is Rs 1,680.34 crore for 28.26 lakh quintals, with 55,355 registrations, up to Nov 2025. e-NAM trade was Rs 596.55 crore in 2024-25 (ES 2025-26). The department website instead says "17 out of 24 functional mandis" (dhpm.jk.gov.in). - On e-NAM by district (2024-25): Pulwama 2, Baramulla 2, Kupwara 2; one each in Anantnag, Kulgam, Shopian, Srinagar, Ganderbal, Budgam, Jammu, Udhampur, Ramban, Kathua and Rajouri (ES Table 4.2.6). - First inter-state e-NAM trade: 11 t of Kashmir apples and pears went to Gultekdi APMC, Pune, in August 2025 (Kashmir Convener, Mar 2026). - 2019 MIS procurement points were Sopore, Parimpora, Shopian and Batengoo (Anantnag) (Tribune, 10 Sep 2019).

2.2 Destination markets

- **Azadpur APMC (Delhi)** is India's largest fruit and vegetable wholesale market and the main price-discovery point for Kashmir apples (SKUAST-K preprint, 2026).
- Other destinations named in 2025–26 reports: Punjab and Haryana mandis (Jalandhar), Maharashtra, Bihar (Chhath demand), West Bengal, and exports to Nepal, Bangladesh and Bhutan (Daily Excelsior, 3 Nov 2026; ETV Bharat, 21 Sep 2025; GK, 29 Sep 2025; FreshPlaza, 20 Oct 2025).
- India's apple exports in MY 2025/26 were 35,633 t: Nepal 17,802, Bangladesh 16,751, Bhutan 751, UAE 298 (USDA FAS, 3 Nov 2026).

2.3 How price discovery works

- **Players.** The Shopian (Aglar) and Sopore APMCs connect growers, orchard contractors (pre-harvest buyers of the standing crop), packing-house operators, commission agents (arhtiyas) and transporters. Commission agents give growers credit in advance and keep a percentage of the sale value. This creates a credit dependency that weakens growers' bargaining power when supply is congested (SKUAST-K preprint, Oct 2026, citing Mir & Mir 2018, Gulati et al. 2018 and GoJK 2021).
- **Consignment model.** Much of the crop is consigned through local forwarding or commission agents to agents in Azadpur and other terminal markets. The price is set at the terminal auction, and freight and commission are deducted from the grower's proceeds. A NABARD draft, reported by DNA (year unclear and dated), estimated about 70% of output is "captive supply" tied to agents' credit (search summary, UNVERIFIED). Commission is often cited at about 6% at Azadpur, but that figure is from a 2010 Himachal report (Tribune 2010, search summary). The current rate in Kashmir is UNVERIFIED.
- **Private channels.** Some private companies run their own warehouses or CA stores and pricing schemes. They bypass the mandis and commission agents and pay premium prices for top grades, with strict grading (USDA FAS, Nov 2026).
- **How prices travel.** Shopian chains adjust to Azadpur prices about twice as fast as Sopore chains. The mean producer-to-terminal spread was about Rs 9.74/kg over 2015–2025, peaking at Rs 46.70/kg during disruptions (SKUAST-K preprint).
- **Grades used in the trade.** A, B and C grades. Premium fruit is packed as "char tahi" (four layers) and "do tahi" (two layers) (Rising Kashmir, 3 Nov 2026). Common varieties: Delicious (Royal/Red Delicious), American (Richared), Kullu (Kulu) Delicious, Maharaji, Hazratbali, and HD varieties such as Gala.

2.4 Per-box vs per-kg selling and carton standardisation (current status)

- **J&K: no official per-kg or universal-carton mandate found** (UNVERIFIED that none exists). In April 2026 growers in Sopore were still demanding standard carton sizes and weights, and an end to low-quality "silicate" boxes. They said they had talked to banks and carton makers (Chenab Times, 14 Apr 2026).
- Box sizes in reports: 14–19 kg (Down To Earth, 18 Sep 2025); 15–16 kg cartons (FreshPlaza/AIKS, 1 Oct 2026); traditional boxes of 16–20 kg against 10 kg HD boxes (Kashmir Life, 1 Aug 2025). CA stores price storage per 17 kg crate (101Reporters, 21 Oct 2026).
- **What this means for the app (my inference, UNVERIFIED).** J&K APMC prices on Agmarknet appear to be box prices converted at 16 kg per box. Shopian's Rs 3,125/q equals Rs 500 per 16 kg box. Zazna's Rs 5,625 and 6,250/q equal Rs 900 and Rs 1,000 per box. Show both Rs/kg and an estimated per-box price, and state the box-weight assumption.
- **Himachal comparison.** Himachal moved to weight-based sale and universal cartons. Reports describe patchy enforcement (HimbuMail, search summary; Tribune, search summary). Implementation dates are UNVERIFIED in this session.

2.5 NAFED Market Intervention Scheme (MIS): history

ITEM	DETAIL	SOURCE
Launch	"Special Market Intervention Price Scheme" for Kashmir apples, announced Sept 2019. Procurement ran 1 Sept 2019 to 2 Mar 2020 through NAFED. Target about 12 lakh MT (about 60% of the crop), estimated fund need Rs 8,000 crore. Payment by DBT within 48 hours. Procurement at Sopore, Parimpora, Shopian and Batengoo.	Tribune, 10 Sep 2019
Rates (Rs/kg, grades A/B/C)	Initially 52 / 36 / 15.75, revised to 58 / 42 / 22, then 60 / 44 / 24	Search summaries of Kashmir Life and Tribune. AIKS also cites 60/44/24 (search summary).
Quantity actually procured	Not found (UNVERIFIED)	–
Government position, Oct 2025	MIS "continues to serve as a benchmark model". No current procurement figures given.	Kashmir Life, 27 Oct 2025
Dec 2025	Horticulture Minister said MIS was being considered if crop insurance did not materialise	GK, 18 Dec 2025
Status, Oct–Nov 2026	No MIS announced. AIKS demanded MIS with minimum prices for A, B and C grades and payment of pending MIS dues, and called an "All India Apple Growers' Protest Day" for 19 Oct 2026.	FreshPlaza/Rural Voice, 1 Oct 2026
USDA view	MIS is an ad hoc support measure aimed mainly at lower-grade, processing apples, and is bureaucratically hard for growers to access	USDA FAS, Nov 2026

2.6 Price ranges, 2024 to 2026 (as reported; dated)

DATE	MARKET / CONTEXT	VARIETY / GRADE	PRICE	SOURCE
2019-20 → 2023-24	Valley wholesale average	Delicious, 16 kg box	Rs 708 → Rs 458 (−35%)	Kashmir Vision, 22 Oct 2026 (underlying source not named, UNVERIFIED)
Sep 2015 – Mar 2025 (mean)	Sopore (Nowpora), Agmarknet	Delicious A / B	Rs 46.84 / 28.81 per kg	SKUAST-K preprint
Sep 2015 – Feb 2025 (mean)	Sopore (Nowpora)	American A / B	Rs 48.69 / 26.31 per kg	same
Sep 2017 – Jan 2025 (mean)	Shopian (Aglar)	Delicious A / B	Rs 47.24 / 30.33 per kg	same
2024	After CA storage	–	Grower profit of about Rs 200–300 per box	101Reporters, 21 Oct 2026
Aug 2025	HD vs traditional	HD fruit, 10 kg box	HD at least Rs 1,600/box; traditional (16–20 kg) at most Rs 1,200/box	Kashmir Life, 1 Aug 2025
Sep 2025 (highway closure)	Valley	Not specified	Box that sold for Rs 1,500–2,000 fetched under Rs 500	Down To Earth, 18 Sep 2025
20 Oct 2025	Sopore	All varieties	US$12–18 per box	FreshPlaza
Late Jan 2026	Parimpora	Apple	about Rs 65/kg	aggregator (search summary, UNVERIFIED)
5 Sep 2026	Early season	Kullu (Kulu)	Rs 1,200–1,400 per box	Kashmir Observer
23 Sep 2026	Fallen fruit ("Giran"), for juice and processing	–	Rs 100–250 per crate	Kashmir Observer
Early Sep → late Sep 2026	Kashmir mandis	15–16 kg carton	Rs 1,100–1,700 falling to Rs 700–1,000 (about Rs 44–67/kg)	AIKS via FreshPlaza, 1 Oct 2026 and Gujarat Samachar, search summary
19 Oct 2026	Shopian mandi	Delicious A / B	Rs 6,250 / 3,125 per q	Kashmir Vision, 22 Oct 2026
Oct 2026	Valley mandis	Good quality	Rs 1,500–2,500 per box; HD apples Rs 150–200/kg in some markets (grower quote)	same
Nov 2026	Mandis (grower quote)	Gala	Rs 2,000–2,500 per box	Tribune, 7 Nov 2026
2 Nov 2026	Sopore	Premium/large (Royal, "three-dana", American)	Rs 1,000–1,300 per box	Daily Excelsior
2 Nov 2026	Sopore	B and C grade (scab, Alternaria, pests)	Rs 200–250 per box	Rising Kashmir; Daily Excelsior
10–11 Nov 2026	Agmarknet-derived snapshot (CommodityMarketLive)	See next rows	Rs per quintal (modal; range)	CommodityMarketLive PDF
11 Nov 2026	Sopore (Nowpora)	Delicious / American	5,100 / 5,800	same
10 Nov 2026	APMC/MAC Shopian	Delicious	3,125 (2,500–3,750)	same
10 Nov 2026	Handwara	Delicious / American	3,437.5 (3,125–5,000) / 3,750 (3,625–4,687.5)	same
10 Nov 2026	Zazna (Ganderbal)	Delicious / Kullu Delicious	5,625 (5,000–6,250) / 6,250 (5,625–6,875)	same
10 Nov 2026	Zaloosa-Charar-e-Sharief	Delicious	4,250 (4,062.5–4,500)	same
10 Nov 2026	Narwal, Jammu	Apple	5,500 (3,000–8,000)	same

DATE	MARKET / CONTEXT	VARIETY / GRADE	PRICE	SOURCE
Sep 2026	Delhi wholesale, imported apples	US Red Delicious	Rs 5,500–5,700 per 20 kg	USDA FAS, Nov 2026
Sep 2026	Delhi wholesale, imported apples	NZ Royal Gala; Turkish Red Delicious; Italian Red Delicious	Rs 4,600–4,800 per 18 kg; Rs 4,200–4,300 per 18 kg; Rs 3,000–3,200 per 14 kg	same

Takeaway for the app: prices vary roughly tenfold by grade within a single day (Rs 200 to Rs 1,300+ per box at Sopore on 2 Nov 2026). Grade and variety have to be first-class fields in the price feed.

2.7 Price data sources and APIs (technical detail)

1. **data.gov.in OGD API: "Current Daily Price of Various Commodities from Various Markets (Mandi)"** - Dataset page: https://www.data.gov.in/resource/current-daily-price-various-commodities-various-markets-mandi. Publisher: Ministry of Agriculture & Farmers Welfare. Underlying data: Agmarknet. - Endpoint: `https://api.data.gov.in/resource/9ef84268-d588-465a-a308-a864a43d0070?api-key=<YOUR_KEY>&format=json&offset=0&limit=100` plus filters, for example `&filters[state.keyword]=Jammu and Kashmir&filters[commodity]=Apple`. Several third-party write-ups confirm the base URL and resource ID (Mineworks; data.gov.in API link, search summary). The exact filter names are UNVERIFIED: copy them from the dataset's "API" tab. Mineworks notes that field names are case-sensitive and differ between datasets. - Key: free. Register at data.gov.in, then My Account, then generate an API key. The public sample key shown on dataset pages is heavily rate-limited (reported as 10 records per call, UNVERIFIED). Use your own key. Mineworks reports a maximum `limit` of about 100 on most datasets and no published rate limit. - Fields: state, district, market, commodity, variety, grade, arrival_date (DD/MM/YYYY), min_price, max_price, modal_price, all in **Rs/quintal**. Prices arrive as strings. - This dataset is a current snapshot. For history, use Agmarknet (the SKUAST-K team downloaded 2012–2025 series from it) or the separate data.gov.in dataset "Variety-wise Daily Market Prices Data of Commodity" (dataset page; its resource ID is not confirmed in this session). - **Which J&K markets report apple** (Agmarknet-derived snapshot for 10–11 Nov 2026, CommodityMarketLive): Sopore (Nowpora) (F&V); APMC/MAC Shopian; Handwara Kupwara (F&V); Market Administrative Committee Zazna; Zaloosa-Chararishrief (F&V); MAC F&V Market Narwal Jammu; Kathua (F&V); Fruit & Vegetable Market Udhampur; Rajouri (Phaliana) (F&V). Aggregators also showed **Parimpora** and **Kulgam** in late 2025 or January 2026 (search summary, UNVERIFIED). Agmarknet start dates: Azadpur January 2012, Sopore/Nowpora August 2015, Shopian/Aglar September 2017 (SKUAST-K preprint). - Data-freshness caveat: a search snippet says the dataset metadata page shows "last updated" in April or August 2025. The Nov 2026 aggregator data suggests the feed is still live. Check before launch.
2. **Agmarknet 2.0** (agmarknet.gov.in). Upgraded in November 2025, with an Agmarknet 2.0 mobile app for both mandi data entry and farmer lookups. 4,367 mandis linked. Gives variety-wise and grade-wise daily price and arrivals (PIB via Global Agriculture, 17 Dec 2025). No official public API found. The site blocks automated fetches (403 in this session).
3. **e-NAM** (enam.gov.in). Has a public live dashboard of commodity-wise trade volumes and mandi-level prices. Its mobile app gives free price information for 247 notified commodities. Nationally: 1,522 mandis (Dec 2025, PIB) or 1,656 mandis (Feb 2026, Lok Sabha reply) (Global Agriculture; Kashmir Convener). No public API documented (UNVERIFIED).
4. **NHB** (nhb.gov.in). Offers daily, weekly, monthly and annual price-arrival reports. The daily bulletin's centre list includes Srinagar, Jammu and Delhi (report page). Coverage of apple and how current the data is are UNVERIFIED.
5. **UPAg portal** (Unified Portal for Agricultural Statistics). India.gov.in points users to "Agmarknet 2.0 Dashboard or the UPAg Portal" for daily prices (search summary). The domain and any API are UNVERIFIED.
6. **J&K Department of Horticulture (Planning & Marketing)** (dhpm.jk.gov.in). The wholesale-rate pages for within and outside the state were empty on 12 Nov 2026. Departmental email: director-hpm@jk.gov.in; officers' phone numbers are listed on the site. The department runs "Kissan Ghar" facilities within and outside J&K (menu items on the site).

3. Logistics and roads

3.1 NH-44 (Srinagar–Jammu) closures and their impact

How often it closes. Journalist Arif Shafi Wani counts 284 closure days since 2019: 47 (2020), 23 (2021), 41 (2022), 58 (2023) and 26 in 2025 up to mid-September (Down To Earth, 18 Sep 2025). About 90% of Kashmir's apple dispatches go along this one corridor (SKUAST-K preprint, citing APEDA 2021). Delays of 3–5 days can cut the grade of American apples by 25–40% (same, citing Mir & Mir 2018).

The 2025 crisis (peak harvest)

DATE (2025)	EVENT	SOURCE
Aug 26–27 and Sep 2–3	Very heavy rain caused landslides and flash floods, mainly in Ramban and Udhampur. Unusual damage near Jakheni and Thard in Udhampur. Closure began on 26 August.	Govt statement via Kashmir Vision, 29 Oct 2025; DTE; ETV Bharat, 21 Sep 2025
by Sep 10	NHAI partially reopened the road	Govt statement (above)
Sep 15–16	Protests at Jablipora mandi and in Srinagar ("Save Apple, Save Kashmir")	DTE
Sep 16–17	Road reopened. Heavy vehicles allowed from Srinagar to Jammu on 17 Sep. Traffic "fully restored" on 17 Sep.	DTE; Roshan Kashmir, 16 Sep 2025, search summary
Impact	More than 2,000 apple trucks stranded, 700–800 of them from Jablipora, each carrying 1,200–1,300 boxes. About half of the stranded fruit was damaged (DTE). Government figure: over 22,000 MT (about 2,200 trucks) stranded as of 17 Sep, about 1% of produce. 45,922 fruit trucks left J&K via NH-44 and Mughal Road. Traders' loss estimates: Rs 500–700 crore (ETV Bharat, search summary); Rs 600–700 crore (Kashmir Vision, 22 Oct 2026).	as cited
Crop damage and compensation	431.091 ha of orchards with more than 33% damage. Compensation admissible under SDRF/NDRF norms: Rs 152.37 lakh, of which Rs 12.28 lakh had been paid by Oct 2025.	Govt via Kashmir Vision, 29 Oct 2025
Response	Control room at Qazigund. Mughal Road made fully operational as an alternative. 157 RTC trucks deployed. Rail parcel services started (section 3.4).	Govt statement; ETV Bharat, 21 Sep 2025; Kashmir Reader, 30 Oct 2025

Closures in 2026 so far (none found during the peak apple season)

DATE (2026)	EVENT	SOURCE
~26 Mar	Landslide closed the highway between Karol Bridge and Chanderkote (Ramban); hundreds of vehicles stranded	Greater Kashmir
6–7 Apr	Landslide at Karol (Ramban); traffic halted both ways	The Right News, 7 Apr 2026
~24–26 Jul	Three-day closure after a landslide near Dewal (Udhampur), followed by a clearance closure at Gangroo (Ramban–Banihal). About 1,700 trucks stranded.	Curly Tales, 27 Jul 2026
~late Oct – early Nov	The Tulbul bridge near Sopore was closed about 10 days for repairs, disrupting mandi traffic (a local road, not NH-44)	Rising Kashmir, 3 Nov 2026

In April 2026 growers also asked for two-way traffic during the Amarnath Yatra and for tolls on produce going to Sopore to be removed (Chenab Times, 14 Apr 2026).

3.2 Mughal Road (NH-701A, Shopian–Poonch)

- This is the main alternative for fruit trucks when NH-44 shuts. In 2025 it was "made fully operational" (Govt via Kashmir Vision). One search summary reported 2,800 trucks cleared over Mughal Road in two days in mid-September 2025 (Roshan Kashmir, search summary, UNVERIFIED).
- Current rules (advisory for 6 Nov 2026): LMVs, then HMVs with no more than 10 tyres, in both directions, subject to weather and GREF clearance. Hours 7 am to 6 pm, from Chandimarh/Baffliaz towards Shopian and from Heerpora towards Poonch (DM Shopian order dated 3 Jul 2026) (Kashmir Life, 6 Nov 2026; JK AdWorld, 5 Nov 2026).
- It closes regularly after snowfall in winter, for example a closure reported in January 2026 (Kashmir Life, search summary; the exact date is UNVERIFIED).

3.3 Where official road status is published

- **X (Twitter): @Traffic_hqrs, "J&K Traffic Police".** Posts timed NH-44 updates (for example "Traffic update at 0930 hrs") and next-day traffic plans. Its bio says the account is not monitored around the clock and asks people to contact the TCUs in emergencies (search summary of the profile). Active as of 12–13 Nov 2026.
- **Daily advisories** are issued by Traffic Police Headquarters J&K with TCUs Jammu, Srinagar and Ramban, and republished by Kashmir Life, Rising Kashmir, Greater Kashmir and JK AdWorld. A typical advisory covers NH-44, Mughal Road, the SSG road (Srinagar–Sonamarg–Gumri) and NH-244 (Kishtwar–Sinthan–Anantnag), with vehicle-type windows and tanker restrictions (Kashmir Life, 6 Nov 2026).
- **Traffic Control Unit (TCU) helplines**, as printed in the 6 Nov 2026 advisory (Kashmir Life). Some digits differ between outlets (JK AdWorld), so confirm by calling before you put them in the app:
- TCU Jammu: 0191-2459048, 0191-2740550, 9419147732, 103
- TCU Srinagar: 0194-2450022, 0194-2485396, toll-free 1800-180-7091, 103
- TCU Ramban: 9419993745, toll-free 1800-180-7043
- Udhampur: 8491928625
- PCR Kishtwar: 9906154100. PCR Kargil: 9541002330 / 0541-002331 (UNVERIFIED, conflicting transcriptions)
- A J&K Traffic Police website with a machine-readable road-status page was **not found** (UNVERIFIED). A third-party travel site lists helpline 1800-180-7145, also UNVERIFIED.
- The IMD API index lists **Highway Nowcast Warning** and **Highway Warning (5 days)** products (IMD API reference). IMD Srinagar also publishes a "Highway weather forecast" PDF (IMD Srinagar). These could feed a road-risk indicator.

3.4 Rail: the Kashmir rail link (USBRL) and apple parcel and freight services

The line - The Udhampur–Srinagar–Baramulla Rail Link is 272 km long and cost Rs 43,780 crore. It has 36 tunnels (119 km) and 943 bridges. The Chenab bridge is 1,315 m long. Inaugurated June 2025. The Katra–Srinagar Vande Bharat started in June 2025 and was extended to Jammu Tawi in April 2026, cutting Jammu–Srinagar to under five hours (IANS, 17 Oct 2026). The flag-off date of 6 June 2025 rests on reports of a Friday flag-off (Vajiram, search summary). - Freight: the first freight train (cement) reached the new **Anantnag Goods Shed on 9 Aug 2025**. Since then more than 1.5 lakh t of cement has arrived, and **at least six goods trains a day** (three each way) run as of Oct 2026. **More than 20,000 t of apples** have moved by rail since Sept 2025. By rail apples reach Delhi in **20–22 hours**, against 30–35 hours by road (Greater Kashmir, 13 Oct 2026).

Apple services: timeline

DATE	SERVICE	DETAIL	SOURCE
11 Sep 2025	First apple parcel vans	Two parcel vans (about 23 t each), one to Jammu and one to Adarsh Nagar (Delhi)	Tribune, 21 Sep 2025
From 13 Sep 2025	Daily timetabled parcel train **Budgam → Adarsh Nagar (Delhi)**	8 parcel vans. Departs Budgam 06:15, arrives about 05:00–05:30 the next day. Stop at Bari Brahmana. Officials put the freight at **Rs 2.41/kg** Budgam–Delhi; Kashmir Life reported about Rs 3/kg, with fruit owners doing their own loading. A trader estimated 10 t at Rs 1.25 lakh by truck against about Rs 60,000 by train. Booking online for full vans, offline for smaller lots.	Daily Excelsior, 12 Sep 2025; KNS (search summary for Rs 2.41/kg); Kashmir Life (search summary for Rs 3/kg); Rising Kashmir (search summary for booking); WION
15 Sep 2025	LG flagged off an 8-van train, Srinagar/Budgam to Adarsh Nagar	Each van about 23 t (about 184 t per train)	Tribune, 21 Sep 2025
Sep 2025	Covered wagons (BCN) from **Anantnag Goods Shed**, and 21 container wagons supplied at Bari Brahmana (about 76 containers of 40 kg each per wagon)	More than 6,400 t of apples moved in about the first 10 days. A Baramulla parcel terminal was "planned soon". A 14-coach Sopore train was due to start loading about 21 Sep 2025.	Tribune, 21 Sep and 22 Sep 2025; Kashmir Observer, 20 Sep 2025
2025 season total	About **14,000 MT (about 9 lakh boxes)** by rail. During the blockade, 1,25,376 boxes worth Rs 10.03 crore went from Budgam and Anantnag to Jammu and Adarsh Nagar.	Govt (Kashmir Reader, 30 Oct 2025)	
17 Apr – 31 May 2026 (trial)	**JPP-RCS** (Joint Parcel Product – Rapid Cargo Service), Budgam → Adarsh Nagar	8 parcel vans and 1 SLR. Budgam 06:15 → Delhi 05:00 the next day; return Delhi 05:00 → Budgam 10:45 the next day. Stops at Bari Brahmana and Ambala Cantt. Extension depended on demand. Rates were not published.	Kashmir Life, 16 Apr 2026

DATE	SERVICE	DETAIL	SOURCE
Aug 2026	Northern Railway on X: "The Cargo Parcel Train from Budgam to Adarsh Nagar, Delhi is operational, and not discontinued". Farmers asked for clearer schedules, booking and capacity information.		Kashmir Despatch
29–30 Oct 2026	Proposal: **50% concession on return freight** under JPP-RCS. Booking both directions in advance costs 1.5 times the normal JP-scale rate for the round trip; booking and payment at the originating station. Discussed with Horticulture (P&M) officers (Parimpora, Bandipora, Anantnag) and aggregators. Exact rates and start date were not given.		Kashmir Observer (PTI), 30 Oct 2026

Gap: 2026 apple tonnage by rail, current JP-scale rates per van, and refrigerated vans were not found (UNVERIFIED). In Sept 2025 the Chief Minister asked for refrigerated cargo trains (Daily Excelsior).

3.5 Truck freight rates, Kashmir to Delhi and North India

PERIOD	RATE	SOURCE
Normal season (pre-crisis)	Rs 70–100 per box, Sopore to Delhi	GK, 29 Sep 2025 (Sopore Mandi Fruit Growers Association)
Sep 2025 crisis	Rs 180–230 per box (private trucks); Rs 240–250 (growers); Rs 270–300 (Sopore, late Sept)	ETV Bharat, 21 Sep 2025; Kashmir Observer, 20 Sep 2025; GK, 29 Sep 2025
Sep 2025 caps	RTC capped at Rs 90–130 per box to Delhi, Haryana and Jalandhar. The Sopore mandi association temporarily fixed Rs 130 (Delhi) and Rs 120 (Punjab). The Transport Dept said a fair fare was Rs 100–120 per box.	ETV Bharat; Kashmir Observer
Per truck	10 t to Delhi about Rs 1.25 lakh (trader, Sep 2025). In Oct 2018, 10–11 t (550–600 boxes) to Azadpur cost Rs 70,000–80,000, "60% above routine".	Daily Excelsior, 12 Sep 2025; GK, 19 Oct 2018
Nov 2026	Freight "rose from Rs 110–115 to Rs 125–140" depending on vehicle and destination. The unit is not stated; it is probably per box (UNVERIFIED).	Daily Excelsior, 3 Nov 2026
Local haul to CA store	About Rs 100–110 per 20 kg box, Dachnipora to Lassipora (about 32 km) (one grower's account)	101Reporters, 21 Oct 2026

Truck capacity in reports: 1,200–1,300 boxes per truck (DTE) and 3,000–3,500 boxes per truck, probably large trailers (Rising Kashmir). Normal transit Kashmir–Delhi is 4–5 days by truck, according to a trader (Daily Excelsior, 12 Sep 2025), or 2 days by a driver's account rising to 5–6 days in the crisis (GK, 29 Sep 2025).

3.6 Controlled-atmosphere (CA) and cold storage

METRIC	VALUE	DATE	SOURCE
CA capacity (cumulative)	1.93 LMT (2021-22), 2.10 (2022-23), 2.70 (2023-24), 2.70 (2024-25), **2.92 LMT** (2025-26 to Nov 2025); 3.07 LMT targeted	2021–25	ES 2025-26, Table 4.2(ii)
Requirement	**6.00 LMT**, so the gap is about 3.08 LMT	2025-26	ES 2025-26; 101Reporters
Main CA clusters	Lassipora industrial estate (Pulwama) and Aglar (Shopian)	Oct 2025	Kashmir Life, 30 Oct 2025
Subsidy	MIDH: 50%, up to Rs 9 crore, for a 5,000 MT CA store, plus a UT capex top-up	Oct 2025	same
Budget 2026-27	Rs 600 crore subsidy proposed for **40 new CA stores**, focused outside the Pulwama–Shopian belt	2026	101Reporters, 21 Oct 2026
CA rental	**Rs 35 per 17 kg crate per month**, about Rs 180–220 per crate for 5–6 months, higher at peak	2026	same
Demand in 2026	Units were fully booked months in advance. Operators now require proper grading before accepting fruit.	Sep 2026	Kashmir Observer, 23 Sep 2026
Number of CA units and their owners	**Not found (UNVERIFIED)**	–	–

4. Weather and alerts

4.1 IMD Meteorological Centre Srinagar (mausam.imd.gov.in/srinagar)

- Products: district-wise and sub-division warnings; district-wise and station-wise nowcasts; 5-day J&K forecast; district forecast and "impact-based district-level forecast" PDFs; extended-range forecast; **Agromet bulletin (PDF)**; **highway weather forecast (PDF)**; AWS data link; radar and satellite; Mausamgram; special Amarnath and Vaishno Devi Yatra pages.
- Social channels: X **@metsrinagar**, Telegram **metsrinagar**, Facebook **jammuandkashmirweather**. All are listed on the IMD Srinagar page.
- Historical data: IMD Pune Data Supply Portal, dsp.imdpune.gov.in (IMD home).

4.2 Agromet advisory services

- IMD runs Agromet Advisory Services through **Gramin Krishi Mausam Sewa (GKMS)**. It has a state-level advisory page (`responsive/agromet_adv_ser_state_current.php` on mausam.imd.gov.in) and a GKMS SOP (IMD home). District AAS bulletins for J&K districts, and whether District Agromet Units sit at Kashmir KVKs, are UNVERIFIED in this session. Check the GKMS pages and SKUAST-K.
- The IMD Srinagar page carries an "Agromet bulletin" PDF (IMD Srinagar).
- J&K's own **Kisan Khidmat Ghar** application, built with BISAG, does "predictive analysis of weather conditions and allied advisories" (ES 2025-26).

4.3 Official apps (links from the IMD home page)

APP	PURPOSE	ANDROID	IOS
Mausam	IMD forecasts, nowcasts and warnings	https://play.google.com/store/apps/details?id=com.imd.masuam	https://apps.apple.com/us/app/id1522893967
Meghdoot (Agro)	District agromet advisories for farmers	https://play.google.com/store/apps/details?id=com.aas.meghdoot	https://apps.apple.com/in/app/meghdoot/id1474048155
Damini	Lightning alerts	https://play.google.com/store/apps/details?id=com.lightening.live.damini	App Store ID 1502385645

IMD's dissemination note also lists RAIN ALARM and UMANG (IMD API doc).

4.4 Recent damaging weather events for apple

DATE	EVENT	IMPACT	SOURCE
3–4 Nov 2018	Early (untimely) snowfall while trees still had leaves	Branch breakage in south Kashmir orchards; large losses reported	**UNVERIFIED in this session** (not re-checked)
~7 Nov 2019	Early heavy snowfall	Widespread orchard damage	**UNVERIFIED in this session**
Jun 2025	Hailstorm in south Kashmir	Orchards damaged	LocalSamosa, 23 Feb 2026
Summer 2025	Long dry spell and abnormally high temperatures	Undersized, lighter fruit and lower returns	GK, 27 Jul 2025; GK, 18 Dec 2025
26–27 Aug and 2–3 Sep 2025	Extreme rain, floods and landslides	431.091 ha of orchards over 33% damaged, plus the NH-44 closure (section 3.1)	Govt via Kashmir Vision
Feb 2026	Unusually warm February	Buds swelling too early	Kashmir Horizon, 3 Mar 2026, search summary
Spring–summer 2026	Repeated hailstorms, especially Shopian and Kulgam; dry August	Damage during fruit development; worries about size and colour	Kashmir Observer, 5 Sep 2026
~Aug 2026	Hail, heavy rain and strong winds; south Kashmir hit hardest	Loss estimate Rs 300–400 crore	India TV, 19 Aug 2026, search summary (UNVERIFIED)
2026 season	Wet weather drove a **scab** outbreak, along with Alternaria, leaf miner and aphids	50–60% of fruit reaching hubs graded lower (dealer estimates). B/C grades sold at Rs 200–250 per box in Sopore.	Whalesbook, 1 Nov 2026; Rising Kashmir, 3 Nov 2026
1980–2024 trend	Winter maximum temperatures up 0.5 °C (Kokernag) to 1 °C (Kupwara); weaker snowfall during Chillai Kalan	Loss of natural "cold storage"; more reliance on CA	101Reporters, 21 Oct 2026

Related: Kashmir uses about 3.69 kg of pesticide per hectare against a national 0.22 kg/ha, with sprays now often more than a dozen a season. A SKUAST study found residues above FSSAI, EU and Codex limits (ETV Bharat, 12 Nov 2026). This is relevant to spray-advisory features.

4.5 Weather APIs an app can use

IMD API platform. Portal: api.imd.gov.in. Reference: api_reference.html. Base URL: `https://api.imd.gov.in/api/v1/`. - Endpoints documented: - `cityforecast?id=<station>` gives a 7-day city forecast. - `current_wx?id=<station>`. - `districtnowcast?id=<district id>` gives categories with a colour code (1–4: green to red). - `districtwarning?id=<obj id>` gives Day 1–5 warning codes and colours. - `districtrainfall`. - `state_district_rainfall_forecast` gives 5-day district rainfall. - `aws_data?id=<call sign>` or `sid=<state id>`. - `stationnowcast`. - `sunmoon?lat=&lon=`. - Mapping endpoints: `cityforecast_mapping`, `aws_data_mapping`. The index also lists Highway Nowcast Warning, Highway Warning (5 days), Lightning, Radar and Agromet Advisory, but their details are not on the page. - Access: the portal has account registration. IMD's dissemination note says organisations wanting the APIs "following IMD's terms and condition" should contact the nodal officer, Dr Sankar Nath, Sc-E, IMD New Delhi (sankar.nath@imd.gov.in) (IMD API doc). IP whitelisting, cost and rate limits are not stated on the pages read (UNVERIFIED). Srinagar's station ID is 42027 by WMO convention (UNVERIFIED; check `cityforecast_mapping`).

Open-Meteo. - Free tier, non-commercial only: 600 calls/min, 5,000/hour, 10,000/day, 300,000/month, no uptime guarantee. - Commercial plans: Standard (1M calls/month), Professional (5M) and Enterprise (50M+), using `customer-api.open-meteo.com` with an `apikey` parameter. Prices are not published on the page. - Data under CC BY 4.0, with attribution required (pricing; terms). - "Commercial" includes apps with ads or subscriptions.

Recommended pattern: use IMD district warnings and nowcasts for official alerts (hail, heavy rain, snow), and Open-Meteo or another model API for hourly forecasts at orchard level. Label each layer with its source.

5. Government schemes and support

5.1 Holistic Agriculture Development Programme (HADP), J&K

- **29 projects, Rs 5,013 crore, over five years.** Expected results: about Rs 28,000 crore added to GSDP, about 2.88 lakh jobs, about 19,000 enterprises, about 13 lakh families helped and about 2.5 lakh youth skilled. **Rs 1,028.21 crore** is set aside for horticulture projects (ES 2025-26). HADP launched in April 2023 (Kashmir Ahead, 17 Apr 2025).
- **Kisan Saathi portal** is the HADP single window. As of 26 Oct 2026: 6,97,728 applications, 4,79,895 approved, **1,15,295 pending**, 4,33,215 farmers registered. Horticulture: 66,818 received, 41,778 approved, 12,503 pending (Rising Kashmir, 27 Oct 2026). The HADP website is **hadp.jk.gov.in**; hadp.in redirects there (checked 12 Nov 2026). The exact Kisan Saathi URL is UNVERIFIED.
- Other digital tools launched on 16 Apr 2025: AI-powered Credit Scheme Advisor app, online DPR generator, KS Output Tracking app, JKCIP dashboard. Also the **Daksh Kisan** learning platform (121 free courses; 10,765 certificates) (Kashmir Ahead). The domain dakshkisan.jk.gov.in comes from a search summary and is UNVERIFIED.
- HADP claims **300 FPOs** have been set up (April 2025, same source).
- **JKCIP** (J&K Competitiveness Improvement Project): Rs 1,800 crore over 7 years in 90 blocks, aiming to reach about 3 lakh families, with its own portal (ES 2025-26).

5.2 High-density plantation subsidy

SCHEME	ASSISTANCE	SOURCE
Modified High Density Plantation Scheme (J&K)	**50% subsidy** on a crop-wise per-hectare budget. Bank finance of up to 80% of project cost can be arranged.	Rajya Sabha reply via Kashmir Life, 24 Jul 2026
Same scheme, grower-level split	50% subsidy, up to 40% optional bank loan (interest deferred 3 years, repayment from year 4), 10% from the farmer. Runs March 2021 to March 2026, target 5,500 ha. Extra support for micro-irrigation, virus-free planting material and anti-hail nets.	Kashmir Life, 1 Aug 2025
MIDH and NHB (central)	40% of the eligible cost in general areas, **50% in Himalayan states/UTs including J&K**	Kashmir Life, 24 Jul 2026
Rupee rate per hectare	Rs 1.75 lakh/ha subsidy on a 50:50 pattern, as of 2021. **The current per-hectare budget was not found (UNVERIFIED).**	101Reporters, 17 Aug 2021
Cost benchmarks	A 2026 study of 200 growers put HD cost at Rs 3,51,759 per kanal over the first five years, more than half of it in year 1. One grower spent about Rs 18 lakh on 12 kanals. (1 ha is about 20 kanals.)	Kashmir Vision, 22 Oct 2026; Kashmir Life, 1 Aug 2025

Whether the Modified HDP scheme was extended after March 2026 is UNVERIFIED.

5.3 Crop insurance for apple (PMFBY/RWBCIS): status as of Nov 2026

Bottom line: apple growers in J&K still have no crop insurance. PMFBY and RWBCIS were formally adopted in J&K in 2016 but have never been implemented for horticulture (GK, 18 Dec 2025).

DATE	DEVELOPMENT	SOURCE
Nov 2024	RWBCIS bids invited for Kashmir (apple, saffron) and Jammu. The lowest Kashmir bid (Iffco-Tokio) quoted 15.15–29.95% premium.	GK, 27 Jul 2025
May 2025	State Level Coordination Committee (Chief Secretary) cancelled the tender over cost and decided to re-tender	same
Oct 2025	Govt: RWBCIS to be implemented for apple, saffron, mango and litchi; tendering in process	Govt via Kashmir Vision
16 Sep 2025	Centre set up a sub-committee to standardise RWBCIS term sheets for apple, banana, grape, mango and pomegranate	Kashmir Life, 22 Sep 2026
4 Feb 2026	Centre asked the UT to prepare a fresh tender for all 20 districts under revised PMFBY/RWBCIS guidelines, from **Kharif 2026**	same
16 Mar 2026	J&K Finance Department directed a fresh tender. The next tender block is Kharif 2026 to Rabi 2028-29.	same
24 Apr 2026	Govt said insurance would start in 2026. Perils: excess heat, hailstorm, heavy rain, snowfall. Lowest bidders were AIC (Kashmir) and Tata AIG (Jammu).	ETV Bharat
Sep 2026 (Assembly)	**Not implemented.** AIC quoted 10–39% (traditional apple 34% in Budgam, Bandipora, Baramulla and Kupwara). 2025-26 bid block: sum insured Rs 6,594.93 crore, total premium Rs 1,280.87 crore, farmers' share Rs 329.75 crore. Both apple term sheets are final, but fresh tenders wait for the Centre's revised guidelines. No timeline given.	Kashmir Life, 22 Sep 2026
23–24 Sep 2026	Minister Javid Dar: "fully committed" to WBCIS, but the scheme is **on hold** awaiting the Government of India's revised norms. It will roll out "immediately" once they are issued.	Kashmir Observer, 24 Sep 2026

PMFBY for other crops covers all 20 districts. In 2024-25 it insured 2.08 lakh farmers on 80,884 ha, with claims of Rs 26.92 crore paid to more than 94,000 farmers. In Kharif 2025 it covered 2.015 lakh farmers on 75,735 ha (ES 2025-26). The Sept 2026 Assembly reply says PMFBY covers hailstorm damage to paddy and maize (Kharif) and wheat and oilseeds (Rabi), and that all 13.985 lakh farmer families are eligible (Rising Kashmir, 25 Sep 2026).

5.4 Credit, helplines, extension and other support

- **Kisan Credit Card.** 11.39 lakh KCCs issued (8.78 lakh crop, 2.61 lakh animal husbandry), with loans of Rs 10,290 crore up to Dec 2025. Cumulative KCCs at the end of 2024-25 by district: Baramulla 1,17,628; Kupwara 1,02,484; Anantnag 95,040; Budgam 77,372; Pulwama 66,198; Kulgam 57,802; Bandipora 51,104; Shopian 46,367; Ganderbal 28,301; Srinagar 12,279 (ES 2025-26). Growers have asked for KCC loan waivers (Chenab Times, Apr 2026).
- **Kisan Call Centre**: toll-free **1800-180-1551**, the standard national number of the Ministry of Agriculture (not re-checked this session, so UNVERIFIED).
- **KVKs.** SKUAST-Kashmir runs Krishi Vigyan Kendras in the Kashmir districts (background knowledge; the list and contacts are UNVERIFIED). SKUAST-K drives orchard research and advisories; its HADP work claims an 80% cut in pesticide use in trials (ETV Bharat, 12 Nov 2026).
- **Soil Health Cards** issued in 2024-25: 80,256 across J&K. Kashmir districts: Baramulla 9,924; Anantnag 6,782; Kupwara 6,518; Budgam 6,007; Pulwama 5,900; Kulgam 5,814; Bandipora 5,378; Shopian 5,233; Ganderbal 3,248; Srinagar 2,586 (ES 2025-26, Table 4.1.9).
- **FPOs and cooperatives.** HADP claims 300 FPOs (April 2025). ES 2025-26 reports 61 PACS developed as FPOs under NCDC, and 45 PACS plus 61 FPOs (7,397 farmers) registered on the NCCF portal. J&K ranks first nationally in computerising all 537 PACS. The number of apple-specific FPOs is UNVERIFIED.
- **e-NAM**: see section 2.1.
- **Other.** PM-KISAN paid Rs 3,907.49 crore to 9.19 lakh farmers in 2024-25. The Agriculture Infrastructure Fund has sanctioned 325 projects worth Rs 625 crore to Dec 2025 (ES 2025-26). The DHPM website lists a "Packaging and Grading Scheme" and PMFME (dhpm.jk.gov.in).

6. Labour (harvest, pruning, spraying)

What is sourced: - **About 400 person-days of work per hectare of orchard per year** (Govt Assembly reply, Sept 2026) (Rising Kashmir, 25 Sep 2026). At 1.73 lakh ha that implies roughly 6.9 crore person-days a year for apple alone (my arithmetic, indicative only). - Seasonal tasks named: pruning, spraying, harvesting, grading, packing, then transport, mandi and storage work (Gadyal Kashmir, 16 Sep 2026). - Growers report **rising labour costs**, along with packaging and transport (Kashmir Observer, 5 Sep 2026; Whalesbook, 1 Nov 2026). Spray rounds have risen from about 4 to more than 12 a season, which means more labour (ETV Bharat, 12 Nov 2026). - Cost of living for farm labour is high. In Dec 2025 J&K's CPI for agricultural labourers (General, base 2019) was 146.48, the highest of nine comparison states. The food index was 144.14, second after Haryana. The all-India figures were 137.12 and 137.79 (ES 2025-26). - HD orchards cut the need for some labour-heavy work (mechanised spraying, trellis systems) and are part of the government's mechanisation push (ES 2025-26 "Way Ahead").

Not verified in this session (search budget exhausted): - Typical **daily wages** in 2025–26 for local and migrant harvest, pruning or spraying labour. - The size and origin states of the migrant workforce. - 2025–26 news on labour shortages. - Security incidents that hit non-local labour and truckers. Background knowledge says such incidents occurred in October 2019 and October 2024; this is UNVERIFIED.

The official source for wages is the Labour Bureau's monthly "Wage Rates in Rural India" (WRRI), which has state-level occupation rates including horticulture workers. Its coverage of J&K is UNVERIFIED. For the app, collect wage data through the labour-booking flow itself, by district and task.

7. Recent news (2025–2026) affecting growers

7.1 Timeline

DATE	DEVELOPMENT	SOURCE
Mar 2025	Govt put 2024-25 output at 20.56 LMT, later revised to 21.10 LMT. CA capacity 2.70 LMT.	Kashmir Life, 6 Mar 2025
May 2025	Crop insurance tender scrapped (premiums too high)	GK, 27 Jul 2025
Jun 2025	USBRL opened; hailstorm in south Kashmir	IANS; LocalSamosa
Jul 2025	Tribune reported the Centre raised the apple minimum import price from Rs 50 to Rs 80/kg, citing a BJP spokesperson. **Conflicts with USDA (Nov 2026)**, which still describes the Rs 50/kg CIF floor. UNVERIFIED.	Tribune, 4 Jul 2025
9 Aug 2025	First freight train (cement) reached Anantnag	GK, 13 Oct 2026
26 Aug – 17 Sep 2025	NH-44 closure in peak season: rotting fruit, protests, freight rates tripled. Traders estimated losses of Rs 500–700 crore.	Section 3.1
11–15 Sep 2025	First apple parcel trains Budgam/Anantnag → Delhi; Rs 2.41/kg	Section 3.4
Oct 2025	Govt said about 1% of the crop was stranded and 14,000 MT moved by rail. 2025 crop estimated at 22.15 LMT.	Kashmir Vision; Kashmir Reader
Jan 2026	**India–EU FTA** concluded. EU apples: 50,000 t/yr at 20% duty with an MIP of Rs 80/kg, rising to 1 lakh t over 10 years.	Business Standard, 28 Jan 2026, search summary; Kashmir Observer/PTI, 7 Feb 2026
7 Feb 2026	**India–US interim framework**: US apples at 25% duty (down from 50%), quota-based, MIP Rs 80/kg (so nothing below about Rs 100/kg landed). Quota size not disclosed. To be signed by mid-March. Strong protest from growers in Kashmir and Himachal.	Kashmir Observer/PTI; Kashmir Reader, 8 Feb 2026, search summary
Mar–Apr 2026	NH-44 landslide closures at Ramban. JPP-RCS cargo-train trial (17 Apr – 31 May). Vande Bharat extended to Jammu.	Sections 3.1 and 3.4
late Apr 2026	**India–New Zealand FTA** signed (reported 27–29 Apr 2026). Growers and Congress raised alarm.	Business Standard, search summary
Jul 2026	Three-day NH-44 closure near Dewal; about 1,700 trucks stranded	Curly Tales
Aug 2026	Hail and storm losses estimated at Rs 300–400 crore (UNVERIFIED)	India TV, search summary
Sep 2026	Assembly session: insurance still not implemented; 7.5 lakh families; 1,103.79 ha HD in Kashmir. Early prices were encouraging (Kullu Rs 1,200–1,400 per box); CA stores fully booked.	Sections 1.4, 1.5, 2.6 and 5.3
Late Sep – Oct 2026	Prices fell (cartons Rs 700–1,000). AIKS demanded MIS and called the 19 Oct "All India Apple Growers' Protest Day".	FreshPlaza, 1 Oct 2026
17–20 Oct 2026	**India–NZ FTA in force from 20 Oct 2026** (DGFT Public Notice 33/2026-27, 17 Oct). Apple TRQ at 25% in-quota duty, minimum CIF US$1.25/kg, import window 1 Apr – 31 Aug. Quota: nil in Year 1 (to 31 Dec 2026), 35,000 t in Year 2 (2027), 45,000 t by Year 6. Other reports say 32,500 rising to 45,000 t.	Swarajya; FreshPlaza; Fruitnet, search summary
29–30 Oct 2026	Railways proposed a 50% return-freight concession for Kashmir fruit under JPP-RCS	Kashmir Observer
1–3 Nov 2026	Scab-hit crop: 50–60% lower grades. Sopore premium Rs 1,000–1,300 per box against Rs 200–250 for B/C. Chhath and Diwali demand.	Section 2.6
3 Nov 2026	USDA: India MY 2026/27 output forecast 2.59 MMT; imports forecast up 19% to 511,000 t	USDA FAS IN2026-0056
12 Nov 2026	Pesticide-residue concerns; SKUAST pesticide-free Gala trial in Pulwama (about 450 q, about Rs 1 crore revenue)	ETV Bharat

7.2 Import competition and trade policy (current state)

- **Duty and floor price as of 3 Nov 2026.** Apples (HS 0808.10) pay **50% basic customs duty** with no quantitative restriction, except under the India–NZ FTA. Under DGFT Notification 5/2023 (8 May 2023), imports with a CIF value at or below **Rs 50/kg** are prohibited; Bhutan is exempt. The 20% retaliatory duty on US apples was removed in Sept 2023 (USDA FAS, Nov 2026). USDA does not mention any US or EU quota in force, which suggests those concessions were not yet operating on 3 Nov 2026 (my inference, UNVERIFIED).
- **Imports by origin** in tonnes, MY 2023/24 → 2024/25 → 2025/26 (USDA FAS, Nov 2026):

ORIGIN	2023/24	2024/25	2025/26	SHARE 2025/26
World	544,024	510,930	428,750	–
Iran	138,774	138,994	69,988	16%
South Africa	39,249	35,252	54,319	13%
New Zealand	31,629	36,173	53,622	13%
Chile	24,773	34,618	46,991	11%

ORIGIN	2023/24	2024/25	2025/26	SHARE 2025/26
United States	37,246	34,064	30,142	7%
Afghanistan	37,937	49,886	28,013	7%
Turkey	123,213	94,956	14,370	3%

- Producer groups allege Iranian apples are relabelled as Afghan to use the Afghanistan duty route. Growers have repeatedly asked for a ban (USDA, same report). Kashmir traders have resolved to stop accepting Turkish apples, and the Shopian association wants Iranian imports banned (LocalSamosa, 23 Feb 2026).
- The J&K government told the legislature that NZ apple imports were under 2% of J&K output: 0.97% in 2023-24 and 1.46% in 2024-25 (Roshan Kashmir, 30 Mar 2026, search summary).

8. What I could not verify, and open questions

1. **Labour wages and shortages, 2025–26**: no sourced numbers (section 6).
2. **Number, ownership and locations of CA units** beyond the Lassipora and Aglar clusters; current CA occupancy.
3. **Untimely snow events of Nov 2018 and Nov 2019**: dates and losses not re-checked.
4. **data.gov.in API behaviour**: exact filter keys, sample-key limits and how current the feed is. I did not call the API (no key; using the key found in search results was blocked). Whether **Parimpora, Kulgam, Pulwama (Prichoo/Pachhar) and Jablipora** report apples regularly.
5. **Agmarknet 2.0 / UPAg / e-NAM APIs**: no public API documentation found.
6. **Current rail tariff** (JP scale per parcel van) and 2026 rail tonnage. The Rs 2.41/kg figure is from Sept 2025, and another report gives about Rs 3/kg.
7. **Current Modified HDP per-hectare budget**, and whether the scheme (March 2021 – March 2026) was extended.
8. **Final 2025-26 apple production** and an official 2026 crop estimate.
9. **India–US and India–EU apple concessions**: signing and entry-into-force dates; the size of the US quota.
10. **The reported July 2025 MIP hike to Rs 80/kg** conflicts with USDA's November 2026 description of the Rs 50/kg floor.
11. **TCU phone numbers**: some digits differ between outlets. No official Traffic Police website road-status page was found.
12. **NAFED MIS**: quantities actually procured in 2019–21, and any dues still unpaid.
13. **KVK list, Kisan Call Centre number and Meghdoot district coverage for Kashmir**: from background knowledge or IMD links; not re-checked.

9. Key primary and official sources

- J&K Economic Survey 2025-26 (Planning, Development & Monitoring Dept, Feb 2026): https://cms.neva.gov.in/NeVA/JK/FileStructures//Notices/e31e606a-b6e0-4592-8852-14d46f50833f.pdf
- USDA FAS, Fresh Deciduous Fruit Annual, India, IN2026-0056 (3 Nov 2026): https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Fresh+Deciduous+Fruit+Annual_New+Delhi_India_IN2026-0056.pdf
- SKUAST-K preprint "Corridor Disruption and Supply Chain Vulnerability: A NASCDI-NARDL Analysis of Kashmir's Apple Sector" (Research Square, posted 29 Oct 2026; not peer-reviewed): https://doi.org/10.21203/rs.3.rs-11006590/v1
- data.gov.in mandi price dataset: https://www.data.gov.in/resource/current-daily-price-various-commodities-various-markets-mandi
- IMD API reference: https://api.imd.gov.in/public/api_reference.html. IMD Srinagar: https://mausam.imd.gov.in/srinagar/
- J&K Traffic Police on X: https://x.com/Traffic_hqrs
- J&K Horticulture Planning & Marketing: https://dhpm.jk.gov.in/
- NHB price and arrival reports: https://nhb.gov.in/
- Open-Meteo terms and pricing: https://open-meteo.com/en/terms and https://open-meteo.com/en/pricing

Appendix C · Technology, apps, competitors, datasets and regulations

Desk research notes, November 2026. Tags: [V] read at source (in Appendix D: confirmed by an independent verifier); [S] from a search summary or not re-confirmed; UNVERIFIED = lead only.

Research date: 2026-11-12. Prepared for: the "orchard doctor" yes/no diagnosis wizard (later photo AI), agro-input shop, lessons, mandi/road/weather info and services marketplace.

How to read this document

Every item has a source link. Evidence tags:

- **[V]**: I opened the linked page, file or API in this session and read the fact there.
- **[S]**: The fact comes from a search-engine summary of the linked page. I did not open the page myself, so check it before you rely on the fact.
- **[UNVERIFIED]**: The fact comes from memory or an unconfirmed lead, or the source could not be reached. Treat it as a lead only.
- **Analysis** marks my own recommendations, not sourced facts.

Method and limits

- Google Play numbers come from the Play Store page HTML for each app, fetched directly on 2026-11-12 with `hl=en&gl=IN`. I read the `ratingValue`, `ratingCount`, the "Downloads" badge and "Updated on". Only the three "most relevant" reviews Google serves without JavaScript were visible for each app, so the review notes are anecdotes, not a systematic analysis.
- Partway through, the session's shared web-search budget ran out. After that I used only direct fetches of known URLs. Several Indian government sites refused automated access during the session: consumeraffairs.nic.in, ppqs.gov.in, nrsc.gov.in, egazette.gov.in, indiacode.nic.in and skuastkashmir.ac.in. As a result, parts of sections 6 (orchard technology) and 7 (regulations) are thinner and are flagged. Section 9 lists what is still open.

---

0. Top picks to integrate first (Analysis)

PRIORITY	WHAT	WHY	LICENCE OR COST NOTE
1	**Bhashini APIs** (Kashmiri ASR, NMT, TTS) plus **AI4Bharat IndicTrans2** (kas_Arab and kas_Deva); Sarvam-Translate as a commercial alternative for translation	Bhashini is the only production API found that lists Kashmiri for speech (ASR and TTS); Sarvam-Translate also lists Kashmiri for text. Bhashini's Kashmiri TTS is listed only in **Devanagari** script, so you will need transliteration or your own TTS for Perso-Arabic.	Bhashini is a government platform (terms not checked). IndicTrans2 is MIT.
2	**Open-Meteo** (commercial plan) plus the **IMD API** (IP whitelisting) plus **ERA5** history	Hyperlocal forecasts to drive the spray-window and scab infection-period logic.	Open-Meteo data is CC BY 4.0, and commercial use needs a paid plan. ERA5 is CC BY 4.0 (since 2 July 2025).
3	**Training data for the photo AI**: Kashmir Indigenous dataset (7,569 images), ICAR-CITH Srinagar dataset, AppleLeaf9, PlantVillage (Mendeley CC0 copy)	Kashmir-specific field images with permissive licences.	All CC BY 4.0 or CC0. Avoid the NC-licensed sets (MinneApple, Fuji-SfM) for a commercial model.
4	**LiteRT** (formerly TensorFlow Lite) on-device model, MobileNetV3-class	Works offline in orchards. PlantVillage-only models generalise poorly to field photos, so collect your own field images with DPDP-compliant consent.	Apache-licensed runtime (licence not checked here).
5	**Agmarknet 2.0 / data.gov.in** mandi prices, plus a partnership with J&K's **Kisan Khidmat Ghar**	Prices and arrivals. KKG already offers orchard-wise weather and prices in J&K.	data.gov.in data is under GODL-India, which allows commercial use with attribution.
6	**SKUAST-K spray schedule / Package of Practices** as the advisory rule base	Every serious Kashmir competitor (Farmit by Qul, Applefy, AppleCare Kashmir) bases its advice on it.	Copyright and permission to reuse not checked [UNVERIFIED].
7	**NHB National Nursery Portal** (accredited and licensed nurseries, searchable by crop "Apple")	Lets the shop verify sapling suppliers.	Public portal.

---

1. Existing farmer apps and what to learn from them

1.1 Government and public apps (India)

All Play Store figures are [V], captured 2026-11-12.

APP (LINK)	WHAT IT DOES	PLAY RATING (RATINGS) / DOWNLOADS / LAST UPDATE	NOTES FROM REVIEWS AND STUDIES
Kisan Suvidha (MoA&FW) Play listing, MyGov page	Six areas: weather, dealers, market prices, plant protection, agro-advisory and Kisan Call Centre [S].	**The listing returns HTTP 404 on Google Play** [V]. It appears to have been withdrawn.	A 2023 Tamil Nadu study found farmers adopted 70% of technologies via Kisan Suvidha, versus 64% (Farmer Mobile App) and 60% (e-NAM) [S] (PDF). An older MyGov comment said the weather and price tabs were not updating [S]. A critical commentary noted about 2.75 lakh downloads at the time [S] (TNM).
Meghdoot (IMD, IITM, ICAR; built by ICRISAT) Play	District- and crop-wise agromet advisories from AMFUs every Tuesday and Friday, plus forecast and historic weather [V].	**3.0** (1,266) / 500K+ / Aug 13, 2026	Visible reviews: slow; "sarkari-office" UX; district-level only (not town); forced sign-up; registration form bugs (Oct 2026) [V]. A MoES parliamentary reply describes its purpose [S] (PDF).
Mausam (IMD) Play	Observed weather, forecasts, radar and warnings from mausam.imd.gov.in [V].	**3.7** (4,617) / 1M+ / Oct 27, 2026	Recent 1-star reviews: no data shown for the entered location; menus empty; fails if the "Fused Location" permission is refused [V].
Damini: Lightning Alert (IITM Pune/ESSO) Play	GPS lightning alerts within 20 and 40 km, plus safety dos and don'ts [V].	**4.2** (315) / 100K+ / Aug 30, 2026	Mixed reviews: one reports a strike with no warning; others praise it [V].

APP (LINK)	WHAT IT DOES	PLAY RATING (RATINGS) / DOWNLOADS / LAST UPDATE	NOTES FROM REVIEWS AND STUDIES
PM-KISAN app / Kisan e-Mitra chatbot Play, Wadhwani AI	Scheme chatbot with text, FAQ-click and voice input; 11 languages; 49 query categories; translation via Bhashini's IndicTrans2; NLP intent detection, then LLM answers filled into templates with beneficiary data [V].	PM-KISAN app: **3.8** (85,019) / 10M+ / Apr 25, 2026	2.69 million queries in 186 days and about 13,000 queries a day (Wadhwani AI, page dated 6 Nov 2025) [V]. Launched 21 Sep 2023 [S] (AIR). **Kashmiri and Urdu are not among the 11 languages** [V].
e-NAM (SFAC) Play	Remote bidding for traders; mandi list, arrivals and prices for farmers; the app has limited features compared with the web version [V].	**3.1** (1,947) / 500K+ / Mar 9, 2026	Visible reviews: login failures, wrong prices, painful registration ("480 clicks" to enter a birth date) [V]. e-NAM transaction value in J&K (per the HADP article) rose from ₹14 lakh (2020–21) to over ₹670 crore [V] (FreshPlaza, 2 Feb 2026).
Agmarknet 2.0 (MoA&FW) Play	Mainly a data-entry app for officials at APMC yards. Data is published on agmarknet.gov.in [V].	**3.4** (27) / 10K+ / Sep 4, 2026	Reviews: historic prices removed, crops missing, "not working properly" [V]. The portal was upgraded to 2.0 in November 2025 and 4,367 mandis are linked [S] (india.gov.in, AgriTimes).
Kisan Sarathi (MeitY, ICAR, DIC, IASRI) Play	Raise queries to experts with images, call KCC/KVK experts, multimedia advisories from KVKs in local language, weather, prices and schemes [V].	**4.1** (276) / 100K+ / Oct 27, 2026	A model for "ask a KVK expert" escalation.
Bharat VISTAAR (MoA&FW) Play	AI "digital public infrastructure": ICAR advisories, IMD forecasts, Agmarknet prices, Soil Health Card advice, NPSS pest advisories [V]. Voice helpline **155261** ("Bharati"); Hindi and English first; Phase I launched 17 Feb 2026; ₹150 crore allocated for 2026-27 [S] (The Week, FPJ).	**3.7** (47) / 10K+ / Feb 18, 2026	Reviews: "very little information", photo upload broken (Oct 2026) [V]. **Analysis:** it is a likely future data or API partner and also a competitor for generic advice.
MahaVISTAAR-AI (Govt of Maharashtra) Play	State AI advisory: weather, prices, pest ID, agromet advisories, warehouse and CHC information [V].	**3.8** (2,487) / 1M+ / Sep 24, 2026	The state-level template that Bharat VISTAAR follows.
NPSS: National Pest Surveillance System (DPPQS and ICAR-NRCIPM) Vikaspedia	Photo-based pest ID: 61 crops for identification, advisories for 15 crops, 6 languages; launched 15 Aug 2024 [S].	Not checked	Whether it covers apple is UNVERIFIED.
Kisan Khidmat Ghar (J&K, built by BISAG-N/MeitY) Play	**Orchard-wise weather prediction**, farm inputs, farm guide, schemes, skilling, expert support, chatbot [V]. Under HADP it supplies live wholesale price data; FreshPlaza's summary also mentions price and pest forecasting [V]/[S] (FreshPlaza).	**3.4** (151) / 50K+ / Aug 20, 2026	Reviews: registration and login failures; usernames with digits or "#" in passwords rejected; long "Krishi Udyami" usernames break login [V]. **This is the J&K government's own digital channel to apple growers.**
Horticulture Procurement MIS (HPMIS), Himachal Play, news	End-to-end digital apple Market Intervention Scheme: one-time farmer registration, digital receipts by SMS, DBT payment, real-time tracking. Over 17,000 farmers registered and over 8,000 MT procured; 213 collection centres; procurement 1 Aug–31 Oct 2026 [V].	**3.7** (18) / 10K+ / Sep 30, 2026	A model for the digital-receipt and DBT flow.

1.2 Private agri-input and advisory apps (India)

APP (LINK)	WHAT IT DOES	PLAY RATING (RATINGS) / DOWNLOADS / LAST UPDATE [V]	REVIEW AND STUDY NOTES
Plantix Play, apple library, API toolkit	Photo crop diagnosis, treatment, community, input retail via "Plantix Partner" dealers [V]. **Its apple library includes Apple Scab, Marssonina Blotch, Alternaria, Powdery Mildew, Woolly Aphid, San Jose Scale, European Red Mite and Codling Moth**: 19 fungal, 27 insect and 4 mite problems plus 10 deficiencies [V].	**4.1** (143,107) / 10M+ / Oct 16, 2026	Visible reviews praise fast, accurate diagnosis; an older one complains about limited crop options [V]. An AgEcon study found it the most effective of the apps compared ("informative, simple, easy to use") [S] (AgEcon). **It is the main photo-AI competitor and already covers Kashmir's key apple diseases.**
AgroStar Play	Agronomy advice plus input e-commerce; "missed call" access; operates in Gujarat, Maharashtra, Rajasthan, MP and UP [V].	**4.2** (79,832) / 5M+ / Oct 30, 2026	Visible reviews: notifications fail to open, cannot change language or location, missing prices, no call-back after a missed call [V].
BigHaat Play	Input e-commerce (seeds, fertilizers, pesticides, machinery), fertilizer calculator, expert advice [V].	**4.3** (14,878) / 1M+ / Nov 2, 2026	Recent 1-star reviews: wrong product delivered, returns ignored, items out of stock while listed in stock, weak IVR support [V]. **Lesson: fulfilment and stock accuracy drive ratings.**
Gramophone Play	Advisory, inputs with home delivery, community of 5 lakh+ farmers [V].	**4.4** (7,175) / 500K+ / **last updated Jun 19, 2023**	The listing looks stale; whether a newer app replaced it is UNVERIFIED.
DeHaat Kisan Play	Inputs, a "Crop Doctor" photo answer "in under 30 minutes", satellite farm monitoring, crop calendar, nearby DeHaat centres [V].	**3.8** (1,280) / 100K+ / Jul 15, 2026	Reviews: praise for doorstep centres and credit; complaints that English mode still shows non-English crop names, and bugs [V].
Fasal Play, site	Horticulture farm app plus "Fasal Sense" IoT; disease and pest predictions; irrigation automation (FasalJet). **Apple is listed** among its crops; its sensors are soil moisture, soil temperature and a weather station. Leaf-wetness sensing is not mentioned on the site [V].	**4.5** (222) / 10K+ / Nov 12, 2026	Older 1-star reviews (2020–22): crashes, unresponsive support [V]. A Kashmir deployment was not confirmed [UNVERIFIED].
Cropin Grow (formerly SmartFarm Plus) Play	Enterprise (B2B) farm digitisation. Company says it serves "100+ enterprises across 92 countries" [V].	**3.7** (121) / 10K+ / Nov 9, 2026	A B2B model, not farmer-facing.

1.3 AI chat assistants

ASSISTANT	FACTS	LINK
FarmerChat (Digital Green)	Text, voice and photo questions [V]. Play: **3.9** (2,087) / 1M+ / Oct 13, 2026 [V]. A 2024 paper reports 15,000+ farmers and 300,000+ queries [V]. An Oct 2024 Hugging Face case study reports 20k+ farmers, 340k+ queries, 6+ languages and 50 value chains, and benchmarks LLMs on faithfulness: Gemini-1.5-Pro 91%, Gemini-1.5-Flash 89% (chosen; 4.5% unanswered), GPT-4-turbo 88%, Llama-3-70B 78% [V]. Digital Green says it has "over a million farmers" (Rural21, Mar 2026). The IFPRI–Digital Green user testing (2025–27) has not published results yet [V].	Play, arXiv 2409.08916, HF blog, Rural21
KissanAI (Titodi Infotech)	Play lists 9 Indic languages (Gujarati, Marathi, Tamil, Telugu, Kannada, Malayalam, Punjabi, Bangla, Hindi); **no Kashmiri or Urdu** [V]. **4.0** (99) / 10K+ [V]. It publishes "Dhenu" agri-LLMs (1B–8B; Llama 3.1/3.2 based) on Hugging Face, with no licence shown on the org page [V].	Play, HF
Jugalbandi (Microsoft, AI4Bharat, OpenNyAI)	WhatsApp bot for government schemes, typed or spoken; 10 languages; 171 schemes (May 2023) [V].	Microsoft

1.4 Lessons from section 1 (Analysis)

1. Government apps lose trust through **broken registration and login**, empty data screens and stale prices (Meghdoot, Mausam, e-NAM, Agmarknet 2.0, Kisan Khidmat Ghar). Allow use **without registration**, use OTP-only login, and show a fallback when an API returns nothing.
2. Never hard-fail when **location permission** is denied. Offer a district or village picker (Mausam complaint).
3. **Language** must be switchable independently of location (AgroStar and DeHaat complaints). None of the national apps checked offers Kashmiri.
4. E-commerce ratings collapse on **wrong items, stock errors and unanswered returns** (BigHaat).
5. **Expert escalation** within an hour is valued (Orchardly reviews, section 2). DeHaat promises 30 minutes.
6. Weather-only apps rate poorly. Value comes from translating weather into **actions** (spray or don't spray, frost or hail alerts).

1.5 Global apple-specific decision tools

TOOL	WHAT IT DOES	BUSINESS MODEL	LINK
RIMpro (NL)	Simulates scab ascospore maturation, release and infection from station weather plus a forecast up to 10 days; needs a green-tip biofix [S]. VT guidance: spray at **≥100 RIM** if scab was present last year and **≥300 RIM** if not; RIM/100 ≈ % of the season's ascospores causing infection [V]. FiBL/BIOFRUITNET recommends RIMpro or Simscab for timing organic sprays; sulfur works only above 12 °C [V].	Subscription: about €220/yr including one station plus about €165 per extra station (VT page) [V]; older figure about €250/yr [S].	rimpro.cloud, VT, MSU, FiBL PDF
Cornell NEWA	Apple scab tool (ascospore maturity, discharge, infection events), fire blight (Cougarblight and EIP/"Maryblyt-like"), carbohydrate thinning model [V]/[S]. VT note: with NEWA alone, a "major" event is >15% ascospore discharge [V]. A NEWA station licence for feeding RIMpro costs $50 [V].	Free public network	newa.cornell.edu/apple-scab, thinning
WSU Decision Aid System (DAS) and AgWeatherNet	11 insect, 7 horticultural and 5 disease models driven by AgWeatherNet stations; codling moth (95%) and fire blight (85%) are the most-used models (2013 survey) [S]. The WSU tree-fruit site was "temporarily unavailable" on 2026-11-12 [V].	Public (cost UNVERIFIED)	WSU DAS, 2013 survey
Trapview (EFOS, Slovenia)	Camera pheromone traps; ML identification over 90% accuracy (vendor/EU summary); thousands of traps in 40+ countries [S]. Croatian and South Tyrol trials found it equivalent to Delta traps for codling moth [S]. **No India deployment found.**	Hardware plus subscription	CORDIS, FAO observatory, MDPI review
FruitScout	Smartphone images of each tree: trunk cross-section, then buds, flowers and fruitlets, giving thinning recommendations (based on the Cornell crop-load model) [S].	Commercial	Fruit Growers News, Good Fruit
Aerobotics	Smartphone fruit sizing and forecasting ("TrueFruit Size"); claims packouts up to 5% higher [V].	Commercial	aerobotics.com
Tevel	Tethered "flying autonomous robots" picking apples and other fruit; Series C of $18M (Dec 2025) [S]; deployments in Italy, US, Chile and Israel [S]; slower than humans [S].	Robot-as-a-service (UNVERIFIED)	tevel-tech.com, Fruitnet
Croptracker (Ontario)	Traceability to block and employee, spray records, harvest and labour tracking, packhouse QC, Crop Load Vision [V].	SaaS (prices not public)	croptracker.com
CSIR Apple Orchard Water Use (South Africa)	FAO-56 ETo-based forecast of orchard water needs 1–7 days ahead from canopy cover, height, soil and cover crop [V].	Free app	Play

2. Competitive landscape: Kashmir and Himachal apple apps and startups

Coordinator request. The "Gaps our app could fill" notes are Analysis.

2.1 Orchardly (Fruitbee Agritech Pvt Ltd, Srinagar). This is the same team ThePrint/ANI called "Orchadly".

- **Offers:** Weather alerts, pest and disease prediction, irrigation and fertigation, spray advisories, soil testing, agronomy, inputs and own stores, drone services, pruning and orchard management, procurement and **QR-based traceability**, nurseries and genetics, R&D [V] (Startuppedia). Runs **40 community micro-weather stations**; it dropped per-farm stations because small growers would not pay for them [V]. In the 2022 ThePrint/ANI story the founders (Ehsan Quddusi of Tral, Izhan and Uzair Javaid of Shopian) described 30 micro-weather zones, free soil testing, "Orchadly Mandi" rate alerts, a planned "Orchadly procure" and barcode traceability, with 6,000 farmers in six months [V] (ThePrint, ANI).
- **Traction:** Company claims 15,000+ growers and about 13,500 acres; revenue ₹15.24 cr (FY25) and ₹19.04 cr (FY26), EBITDA positive, bootstrapped, 20+ staff [V] (company figures). Tracxn reports ₹13.4 cr for FY25, which conflicts with the company figure [S] (Tracxn). Named Top Emerging Startup at the J&K Startup Awards, Jan

2026 [S] (Greater Kashmir). Play: **4.2** (441) / 10K+ [V] (Play). YourStory, July 2026: "The $2 Million Startup Building an Operating System for India's Apple Industry" (link; returned 403 to my fetch).

- **Gaps we could fill:** Reviewers ask for **faster expert responses** [V]. No Kashmiri-language voice is mentioned. There is no transparent self-serve diagnostic wizard; advice is expert- and service-led. Orchardly is vertically integrated (it sells its own inputs), so a **neutral marketplace** with price comparison and an open services marketplace is a different position.

2.2 Farmit by Qul (Qul Fruits, Kashmir)

- **Offers:** Personalised spray schedules from orchard age, density, yield history and stage, "based on Qul's SKUAST-K validated Package of Practice". **AI chat in Urdu and English**, voice input in Urdu or English, **crop-stage identification from two photos**, a weather check before each spray recommendation, **automatic blocking of insecticide advice during full bloom**, activity logging and export, offline caching, and "privacy-first AI" (name and phone never sent to the AI service) [V] (Play, App Store).
- **Traction:** Play 1K+ downloads, no rating yet, updated Jul 23, 2026 [V]. Qul (founded 2009, CEO Khuram Mir) says it works with 15,000+ growers across 2,500 acres, has 14 post-harvest hubs ("Qulgahs") and raised ₹60 cr from Incofin and Fielding. It describes Farmit as a "re-launch" and is targeting 100,000 farmers in 5 years [V] (Global Indian). Qul is also planting about 1.5 million high-density trees in 2026 [S] (FreshPlaza).
- **Possible predecessor: FarmIT** (Kashmir Life, Feb 2022) was built by a five-person team (Mursaleen, Ashraf, Adnan; pathologist Ruqaiya; orchard manager Iqra). It offered orchard mapping, photo diagnosis against preloaded images, video guidance, experts and weather alerts. Prescriptions were checked by pathologists before release. Android only; no user numbers given [V] (Kashmir Life). **Whether FarmIT and Qul's Farmit are the same product is UNVERIFIED.**
- **Gaps we could fill:** No Kashmiri language (Urdu and English only). Farmit is tied to Qul's HD-orchard business, so it fits traditional orchards and non-Qul growers less well. No marketplace for labour or transport is mentioned.

2.3 Apple Doc (Dr Basharat Ahmad Bhat, incubated at SKIIE/SKUAST-K)

- **Offers:** An "AI-driven decision support system" with hourly and real-time weather advisories, spray and fertiliser timing, irrigation by soil moisture, soil-test-based advice, satellite-linked orchard data, expert chat, video calls or visits (KVK experts for remote areas), SMS updates that work offline, and links to input suppliers, schemes and banks [V] (Greater Kashmir, Business Kashmir, Kashmir Reader). Funded with ₹50 lakh under BIRAC-BIG; launched 7 Mar 2022 [V]. Planned drone precision spraying and expansion to rice, cherry and walnut [V].
- **Traction:** No user numbers were published. **I could not find a current Play Store listing** in Play searches on 2026-11-12, so its current availability is UNVERIFIED.
- **Gaps:** It may be dormant, and no published evidence shows the image-based AI works. The market for SKUAST-linked decision support is open.

2.4 Kash Crop (Hazik Hussain, Norpora, Tral)

- **Offers:** GPS plus an in-app AI assistant; weather, plant-protection product information, disease identification, market rates, soil testing, customer care; apple first [V] (FreshPlaza, 4 Dec 2025, ETV Bharat). Release was announced for New Year's Day 2026.
- **Traction:** No listing found in Play searches on 2026-11-12 [V], so its status is UNVERIFIED. Rising Kashmir credits a "Haziq Hussain" as developer of SKUAST-K's new Plant Health Clinic app (May 2026) [V]. **Whether this is the same person is UNVERIFIED.**
- **Gaps:** It is an early-stage solo project. It shows that local developers are building similar feature sets.

2.5 SKUAST-K e-Plant Health Clinic (Directorate of Extension)

- **Offers:** Case creation with crop photos and location, expert review, **live video consultation with SKUAST-K scientists**, messaging, feedback and reopening cases. The store text calls AI guidance "decision support". Launched 18 May 2026 by the VC [V] (Rising Kashmir, Play).
- **Traction:** Play shows **10+ downloads** [V].
- **Gap or partnership:** This is the natural **expert-escalation partner** for our yes/no wizard. Analysis: integrate a "send case to SKUAST-K clinic" hand-off rather than competing with it.

2.6 HADP apple value-chain restructuring and Kisan Khidmat Ghar (J&K government)

- **Facts** [V] (FreshPlaza, 2 Feb 2026):
- **HADP Project-21** (designer plants for high density): 10,000+ applications, 4,500 approvals, 1,500+ units in two years.
- Nurseries moving to rootstock banks and mother orchards; small grading units near production areas.
- Investment in CA storage and reefer transport; first rail apple consignment from the Valley in 2025.
- Output above 2.1 million t.
- **Kisan Khidmat Ghar** provides live wholesale prices.
- **Gaps:** The KKG app has registration and login problems (reviews) [V]. Analysis: our app can **complement or feed** KKG and HADP rather than replicate them, for example by linking HADP subsidies, designer-plant suppliers and grading units into lessons and the services marketplace.

2.7 Other Kashmir and Himachal items

ITEM	FACTS	LINK
Applefy	Live spray windows and weather, a growth-stage checklist, multiple orchards, a marketplace for saplings, sprays, fertilizer and tools. Follows the "SKUAST Integrated Management Guide for Apple Orchards, Kashmir". Android only; SMS OTP login; team not named [V]. No Play listing found by search [V].	applefy.in/faq
AppleCare Kashmir (GitHub, open source)	FastAPI plus Flutter. A MobileNetV3-Large classifier for 6 classes (scab, Alternaria leaf blotch, powdery mildew, black rot/canker, cedar apple rust, healthy) trained on "Kashmir indigenous and PlantVillage" data; reports 98.04% test accuracy and about 15 ms in ONNX. SKUAST-K stage advisories with doses per 100 L; spray-safe weather windows for 10 districts; **voice advisories in Urdu, Kashmiri and English**; rejects non-plant images. **No licence file** [V].	GitHub
Kashmir Apple Market & Orchads	Market rates and orchard information. Developer "Financial Knowledge Pro". **4.7** (12) / 500+ [V].	Play
Orchive: Smart Apple Farming AI	Planting-density calculator, elevation insights, variety and pollination guide, "Orchive Vision" image analysis, satellite insights. Developer Sawan Chauhan (location not stated). **5.0** (7) / 100+ [V].	Play

ITEM	FACTS	LINK
Himachal agromet advisories	Dr YSP UHF Nauni and CSK HPKV run 4 AMFUs with IMD. About **50 WhatsApp groups reach about 5,000 farmers**; frost and fog alerts for apple orchards [V]. Nauni asks growers to email symptom photos to hodmpp@uhf.ac.in [S].	Tribune, 15 Jul 2025, News Himachal
Plantix	Already covers apple scab, Marssonina and Alternaria (section 1.2).	library

Overall gap map (Analysis). None of the Kashmir products found combines all of the following:

1. **Kashmiri voice** in both Perso-Arabic and Devanagari scripts. Only the AppleCare prototype mentions it.
2. A **transparent yes/no diagnostic wizard** with confidence levels and escalation to the SKUAST-K clinic.
3. A **neutral input marketplace** that checks licences.
4. A **services marketplace** for labour, spray teams, drones and transport.
5. **Road status for the Srinagar–Jammu highway and other routes**, plus Delhi and Kashmir mandi price comparison.

3. Datasets

3.1 Apple leaf and fruit disease image datasets

DATASET	CONTENT	LICENCE	LINK	NOTES
Indigenous Dataset for Apple Leaf Disease Detection and Classification (Yatoo & Sharma; Kashmir)	**7,569 field images**, Healthy / Alternaria / Apple-Mosaic; expert-reviewed; Keras-augmented [V]. Data in Brief vol. 53 (2024) [S].	**CC BY 4.0** [V]	Mendeley DOI 10.17632/9m2dcb5mmr.3, article	Kashmir field conditions. No scab or Marssonina class. Includes augmented images, so de-duplicate before you split into train and test.
Apple Leaf Diseases Image Dataset of ICAR-CITH (Sajad Un Nabi, Ronit Jaiswal)	Alternaria leaf blotch, powdery mildew and mosaic from **ICAR-CITH experimental fields, Srinagar**; scab images taken from Kaggle; 11 Aug 2025 [V]	**CC BY 4.0** [V]	Mendeley DOI 10.17632/gm6mfz8fz6.1	Image counts not shown on the page. Check the origin and licence of the scab images.
D-KAP Kashmiri Apple Plant Disease Dataset (H. Sharma, D. Padha, N. Bashir)	Scab, rot and Alternaria leaf blotch plus healthy; about 419 images [S]	**ODbL (database) + DbCL (contents)** per Kaggle metadata [V]	Kaggle	ODbL is share-alike for the database.
AppleScab-LT (Hilal, Sofi, Goel; SKUAST-K and a Shopian orchard, Apr–Sep 2025 [S])	21 longitudinal leaf sequences, 2,101 images, polygon annotations, severity [V]	Not stated; hosting not stated [V]	arXiv 2608.14235	Valuable for scab-progression or severity models. Contact the authors.
SKUAST-K orchard dataset (Khan, Quadri, Banday)	Scab, Alternaria, mosaic, **Marssonina leaf blotch** and powdery mildew (1,556 scab, 1,312 MLB images) [S]; CNN accuracy about 97.18% [V]	Availability and licence not stated [V]	arXiv 2007.02980	The only Marssonina-labelled Kashmir set found. It is not public.
Himachal and Uttarakhand set (research article)	"50,000 images" from HP and Uttarakhand apple farms; Marssonina and scab [S]	Not public (UNVERIFIED)	ScienceDirect	Lead only.
PlantVillage (Hughes & Salathé)	54,303 images, 38 classes in TFDS. The original source is no longer available, so TFDS uses a republished copy [V]. Apple classes (scab, black rot, cedar apple rust, healthy) [UNVERIFIED]	Original image terms unclear: the GitHub repo has no licence, and the arXiv page shows CC BY-NC-SA 4.0 for the paper only [V]	arXiv 1511.08060, GitHub, TFDS	**Lab backgrounds. Mohanty et al. 2016: 99.35% held-out accuracy fell to 31.4% on images from other sources** [S] (arXiv 1604.03169).
PlantVillage, augmented copy (Arun Pandian & Gopal 2019)	61,486 images, 39 classes including apple, with augmentation [V]	**CC0 1.0** [V]	Mendeley 10.17632/tywbtsjrjv.1	The cleanest licence route to PlantVillage images.
Plant Pathology 2020 (FGVC7) (Cornell; Thapa et al., *Applications in Plant Sciences* 8(9):e11390)	3,651 images: 1,200 scab, 1,399 cedar apple rust, 187 complex, 865 healthy [S]	**Governed by Kaggle competition rules; data licence UNVERIFIED.** The article is CC BY 4.0 [V].	Kaggle, article	Check the competition "Rules" tab before commercial use.
Plant Pathology 2021 (FGVC8)	18,632 images; adds two diseases [S]	Per competition rules (UNVERIFIED); mirrors list "Unknown" [S]	Kaggle	Same caution.
AppleLeaf9 (Yang, Duan, Wang 2022, *Agronomy* 12:2784)	14,582 images (94% in the wild), 9 classes: healthy, Alternaria leaf spot, brown spot, frogeye, grey spot, mosaic, powdery mildew, rust, scab [V]	**CC BY 4.0** [V]	GitHub	Built from PlantVillage, ATLDSD, PP2020 and PP2021, so the upstream terms may still apply.
ATLDSD, Apple Tree Leaf Disease Segmentation (Feng & Chao 2022)	About 1,641 image and mask pairs with pixel-level masks [S]	UNVERIFIED	Science Data Bank DOI 10.11922/sciencedb.01627	Useful for lesion-area severity scoring.
A Dual-Dataset Framework for Apple Leaf Analysis (pest lifecycle and disease)	Not inspected	UNVERIFIED	Mendeley r56kb9t3t4	Lead only.

3.2 Fruit detection and yield datasets

DATASET	CONTENT	LICENCE	LINK
MinneApple (Häni, Roy, Isler)	Detection, segmentation and counting benchmark; about 1,000 images with 41,000+ annotated instances per the abstract [S]	**CC BY-NC-SA 3.0 US** per the DRUM record [S]. The code repo is MIT [V].	GitHub, DRUM DOI 10.13020/8ecp-3r13

DATASET	CONTENT	LICENCE	LINK
Fuji-SfM (GRAP, Univ. Lleida)	Mask R-CNN training images, SfM images of 11 Fuji trees, 3D point cloud with apple ground truth (Spain, 2017) [S]	**Conflict:** Zenodo metadata says CC BY 4.0, but the bundled **LICENSE.txt says CC BY-NC-SA 4.0** [V], and the GRAP page says research and education only [S]. Treat it as **non-commercial**.	Zenodo 3712808, Data in Brief
Other GRAP sets (KFuji RGB-DS, PFuji-Size)	Not inspected	UNVERIFIED	GRAP publications

3.3 Weather

SOURCE	FACTS	LICENCE AND TERMS	LINK
IMD API	Endpoints under `api.imd.gov.in/api/v1/` [V]: `cityforecast`, `cityforecastloc`, `current_wx`, `districtnowcast`, `stationnowcast`, `aws_data`, `districtwarning`, `districtrainfall`, `sunmoon`, among others. Agromet advisory, radar and lightning are also listed.	**IP whitelisting** via a request form and confirmation email [S]. The newer portal mentions JWT and key-based access [S]. Rate limits not stated.	API reference, whitelisting form
IMD gridded data	0.25° daily rainfall 1901–2024 (135×129 grid); NetCDF, GRD and ASCII; cite Pai et al. (2014); IMD disclaims liability [S]	Formal licence not found; some products are "restricted" on IRI [S] (UNVERIFIED)	IMD Pune NetCDF
ERA5 / ERA5-Land (Copernicus C3S)	Global hourly reanalysis	**CC BY 4.0 since 2 July 2025**, replacing the Copernicus licence. Credit "Generated using Copernicus Climate Change Service data [Year]..." [S]	ECMWF forum, access conditions
Open-Meteo	Forecast, historical, ensemble APIs	Data **CC BY 4.0, attribution required even for commercial use** [V]. Free tier is **non-commercial only**: under 10,000 calls/day, 5,000/hour, 600/minute; apps with subscriptions or ads count as commercial [V]. Paid tiers: Standard 1M, Professional 5M, Enterprise 50M+ calls/month (prices only at checkout) [V]. Server is **AGPLv3**, so you can self-host [V].	Terms, Pricing

3.4 Market prices

SOURCE	FACTS	LICENCE	LINK
Agmarknet 2.0	Upgraded Nov 2025; 4,367 mandis linked [S]. Ministry points users to the Agmarknet 2.0 dashboard and UPAg [S].	Official API for 2.0: UNVERIFIED	agmarknet.gov.in, india.gov.in
data.gov.in, "Current Daily Price of Various Commodities from Various Markets (Mandi)"	Daily min, max and modal wholesale prices from Agmarknet [S]. The page snippet says "The API for this resource does not exist... Request API" [S]. Community wrappers exist.	**GODL-India**: worldwide, royalty-free, **commercial use allowed** with attribution; notified Feb 2017 [S]. AIKosh lists the same data under the Open Government License, India [S].	data.gov.in resource, AIKosh, GODL (Wikipedia)
e-NAM	Mandi-wise arrivals and prices in the app [V]. J&K e-NAM value over ₹670 cr [V].	Terms UNVERIFIED	enam.gov.in, Play
Kisan Khidmat Ghar (J&K)	Live wholesale prices from major markets [V]	Partnership needed	FreshPlaza

3.5 Soil

SOURCE	FACTS	LICENCE	LINK
Soil Health Card portal	Per-farmer card PDFs (state, district, village, then name or ID); national nutrient dashboard [S]. Village-level J&K datasets appear on third-party portals (Dataful; India Data Portal CKAN "Soil Nutrient Analysis", updated 26 Aug 2025) [S].	Third-party portal terms UNVERIFIED	soilhealth.dac.gov.in, Dataful, IDP CKAN, AIKosh SHC
SoilGrids 2.0 (ISRIC, 250 m)	pH, SOC, bulk density, texture, CEC, N at six depths. **The REST API is paused with no restoration date.** Use WCS, GeoTIFF or Google Earth Engine instead [S].	**CC BY 4.0** [S]	ISRIC docs
ICAR-NBSS&LUP soil maps	Not checked	UNVERIFIED	(no source gathered)

3.6 Satellite

SOURCE	FACTS	LICENCE	LINK
Sentinel-2 (Copernicus)	10 m multispectral	"Free, full and open"; modified products must carry **"Contains modified Copernicus Sentinel data [Year]"** [S]	Legal notice, CDSE T&C
Landsat (USGS)	30 m	**No restrictions** on use or redistribution; citation requested [S]	USGS FAQ
ISRO Bhuvan / NOEDA, Bhoonidhi	Free registration. AWiFS (56 m) and LISS-III (23.5 m) are free with about a **2-year lag**. LISS-IV was priced per a NIAS brief. A blog reports the LISS-IV archive was opened on Bhoonidhi [S].	NRSC data policy (UNVERIFIED)	Bhuvan free data, Bhoonidhi FAQ, LISS-IV blog

3.7 Kashmiri speech and language datasets

ITEM	FACTS	LICENCE	LINK
Bolbosh (Kashmiri TTS; GAASH Lab)	First open neural Kashmiri TTS (Matcha-TTS, OT-CFM). **MOS 3.63 vs 1.86** for zero-shot Indic baselines [V] (arXiv 2603.07513, Mar 2026).	Dataset CC BY 4.0; GAASH Kashmiri Matcha-TTS model MIT [S]	arXiv, project, HF dataset, HF model

4. AI and language technology for Kashmiri, Urdu and Hindi users

4.1 Language support matrix

PROVIDER / MODEL	KASHMIRI	URDU / HINDI	LICENCE AND ACCESS	SOURCE
Bhashini (MeitY)	**ASR:** `bhashini/ai4bharat/conformer-multilingual-asr`, `bhashini/bodhan/asr-transcribe-flex` and `-core`. **NMT:** `bhashini/iiith/nmt-all`, `ai4bharat/indictrans-v2-all-gpu--t4`, `bhashini/bodhan/Indic-trans-v4`, `bhashini/iitp/coild/nmt` (all source and target); `Bhashini/IIITH/Trans/V1` (target only). **TTS:** `Bhashini/IITM/TTS` "**Kashmiri (Devanagari Script)**", `bhashini/bodhan/indic-tts`. Also transliteration (IndicXlit), printed-text OCR, and audio and text language ID [V].	Yes (multiple)	Government API platform; the docs page was about 9 months old [S]. Usage terms UNVERIFIED.	Bhashini API docs
AI4Bharat IndicTrans2	**kas_Arab and kas_Deva** [S]	Yes	**MIT** checkpoints; training data CC0 / CC BY 4.0 [S]	GitHub, HF
AI4Bharat IndicConformer 600M multilingual (ASR)	Kashmiri (ks) listed among 22 languages; a separate monolingual `indicconformer_stt_ks_hybrid_ctc_rnnt_large` exists [S]	Yes	**MIT**; **gated** on HF (accept terms) [S]	HF, GitHub
AI4Bharat Indic Parler-TTS	Kashmiri only **"unofficial" support** [S]	Urdu officially supported [S]	Apache-2.0 per model card; AIKosh lists MIT (conflict) [S]	HF
Meta MMS	**Language ID only; no ASR or TTS** for `kas` [V]	Urdu ASR and TTS (Arabic, Devanagari and Latin scripts) [V]	**CC-BY-NC 4.0, non-commercial** [S]	MMS coverage table
Google Cloud Translation / Google Translate	**Not on the Cloud Translation supported-language page** (checked 2026-11-12; Dogri, Konkani, Maithili, Manipuri, Sindhi and Urdu are) [V]. The June 2024 expansion added 7 Indian languages, not Kashmiri [S].	Yes	Commercial API	Cloud languages, Google blog
Sarvam AI	**Sarvam-Translate and Sarvam Vision list `ks-IN` and `ur-IN`.** Saaras v3 ASR covers "23 languages"; Kashmiri is likely but not confirmed for v3 [V]. **Bulbul TTS v3 and v4 do not cover Kashmiri or Urdu** [V].	Partly	Commercial API	Sarvam models
Kashmiri ASR quality	A search summary attributes to the Bolbosh paper a best tested Kashmiri ASR WER of 41.2% (66.6% with diacritics) [UNVERIFIED: not visible on the abstract page]			arXiv 2603.07513

Implications (Analysis): - Kashmiri voice output is the weakest link. Bhashini TTS is Devanagari-only, while most Kashmiri readers use Perso-Arabic script, and Bolbosh is research-grade. - Plan for: pre-recorded Kashmiri audio for the fixed yes/no wizard prompts (the most reliable option); Bhashini or IndicTrans2 for dynamic text; Urdu TTS as a fallback; and human-reviewed translations of safety-critical text (doses, PHI). - Avoid MMS in a commercial product because of its NC licence.

4.2 WhatsApp, voice and chat channels in Indian agriculture

- **WhatsApp:** Jugalbandi [V]. Himachal AMFU WhatsApp groups (human-run) [V]. Orchardly grew out of WhatsApp advice to growers [V].
- **Voice helpline:** Bharat VISTAAR on 155261 [S]. Kisan e-Mitra accepts voice input [V].
- **App plus LLM:** FarmerChat, KissanAI, Farmit (Urdu chat and voice) [V].

4.3 Evaluations of LLM farm advisors

EVALUATION	FINDING	LINK
Digital Green LLM-as-judge (Oct 2024)	Binary RAG-faithfulness judge built on atomic facts correlated highly with humans on about 360 questions. Results on about 700 queries: Gemini-1.5-Pro 91%, Flash 89%, GPT-4-turbo 88%, Llama-3-70B 78%. Track **unanswered rate** alongside faithfulness [V].	HF blog
DG-Eval / fine-tuning paper (Feb–Mar 2026)	Separates fact retrieval (LoRA SFT on expert "golden facts") from a delivery "stitching" layer. Atomic-fact recall, precision and contradiction metrics. A smaller fine-tuned model matched or beat frontier models on factual quality at lower cost (qualitative claims; Bihar data) [V].	arXiv 2603.03294
Assorted 2025 studies	Headline accuracies of 91–97% come from small closed benchmarks, for example a Dhenu2 RAG study at 95.2% on 504 questions [S]. **Kisan-BiQAS (KCC data): LLaMA F1 ≈ 0.45; Qwen F1 ≈ 0.15 with ≈58% hallucination** [S]; the exact paper behind this figure was not identified among the results (UNVERIFIED source).	ResearchGate (Dhenu2), Frontiers review
IFPRI and Digital Green user testing (2025–27)	Preliminary: local specificity (crop stage, weather, available inputs) drives trust; voice matters for low-literacy users; women benefit from assisted onboarding; low bandwidth slows image and voice uploads [V].	Rural21

4.4 On-device image models

- **LiteRT** is TensorFlow Lite renamed on **4 Sep 2024**. It is the same runtime with a new package name (Python `ai-edge-litert`), and it now accepts PyTorch, JAX and Keras models. Apps using TFLite through Play Services needed no changes [S] (Google dev blog).
- **PlantVillage Nuru** (cassava, offline) [S] (Frontiers 2020):
- With six leaves examined, symptom-recognition accuracy was 74–88%.
- Field accuracy was 65% (2020), against 40–58% for extension agents and 18–31% for farmers.
- **Generalisation warning:** a model trained on PlantVillage reached 99.35% on held-out images but **31.4%** on images from other sources [S] (arXiv 1604.03169). Lightweight apple models such as CC-MobileViT (98.10%) are reported mostly on curated sets [S] (Frontiers 2025).
- **Analysis:** Collect Kashmir field photos through the wizard with explicit DPDP consent and record the GPS location. Use "unknown / see expert" outputs, and add a non-plant rejector (as AppleCare Kashmir does).

5. Disease and pest forecasting

5.1 Apple scab infection models

- **Mills table** (Mills & Laplante 1944, 1951), **modified by Jones 1980**, with Mills's criteria **revised by MacHardy & Gadoury 1989** (journal citation *Phytopathology* 79:304 is from memory, UNVERIFIED). The Virginia Tech extension page reproduces a simplified Mills table and links the MacHardy & Gadoury PDF. It recommends Mills charts when growers do not use digital models [V] (VT). **The hour-by-temperature thresholds were not transcribed in this session.** Take them from the primary paper (UNVERIFIED here).
- **Low-temperature revision** by Stensvand et al. 1997 (Phytopathology): reference from memory [UNVERIFIED].
- **RIMpro:** see section 1.5.
- **NEWA apple scab:** ascospore maturity, discharge and infection events; also a "potential ascospore dose" threshold of 53 infected leaves per 600 shoots for skipping the green-tip spray in low-inoculum orchards [V] (NEWA).

5.2 Marssonina, Alternaria and mite models

- **Not researched in this session** (search budget exhausted).
- Leads to follow up: weather-based Marssonina coronaria and Alternaria mali infection models from Korea, China, Italy and India (SKUAST-K/ICAR-CITH); degree-day models for European red mite and two-spotted spider mite egg hatch.
- All UNVERIFIED. Plantix lists Marssonina Blotch, Alternaria, European Red Mite and Spider Mites in its apple library [V], but that covers identification, not forecasting.

5.3 Weather-station and IoT networks: Kashmir and Himachal

NETWORK	FACTS	LINK
Orchardly (Kashmir)	40 community micro-weather stations; ML-based preventive and curative advice from weather, soil, crop stage and disease patterns [V]	Startuppedia
Kisan Khidmat Ghar (J&K govt)	"Orchard-wise weather prediction" [V]; pest and disease forecasting linked to local weather (FreshPlaza summary) [S]	Play, FreshPlaza
Apple Doc	Satellite-based weather advisories; "very good accuracy" claimed without data [V]	Business Kashmir
Fasal	IoT units with disease and pest predictions; apple listed; Kashmir not confirmed [V]	fasal.co
Himachal AMFUs (Nauni, Palampur, with IMD)	Frost and fog alerts for apple; WhatsApp distribution; growers acting on advisories report fewer losses (qualitative) [V]	Tribune
SKUAST-K and HADP automatic weather stations for scab forecasting	**UNVERIFIED.** Not found before the search budget ran out.	(none)

Analysis: Implement a Mills/MacHardy infection-period engine using hourly forecast temperature, precipitation and RH, and estimate leaf wetness from RH and rain. Calibrate it against SKUAST-K/ICAR-CITH observations, and label outputs as "risk", not certainty.

6. Orchard technology

6.1 Drones

- **Drone Rules, 2021** (PIB release 26 Aug 2021; replaced the UAS Rules of March 2021) [V] (PIB 1749154):
- **Approvals and paperwork:** forms cut from 25 to 5; fee types cut from 72 to 4; remote pilot licence fee ₹100, valid 10 years.
- **Airspace:** Digital Sky airspace map with green, yellow and red zones; **no permission needed in green zones** (up to 120 m); yellow zone cut to 12 km from airport perimeters.
- **Coverage:** extended to drones up to 500 kg.
- **Licensing:** remote pilot certificate from DGCA-authorised drone schools.
- **Penalties:** maximum ₹1 lakh.
- **Future requirements:** NPNT and geo-fencing to be notified later.
- **Pesticide spraying by drone:** the MoA&FW SOP for drone pesticide and nutrient application and the CIB&RC interim approvals for drone use of registered pesticides are **UNVERIFIED in this session** (sources blocked). Confirm the current status on ppqs.gov.in (CIB&RC) and agriwelfare.gov.in.
- **Kashmir and Himachal drone-spray trials:** UNVERIFIED (not found). Orchardly lists "drone-based services" [V]; Apple Doc planned drone precision spraying [V].
- **Analysis:** Before offering drone services in J&K, check the airspace zones on Digital Sky for each orchard location. Much of J&K may sit in yellow or red zones (UNVERIFIED).

6.2 Remote sensing of orchards

- Sentinel-2, Landsat and Bhuvan: section 3.6.
- In-market satellite features: DeHaat (farm health, moisture, stress) [V]; Orchive ("satellite-based insights") [V]; Orchardly (multispectral monitoring "upcoming") [V].
- ISRO's **CHAMAN** horticulture programme page (nrsc.gov.in/chaman) could not be loaded. Whether CHAMAN mapped J&K apple is UNVERIFIED.

6.3 Hail nets and anti-hail systems

UNVERIFIED. No sources were gathered before the search budget ran out.

6.4 Sprayer types (air-blast, tower, tunnel)

UNVERIFIED. No sources were gathered. Related: the HADP article mentions canopy management and **targeted application** demonstrated in model orchards to cut pesticide use [V] (FreshPlaza).

6.5 Smart pheromone traps

Trapview: section 1.5. A 2022 MDPI review of camera traps for codling moth is a good primer [S] (MDPI). No Indian deployments were found.

6.6 Harvesting robots, platforms and crop-load tools

Tevel (flying pickers), FruitScout (crop load), Aerobotics (sizing) and Croptracker Crop Load Vision: section 1.5. Harvest-assist platforms were not researched (UNVERIFIED).

6.7 Grading, CA storage and logistics (J&K)

HADP [V] (FreshPlaza): - Small-scale grading units near production areas enable grade-based pricing. - Investment in CA storage and refrigerated transport. - First rail apple consignment from the Valley in 2025. - High-density designer-plant units (Project-21).

CA capacity figures, 1-MCP/SmartFresh and dynamic CA were not researched (UNVERIFIED).

6.8 Traceability, QR and blockchain

- Orchardly runs **QR-based traceability** recording sprays, fertiliser and harvest [V].
- Orchardly's 2022 plan included barcode-based supply-chain traceability [V].
- HPMIS (Himachal) issues digital receipts plus DBT [V].
- Croptracker provides traceability to block and employee [V].
- APEDA export traceability (HortiNet) and blockchain pilots: UNVERIFIED.

6.9 GI tags relevant to Kashmir apples

UNVERIFIED. The GI registry list page returned 404 when fetched. Check the GI Registry at ipindia.gov.in for any "Kashmir apple" or variety-level application.

---

7. Regulations that shape the app

7.1 Digital Personal Data Protection Act 2023 and DPDP Rules 2025 (verified from the gazette text)

Commencement [V] (MeitY Rules PDF, G.S.R. 846(E), commencement notification G.S.R. 843(E), PIB 14 Nov 2025): - The Rules are dated 13 Nov 2025. - Rules 1, 2 and 17–21 (the Data Protection Board and related) took effect on publication. - **Rule 4 (Consent Managers) takes effect one year after publication, about 13–14 Nov 2026. - Rules 3 and 5–16, 22 and 23 (notice, security, breach, retention, children, rights and related) take effect 18 months after publication, about 13–14 May 2027.** - Under the Act: sections 3–5, 6(1)–(8), 6(10), 7–17 and others come into force 18 months after publication; section 6(9) and section 27(1)(d) after one year.

Key obligations [V] (rules text):

AREA	REQUIREMENT	PRODUCT IMPLICATION (ANALYSIS)
Notice (Rule 3)	Standalone, plain-language notice with an **itemised list of personal data**, the specified purpose(s) and the goods or services enabled. Give a link and the means to **withdraw consent as easily as it was given**, to exercise rights and to complain to the Board.	Separate consent screens for (a) account, (b) photo diagnosis and AI training, (c) location, (d) marketing. Make the notice available in Kashmiri, Urdu, Hindi and English.
Security (Rule 6)	Encryption, obfuscation or masking, access control, logging and monitoring, backups. **Retain logs and personal data for one year** for detecting unauthorised access. Include security clauses in processor contracts.	Vendor contracts with LLM, SMS and cloud providers.
Breach (Rule 7)	Inform affected users without delay. Inform the **Board without delay, with a detailed report within 72 hours**.	Incident-response runbook.
Retention (Rule 8)	The Third Schedule's 3-year inactivity erasure applies to **e-commerce entities with 20 million (2 crore) or more registered users in India**, with a **48-hour pre-erasure notice**. All data fiduciaries must keep processing logs for at least **one year**.	Below the threshold at launch; plan for it.
Contact (Rule 9)	Publish the DPO's contact, or that of a person who can answer processing questions.	In-app "Privacy contact".
Children (Rule 10)	**Verifiable parental consent** before processing a child's data (under 18).	Age gate. Youth may use family phones.
Rights and grievances (Rule 14)	Publish how to exercise rights. **Respond to grievances within 90 days.** Allow nomination.	
Penalties (Act Schedule) [V] (Act PDF)	Security-safeguard failure: up to **₹250 crore**. Breach-notice failure: up to ₹200 crore. Children's obligations: up to ₹200 crore. Other breaches: up to ₹50 crore.	

7.2 Consumer Protection (E-Commerce) Rules, 2020 [UNVERIFIED in this session: consumeraffairs.nic.in was unreachable]

Official text: consumeraffairs.nic.in PDF. From memory, to be verified: - Notified July 2020; applies to marketplace and inventory models. - Marketplace duties: a grievance officer who **acknowledges complaints within 48 hours and resolves them within one month**; display of seller details; country-of-origin information; no price manipulation.

Also check (UNVERIFIED): - The CCPA's 2023 Guidelines on Dark Patterns. - IT (Intermediary) Rules 2021 grievance timelines. - Legal Metrology e-commerce declarations. - GST TCS for e-commerce operators. - RBI rules if you handle payments.

7.3 Pesticides

- **Insecticides Act, 1968 and Insecticides Rules, 1971:**
- Selling, stocking or exhibiting insecticides for sale requires a licence from the state licensing officer (section 13 of the Act, from memory) [UNVERIFIED].
- **Whether an amendment expressly permits or regulates e-commerce sale, and on what conditions, could not be verified** [UNVERIFIED]. Confirm with CIB&RC/DPPQS (ppqs.gov.in) and the J&K Department of Agriculture/Horticulture licensing officer.
- **In practice:** national platforms openly list pesticides for online sale: BigHaat, Gramophone, DeHaat and AgroStar [V] (Play descriptions).
- **Pesticide Management Bill, 2020:** would replace the 1968 Act. Introduced in the Rajya Sabha 23 Mar 2020; Standing Committee report 21 Dec 2021 [V] (PRS). Status after 2021: UNVERIFIED.
- **Analysis:** Use a **licensed-dealer marketplace model**. Verify each seller's licence and keep the licence number on file. Sell only CIB&RC-registered products with label claims for apple, and show the label dose and PHI. Copy Farmit's "**no insecticide recommendation at full bloom**" rule.

7.4 Fertilizers

Fertiliser (Control) Order, 1985: dealer registration and authorisation for each source; Aadhaar/PoS-based retail of subsidised fertilizers (urea, DAP) [UNVERIFIED in this session]. The Department of Fertilizers is at fert.nic.in.

Analysis: Limit online sales to non-subsidised specialty fertilizers (water-soluble, micronutrients) through registered dealers until you have legal confirmation.

7.5 Nursery plants and saplings

- **NHB National Nursery Portal** [V] (nhb.gov.in/nnp): **1,553 accredited nurseries** (NHB, CIH, DCCD, DASD), **174 state-licensed nurseries**, 186 meeting minimum quality criteria, 10,168 registered buyers; searchable by crop ("Apple") and location. There is also an NHB **Nursery Rating System**, "Find a Star Nursery" (nhb.gov.in).
- **J&K nursery registration law or rules:** UNVERIFIED (not found).
- **HADP Project-21** "designer plants" scheme and the shift to rootstock banks and mother orchards [V] (FreshPlaza).
- **Clean Plant Programme** (central; approved 2024): UNVERIFIED in this session.
- **Analysis:** List saplings only from NHB-accredited or state-licensed nurseries, and show the rootstock and certification.

7.6 Platforms that book labour or transport

- **Code on Social Security, 2020** [V] (PRS summary). The PRS page summarises the 2020 Bill; I did not compare it with the enacted text (UNVERIFIED that nothing changed):
- Defines **gig and platform workers**.
- Schemes are funded partly by **aggregators at 1–2% of annual turnover, capped at 5% of the amounts paid or payable to gig and platform workers**.
- Aggregators are listed in **Schedule 7, which has nine categories** including ride sharing, food and grocery delivery, content and media, and **e-marketplaces**.
- Gig and platform workers must be registered.
- Wikipedia lists commencement dates of 3 May 2021 and **21 November 2025** [V] (Wikipedia). Exactly which sections took effect on 21 Nov 2025 is UNVERIFIED.
- **Also check (all UNVERIFIED):** the e-Shram registration process (eshram.gov.in); the MoRTH Motor Vehicle Aggregator Guidelines (2020, updated 2025?) for passenger transport; goods-carriage permits for truck booking; OSH Code contract-labour licensing; any J&K-specific rules.
- **Analysis:** If the services marketplace onboards pruning or spray workers or truck drivers as platform workers, expect aggregator obligations. Structure early as a lead-generation directory, or take legal advice.

7.7 Drones

See 6.1: Drone Rules 2021 [V]. The pesticide-application SOP and interim CIB&RC approvals are UNVERIFIED.

---

8. Quick reference: other sources used

- Kashmir Life, "Android Orchards" (FarmIT): https://kashmirlife.net/android-orchards-287323/
- ThePrint / ANI "Orchadly" (2022): https://theprint.in/india/mobile-app-that-connects-kashmiri-farmers-to-save-grow-sell-apples/1157350/
- Startuppedia on Orchardly: https://startuppedia.in/startup-stories/kashmiri-entrepreneurs-build-an-os-for-indias-apple-industry-clock-rs-19-cr-in-fy26-12440645
- FreshPlaza HADP (2 Feb 2026): https://www.freshplaza.com/asia/article/9807105/
- FreshPlaza Kash Crop: https://www.freshplaza.com/asia/article/9791367/kashmir-developer-launches-horticulture-support-app/
- Studies on apps: Tamil Nadu 2023 (https://ageconsearch.umn.edu/record/367859/files/Yamunarani41122023AJAEES111385.pdf); MDPI AgriEngineering 2022 evaluation of smartphone apps for Indian farmers (https://mdpi.com/2624-7402/4/3/42) [S, findings not extracted].

---

9. Not verified or not covered (follow-up list)

The shared web-search budget ran out mid-task, and several .gov.in sites blocked automated fetches. The items below need a follow-up pass:

1. **Pesticide e-commerce rules:** any amendment to the Insecticides Rules 1971 on online sale, the licence conditions, and the current status of the Pesticide Management Bill.
2. **FCO 1985 online-sale conditions** and PoS/DBT constraints.
3. **J&K nursery law or rules**; Clean Plant Programme details.
4. **Consumer Protection (E-Commerce) Rules 2020** text and amendments; dark-patterns guidelines; IT Rules grievance timelines.
5. **Motor Vehicle Aggregator Guidelines 2025**, goods-transport booking rules, e-Shram obligations, and which Social Security Code sections took effect on 21 Nov 2025.
6. **Drone pesticide SOP and CIB&RC interim approvals**; Kashmir and Himachal drone-spray trials; J&K airspace zones.
7. **Marssonina and Alternaria infection models**; mite degree-day models; Mills-table thresholds (transcribe from MacHardy & Gadoury 1989).
8. **SKUAST-K and HADP AWS or IoT scab-forecast networks** and their reported results; Fasal deployments in Kashmir.
9. Hail nets and anti-hail guns, sprayer types, CA storage capacity in Kashmir, APEDA traceability, **GI status of Kashmir apples**, ISRO CHAMAN apple mapping.
10. Dataset licences: the Plant Pathology 2020/2021 Kaggle rules, ATLDSD, the original PlantVillage terms, and AppleScab-LT availability.
11. Current availability of **Apple Doc** and **Kash Crop** (not found on Play on 2026-11-12); whether FarmIT (2022) is Qul's Farmit; whether Hazik Hussain (Kash Crop) is the "Haziq Hussain" behind SKUAST-K's clinic app.
12. Bhashini usage terms and SLAs; whether Saaras v3 covers Kashmiri; whether the Kashmiri ASR WER figures appear in the Bolbosh paper.

Appendix D · Agronomy, spray schedule, inputs and open questions

Desk research notes, November 2026. Tags: [V] read at source (in Appendix D: confirmed by an independent verifier); [S] from a search summary or not re-confirmed; UNVERIFIED = lead only.

Research date: 2026-11-13. Part D of the Baghbaan research appendix (follows A, B and C in this folder). It covers the spray schedule, nutrition, harvest and storage, the pest and disease library, input quality and rules, labour and costs, and what is still open.

> **Spray schedule edition: SKUAST-K "Integrated Management Guide for Apple Orchards in Kashmir 2026", document No. AU/DEx/449-BL/01/26** (here "IMG 2026"). No 2027 edition had been found as of 13 Nov 2026. It was read from a third-party Scribd copy of the booklet (same document number, 32 pages), not from an official SKUAST-K URL. The verifier parsed the same copy. Check every row against the printed booklet (district horticulture offices) before release. The app must show "IMG 2026" on every recommendation drawn from it.

How to read this document

Every fact has a tag and a source link.

- **[V]**: confirmed by the independent verifier at the source (or at a better source, which is then the one linked).
- **[S]**: the researcher read it at the source, but the verifier could not open or confirm it. Check before relying on it.
- **UNVERIFIED**: known only from a search-engine summary, or the verifier marked it uncertain and nobody read it at source. Treat it as a lead.
- **Corrected**: the verifier refuted the researcher's claim. Only the corrected fact is given, with the verifier's source.
- **Analysis**: my own reading or recommendation, not a sourced fact.

Doses are given **only** where a fact is tagged [V] or [S]. SKUAST gives doses per **100 L of water**; any per-drum (200 L) figure would be our own doubling, not a SKUAST figure [V] (IMG 2026).

Shorthand for sources used many times:

- IMG 2026: https://www.scribd.com/document/1002728111/Apple-Spray-Schedule-2026-SKUAST-K
- IMG 2025 (No. AU/DE/386BL/01/2025, 27-01-2025): https://www.scribd.com/document/841000027/Spray-Schedule-2025
- Guide 2023 (PDF mirror): https://theorchardmaster.in/wp-content/uploads/2023/08/Advisory-for-Management-of-apple-2023.pdf
- DoE Apr 2026 (SKUAST-K Directorate of Extension monthly message): https://skuastkashmir.ac.in/frmPDF.aspx?FN=April+2026+06-04-2026.pdf
- DoE Aug 2026: https://skuastkashmir.ac.in/frmPDF.aspx?FN=August+2026+03082026.pdf
- REAC 2026 (tentative fertilizer schedule, Annexure-VI): https://skuastkashmir.ac.in/frmPDF.aspx?FN=Conduct+of+58th+59th+Research+and+Extension+Advisory+committee+REACMeeting.pdf
- SKUAST-J HDP PoP: https://skuastjammu.ac.in/newsite/pdf/POP-High-Density-Apple.pdf

1. Spray schedule

1.1 Which edition and who issues it

FACT	TAG	SOURCE
The newest schedule is the IMG 2026, compiled by the SKUAST-K Directorate of Research, No. AU/DEx/449-BL/01/26. It has 15 numbered stages, from delayed dormancy to dormancy. No 2027 edition found (search on 2026-11-13).	[V]	IMG 2026
Issued through the Directorate of Horticulture, Kashmir; printed and published by the SKUAST-K Directorate of Extension. Implementing agencies: SKUAST-K Directorate of Research, Directorate of Extension, KVKs, and the Department of Horticulture, Kashmir.	[V]	IMG 2026
Drafted with inputs from the Department of Horticulture Kashmir and the J&K Directorate of Enforcement. Use and sale of listed products is subject to CIB&RC registration and label claim.	[V]	IMG 2026
27 Mar 2026: Director Horticulture Kashmir said the department and SKUAST distributed schedules and booklets, available at district horticulture offices.	[V]	KNS
The 2025 guide was formally unveiled on 19 Mar 2025 and made downloadable from the department website. No press report of a formal 2026 release event was found.	[V]	Ziraat Times
The guide replaced the older fungicide/insecticide "spray schedule" with an integrated approach (SKUAST-K entomologist).	[V]	Greater Kashmir, 23 Apr 2026
Edition changes: IMG 2025 gave day intervals between sprays (for example fruitlet 10-14 days after petal fall); IMG 2026 drops them and ties fungicide sprays to rain forecasts. The 2023 guide had no tight-cluster or flowering spray and used Dimethoate 30 EC; Dimethoate is gone from IMG 2026 (it still appears in the Aug 2026 message for pin-hole/shot-hole borer).	[V]	IMG 2025, Guide 2023

1.2 Stage table (IMG 2026, plus 2026 monthly messages)

All rows below are [V]. Rows marked UNRESOLVED or TIMING CONFLICT disagree between SKUAST's own 2026 documents: hold the dose or timing back in the app until SKUAST-K confirms it. Stage 13 (harvest) has no spray.

STAGE	TARGET	PRODUCT (CHOOSE ONE WHERE SEVERAL)	DOSE PER 100 L WATER	NOTES	TAG	SOURCE
1 Delayed dormancy	San Jose scale, mite eggs, overwintering pests	Horticultural mineral oil (Orchol, Duratek AgriSpray Oil, HP Spray Oil E/HP HMO, MAK, SERVO)	2 L	Apply before green tip. No fungicide at this stage.	[V]	IMG 2026
2 Green tip	Scab	Mancozeb 75 WP; Captan 50 WP; Propineb 70 WP; Zineb 75 WP; Metiram 70 WG	300 g each	Choose one. Skip if the next 7 days are dry; spray before rain of a full day or more. No insecticide.	[V]	IMG 2026
3 Tight cluster	Scab	Mancozeb 75 WP; Captan 50 WP; Propineb 70 WP; Zineb 75 WP; Metiram 70 WG; Mancozeb 75 WG	300 g each	Choose one. Same rain rule as green tip. No insecticide.	[V]	IMG 2026

STAGE	TARGET	PRODUCT (CHOOSE ONE WHERE SEVERAL)	DOSE PER 100 L WATER	NOTES	TAG	SOURCE
4 Pink bud	San Jose scale and other overwintering pests (only if HMO spray was missed)	Chlorpyriphos 20 EC	100 ml	Need-based.	[V]	IMG 2026
4 Pink bud	San Jose scale, woolly apple aphid, two-spotted mite (only if HMO spray was missed)	Quinalphos 25 EC	100 ml	Alternative to Chlorpyriphos in the April 2026 message; not listed in the 2026 guide at this stage.	[V]	DoE Apr 2026
4 Pink bud	Blossom thrips (2 or more per flower), leaf roller	Thiacloprid 21.7 SC	40 ml	Need-based.	[V]	IMG 2026
4 Pink bud	Scab (primary infection stage)	Zineb 68% + Hexaconazole 4% 72 WP	125 g in the guide; 100 g (0.1%) in the April 2026 message	UNRESOLVED: guide prints 'Hexaconazole 5%' (annexure says 4% 72 WP) and 125 g. Confirm with SKUAST-K before showing a dose.	[V]	IMG 2026
4 Pink bud	Scab (primary infection stage)	Metiram 55% + Pyraclostrobin 5% 60 WG (100 g); Dodine 65 WP (60 g); Dodine 40 SC (90 ml); Tebuconazole 6.7% + Captan 26.9% 33.6 SC (250 ml); Tebuconazole 38.9% SC (40 ml); Chlorothalonil 40% + Difenoconazole 4% 44 SC (180 ml)	as listed	Choose one. In dry weather spray when most flowers are at pink, to avoid early bloom.	[V]	IMG 2026
4 Pink bud	Scab (primary infection stage)	Mancozeb 63% + Carbendazim 12% (WP in table, WG in annexure)	250 (unit missing at this stage; 250 g at later stages)	UNRESOLVED unit and formulation. Confirm before showing.	[V]	IMG 2026
4 Pink bud	Boron nutrition (need-based)	Boron (foliar)	1-1.5 g per litre	Also 3-4 bee hives per hectare at 5-10% bloom.	[V]	IMG 2026
5 Flowering	Scab (only if 10 days since last fungicide AND rain forecast)	Mancozeb 75 WP; Captan 50 WP; Propineb 70 WP; Zineb 75 WP; Metiram 70 WG; Mancozeb 75 WG	300 g each	Spraying not advised in flowering. No insecticide. Pheromone traps 8-10/ha for leaf blotch miner.	[V]	IMG 2026
6 Petal fall (60-70%)	Insects (need-based)	Neem oil (1500 ppm) (300 ml); Chlorpyriphos 20 EC (100 ml); Quinalphos 25 EC (100 ml)	as listed	Choose one.	[V]	IMG 2026
6 Petal fall (60-70%)	Mites (4-5 per leaf)	Etoxazole 10 SC (110 ml); Hexythiazox 5.45 EC (40 ml); Spiromesifen 22.9 SC (40 ml); Cyenopyrafen 30 SC (30 ml)	as listed	Choose one.	[V]	IMG 2026
6 Petal fall (60-70%)	Scab and other diseases (spray before rain)	Difenoconazole 25 EC (30 ml); Flusilazole 40 EC (20 ml); Penconazole 10 EC (50 ml); Fluxapyroxad 75 g/l + Difenoconazole 50 g/l SC (30 ml); Trifloxystrobin 25% + Tebuconazole 50% 75 WG (40 g); Fluopyram 17.7% + Tebuconazole 17.7% 400 SC (50 ml); Trifloxystrobin 10% + Difenoconazole 12.25% + Sulphur 3% SC (90 ml)	as listed	Choose one. Boron 1-1.5 g/L if not given at pink bud.	[V]	IMG 2026
7 Fruitlet	Insects	Quinalphos 25 EC (100 ml); Chlorpyriphos 20 EC (100 ml); Thiacloprid 21.7 SC (40 ml)	as listed	Choose one. Fruit borer areas: pheromone traps 8-10/ha (monitoring), 40/ha (mass trapping).	[V]	IMG 2026
7 Fruitlet	Mites (more than 5 per leaf)	Hexythiazox 5.45 EC; Spiromesifen 22.9 SC	40 ml each	Choose one.	[V]	IMG 2026
7 Fruitlet	Scab (spray before rain)	Zineb 68% + Hexaconazole 4% 72 WP	125 g in the guide; 100 g (0.1%) in 2026 monthly messages	UNRESOLVED, as at pink bud.	[V]	IMG 2026
7 Fruitlet	Scab (spray before rain)	Dodine 65 WP (60 g); Dodine 40 SC (90 ml); Fluxapyroxad 250 g/l + Pyraclostrobin 250 g/l 500 SC (20 ml); Tebuconazole 6.7% + Captan 26.9% 33.6 SC (250 ml); Pyraclostrobin 12.8% + Boscalid 25.2% WG (30 g); Tebuconazole 38.9% SC (40 ml); Mancozeb 63% + Carbendazim 12% WP (250 g)	as listed	Choose one. Heavy set: thin by hand or NAA 5-10 ppm at 10-12 mm fruit.	[V]	IMG 2026
8 Fruit development I	Insects (need-based)	Neem oil (1500 ppm), then Lecanicillium lecanii 1.15% WP 4 days later (300 ml, then 500 g); Chlorpyriphos 20 EC (100 ml); Quinalphos 25 EC (100 ml)	as listed	Do not spray the bio-agent where sericulture is practised.	[V]	IMG 2026
8 Fruit development I	Mites (more than 10 per leaf)	Hexythiazox 5.45 EC; Spiromesifen 22.9 SC	40 ml each	Choose one.	[V]	IMG 2026
8 Fruit development I	Scab and foliar diseases (only if rain forecast)	Mancozeb 75 WP (300 g); Captan 50 WP (300 g); Propineb 70 WP (300 g); Zineb 75 WP (300 g); Ziram 80 WP (200 g); Metiram 70 WG (300 g)	as listed	Choose one. Calcium chloride 3-4 g/L (1st of 3). Second soil fertilizer dose.	[V]	IMG 2026
9 Fruit development II	Insects	Chlorpyriphos 20 EC (100 ml); Quinalphos 25 EC (100 ml); Thiacloprid 21.7 SC (40 ml); Summer spray oil (750 ml)	as listed	Choose one.	[V]	IMG 2026

STAGE	TARGET	PRODUCT (CHOOSE ONE WHERE SEVERAL)	DOSE PER 100 L WATER	NOTES	TAG	SOURCE
9 Fruit development II	Mites	Hexythiazox 5.45 EC (40 ml); Fenazaquin 10 EC (40 ml); Spiromesifen 22.9 SC (40 ml); Propargite 42% + Hexythiazox 2% EC (100 ml); Etoxazole 10% SC (110 ml); Summer spray oil (750 ml)	as listed	Choose one.	[V]	IMG 2026
9 Fruit development II	Scab and foliar diseases (only if rain forecast)	Difenoconazole 25 EC (30 ml); Flusilazole 40 EC (20 ml); Penconazole 10 EC (50 ml); Trifloxystrobin + Tebuconazole 75 WG (40 g); Fluopyram 17.7% + Tebuconazole 17.7% 400 SC (50 ml); Fluxapyroxad 250 g/l + Pyraclostrobin 250 g/l 500 SC (20 ml); Fluxapyroxad 75 g/l + Difenoconazole 50 g/l SC (30 ml); Chlorothalonil 40% + Difenoconazole 4% 44 SC (180 ml); Mancozeb 63% + Carbendazim 12% WP (250 g); Trifloxystrobin 10% + Difenoconazole 12.25% + Sulphur 3% SC (90 ml)	as listed	Choose one. Calcium chloride 3-4 g/L (2nd).	[V]	IMG 2026
10 Fruit development III	Insects (need-based)	Chlorpyriphos 20 EC (100 ml); Quinalphos 25 EC (100 ml); Thiacloprid 21.7 SC (40 ml); Neem oil (1500 ppm), then Lecanicillium lecanii 1.15% WP or Metarhizium anisopliae 1.0% WP 4 days later (300 ml, then 500 g)	as listed	Bio-agents not in sericulture areas. Fruit fly: methyl eugenol traps 25-40/ha (August message: 25/ha from flowering).	[V]	IMG 2026
10 Fruit development III	Mites (more than 10 per leaf)	Fenazaquin 10 EC (40 ml); Cyenopyrafen 30 SC (30 ml); Etoxazole 10% SC (110 ml)	as listed	Choose one.	[V]	IMG 2026
10 Fruit development III	Foliar diseases (only if rain forecast)	Mancozeb 75 WP (300 g); Zineb 75 WP (300 g); Ziram 80 WP (200 g); Ziram 27 SC (600 ml); Chlorothalonil 75 WP (150 g); Mancozeb 75 WG (300 g)	as listed	Choose one. Calcium chloride 3-4 g/L (3rd); potassium sulphate 4-5 g/L if colour needed; last 1/3 of urea.	[V]	IMG 2026
11 Fruit development IV	Alternaria leaf blotch or scab (if present)	Zineb 68% + Hexaconazole 4% 72 WP	125 g in the guide; 100 g (0.1%) in the August 2026 message	UNRESOLVED dose. No insecticide at this stage.	[V]	IMG 2026
11 Fruit development IV	Alternaria leaf blotch or scab (if present)	Hexaconazole 5 EC (50 ml); Myclobutanil 10 WP (70 g); Metiram 55% + Pyraclostrobin 5% 60 WG (100 g); Pyraclostrobin 12.8% + Boscalid 25.2% WG (30 g)	as listed	Choose one. No insecticide at this stage.	[V]	IMG 2026
11 Fruit development IV	Sooty blotch, flyspeck (August 2026 message adds Marssonina)	Mancozeb 75 WP (300 g); Ziram 27 SC (600 ml); Propineb 70 WP (300 g); Ziram 80 WP (200 g); Mancozeb 75 WG (300 g)	as listed	Choose one.	[V]	IMG 2026
12 Pre-harvest	Storage rots (fruit for long-term storage)	Captan 50 WP (300 g); Ziram 80 WP (200 g); Zineb 75 WP (300 g)	as listed	TIMING CONFLICT: 15-20 days before harvest in the guide, 25 days in the August 2026 message. No insecticide.	[V]	IMG 2026
12 Pre-harvest	Storage rots (fruit for long-term storage)	Propineb 70 WP	300 g	TIMING CONFLICT: the guide's own waiting period for Propineb is 29 days, longer than the 15-20 day timing. Do not show without this caveat.	[V]	IMG 2026
12 Pre-harvest	Need-based nutrition and fruit drop	NAA (10 ppm); Potassium sulphate (4-5 g per litre); Calcium chloride (3-4 g per litre)	as listed	NAA against pre-harvest drop; K2SO4 (2nd) for colour; CaCl2 for storage life.	[V]	IMG 2026
14 Post-harvest	Scab / Alternaria inoculum (infected orchards)	Urea	3-4% (3-4 kg)	Spray when 50% of leaves have turned yellow naturally.	[V]	IMG 2026
14 Post-harvest	Insects (need-based)	Chlorpyriphos 20 EC	100 ml		[V]	IMG 2026
14 Post-harvest	Mites (more than 20 per leaf)	Fenazaquin 10 EC (40 ml); Etoxazole 10% SC (110 ml)	as listed	Choose one.	[V]	IMG 2026
14 Post-harvest	Apple blotch leaf miner (where a problem)	Quinalphos 25 EC	100 ml	Listed as an 'essential spray' in those areas.	[V]	IMG 2026
15 Dormancy	Fallen-leaf decomposition (scab inoculum)	Urea on orchard floor	5% (5 kg)	Only if the autumn urea spray was missed. No pesticide.	[V]	IMG 2026
Monthly message (August 2026)	Apple blotch leaf miner	Thiamethoxam 25 WG (50 g); Flubendiamide 39.35 SC (40 ml); Chlorantraniliprole 18.5 SC (100 ml); Thiacloprid 21.7 SC (60 ml); Lambda-cyhalothrin 5% EC (50 ml); Thiamethoxam 12.6% + Lambda-cyhalothrin 9.5% ZC (50 ml); Imidacloprid 6% + Lambda-cyhalothrin 4% SL (50 ml)	as listed	Not in the 2026 guide's stage table; stage fit not stated. 15-day interval stated with Lambda-cyhalothrin.	[V]	DoE Aug 2026

STAGE	TARGET	PRODUCT (CHOOSE ONE WHERE SEVERAL)	DOSE PER 100 L WATER	NOTES	TAG	SOURCE
Monthly message (August 2026)	Fruit borer (incl. codling moth) when adults trapped	Chlorpyriphos 50% + Cypermethrin 5% EC (125 ml); Chlorantraniliprole 18.5% SC (100 ml)	as listed	Monitoring traps 20/ha in August (8-10/ha in April message and guide).	[V]	DoE Aug 2026
Monthly message (August 2026)	San Jose scale / woolly apple aphid (more than 13 SJS crawlers per cm2 of twig, or WAA colonies on terminal shoots)	Chlorpyriphos 20 EC (100 ml); Quinalphos 25 EC (100 ml); Thiacloprid 21.7 SC (40 ml)	as listed	Need-based.	[V]	DoE Aug 2026
Monthly message (August 2026)	European red mite / two-spotted mite (more than 15 per leaf)	Fenazaquin 10 EC (40 ml); Etoxazole 10% SC (110 ml); Cyenopyrafen 30 SC (30 ml)	as listed	Need-based. Stop insecticides/acaricides 14 days before harvest (guide says stop all sprays 3-4 weeks before).	[V]	DoE Aug 2026
Monthly message (April 2026)	Sticker in rainy season (any fungicide stage)	Sandovit (sticker)	50-75 ml	Never with Dodine.	[V]	DoE Apr 2026

1.3 Spray rules (IMG 2026 instructions)

RULE	TAG	SOURCE
Do not mix fungicides, insecticides and spray suspensions; keep 3-4 days between an insecticide and a fungicide spray.	[V]	IMG 2026
If heavy rain falls within 12 hours of a spray made without adjuvant or sticker, repeat it immediately.	[V]	IMG 2026
Do not use the same pesticide in two consecutive sprays.	[V]	IMG 2026
No spray in full bloom. Stop insecticide and fungicide sprays 3-4 weeks before harvest (but see the conflicts below).	[V]	IMG 2026
Always spray before rain. Avoid spraying right after severe rain after 15 June, especially at low altitude.	[V]	IMG 2026
Spray early morning or evening; avoid heat and rain (phytotoxicity). Tell nearby beekeepers the spray time and keep colonies away.	[V]	IMG 2026
Use only SKUAST-K recommended brands and doses; avoid tank-mixing (SKUAST-K DoE, 2 Jul 2026).	[V]	Rising Kashmir
After hail (May 2026 advisory, newspaper report): protective fungicide once weather stabilises, then 0.2% urea 3 days later; if a systemic was applied just before the hail, wait 4-5 days before reapplying.	[V]	Greater Kashmir, 3 May 2026
2023 guide only: no HMO beyond green tip; Captan 50 WP at least 10 days after the oil spray. Not restated in IMG 2026.	[V]	Guide 2023

1.4 Waiting periods (pre-harvest interval, days, IMG 2026 table)

Growers must not harvest, cut grass or let poultry or livestock into the orchard until the waiting period is over [V].

PRODUCT	DAYS	PRODUCT	DAYS
Hexaconazole 5 EC	10	Flusilazole 40 EC	25
Fluxapyroxad 250 g/l SC	14	Fluopyram 17.7% SC	7
Difenoconazole 25 EC	15	Tebuconazole 17.7% SC	6
Pyraclostrobin 250 g/l 500 SC	25	Trifloxystrobin 25 WG	4
Ziram 80 WP	17	Boscalid 25.2 WG	13
Metiram 70 WG	11	Myclobutanil 10 WP	15
Dodine 65 WP	22	Imidacloprid 17.8 SL	11
Penconazole 10 EC	13	Spiromesifen 22.9 SC	9
Captan 50 WP	1 (unusually short; reproduced as printed)	Hexythiazox 5.45 EC	13
Propineb 70 WP	29	Propargite 57 EC	9
Chlorothalonil 75 WP	36	Mancozeb 75	14

Tag [V]; source IMG 2026. Imidacloprid 17.8 SL and Propargite 57 EC appear in this table but in no stage row.

1.5 Conflicts in SKUAST's own 2026 documents (do not show as settled)

ISSUE	IMG 2026 SAYS	OTHER 2026 SKUAST SOURCE SAYS	TAG	SOURCE
Zineb + Hexaconazole at pink bud and fruitlet	"Hexaconazole 5%", 125 g	Annexure (Avtar) and FD-IV row: 4% 72 WP; Apr and Aug messages: 0.1% = 100 g; 2023 guide: 100 g. Treat "5%" as a typo; dose unresolved.	[V]	DoE Apr 2026
Mancozeb 63% + Carbendazim 12% at pink bud	"250" with no unit, called WP	250 g at later stages; annexure calls it WG (SAAF)	[V]	IMG 2026
Pre-harvest storage fungicide timing	15-20 days before harvest	Aug 2026 message: 25 days before harvest; IMG's own PHI for Propineb is 29 days	[V]	DoE Aug 2026

ISSUE	IMG 2026 SAYS	OTHER 2026 SKUAST SOURCE SAYS	TAG	SOURCE
When to stop spraying	All insecticide and fungicide sprays stop 3-4 weeks before harvest	Aug 2026: stop insecticides/acaricides 14 days before harvest	[V]	DoE Aug 2026
Fruit borer pheromone traps	8-10/ha (monitoring), 40/ha (mass trapping)	Aug 2026: 20/ha	[V]	DoE Aug 2026
Fruit fly methyl eugenol traps	25-40/ha	Aug 2026: 25/ha from onset of flowering	[V]	DoE Aug 2026
Pink bud if HMO missed	Chlorpyriphos only	Apr 2026 also allows Quinalphos 25 EC (both 100 ml)	[V]	DoE Apr 2026
Sticker	No sticker recommendation in the instruction list	Apr 2026: Sandovit 50-75 ml/100 L in rainy season, never with Dodine (2025 guide said the same)	[V]	DoE Apr 2026
Mite action thresholds	4-5/leaf (petal fall), >5 (fruitlet), >10 (FD-I, FD-III), >20 (post-harvest)	Aug 2026: >15/leaf	[V]	DoE Aug 2026

Analysis: the app should hold each conflicting value back, show "confirm with your Horticulture Development Officer", and ask SKUAST-K for a written ruling as part of the partnership.

1.6 High-density plantations (HDP)

FACT	TAG	SOURCE
There is **no separate SKUAST-K spray schedule for HDP** in IMG 2026. The chemicals and doses are the same as for traditional orchards. The report's line that the schedule "has high-density variants" is not supported.	[V]	IMG 2026
HDP appears only as cultural notes: at petal fall scout HDP closely and check the lower side of leaves; remove scab-infected leaves before rain ("practical in HDP"); post-harvest, tighten trellis wires (and close anti-hail nets, stake branches against snow).	[V]	IMG 2026
The IMG 2026 fertilizer annexure is labelled "Conventional Orchards". The only HDP fertilizer table is the tentative REAC 2026 one and the SKUAST-Jammu PoP (section 2).	[V]	IMG 2026

2. Nutrition, soil and high-density orchards

2.1 Fertilizer by tree age, traditional (conventional) orchards, g per tree per year

The **official IMG 2026 Annexure-1** is the source to use. A **tentative** SKUAST-K schedule presented to the REAC on 16 July 2026 has the same urea and DAP but lower MOP for years 3-14, and adds vermicompost. Whether REAC adopted it is not known. Do not show the REAC MOP values as the recommendation.

AGE (YR)	UREA	DAP	MOP (IMG 2026)	MOP (REAC TENTATIVE)	FYM KG (IMG 2026)
1	50	25	80	80	10-15
2	100	50	160	160	10-15
3	150	75	250	240	10-15
4	200	100	330	320	10-15
5	250	125	420	400	10-15
6	350	175	580	550	15-25
7	450	225	750	720	15-25
8	550	275	920	870	15-25
9	650	325	1080	1000	15-25
10	750	375	1250	1150	15-25
11	900	450	1500	1450	25-40
12	1050	525	1750	1700	25-40
13	1200	600	2000	1900	25-40
14	1350	675	2250	2100	25-40
15 (IMG: "15 & above")	1500	750	2500	2500	25-40 (40-60 kg from year 16)

Tag [V]. Sources: IMG 2026 Annexure-1 (verifier read it there), REAC 2026. FYM goes on in February-March. IMG 2026 says inorganic doses should follow soil and leaf analysis where available [V]. REAC adds vermicompost 4-15 kg/tree by age [V]. The J&K Horticulture FAQ quotes the same year-1 and 15+ endpoints, but the page could not be opened (UNVERIFIED, FAQ).

2.2 Fertilizer timing and placement

FACT	TAG	SOURCE
Split: at green tip, FYM + 1/3 urea + all DAP + 1/2 MOP; at fruit development I, 1/3 urea + remaining 1/2 MOP; at fruit development III, the last 1/3 urea. Same in the 2023 guide and IMG 2026.	[V]	IMG 2026
Corrected (the 2023 rule "keep 60 cm around the trunk undisturbed; apply 1-2 ft from trunk" is superseded): IMG 2026 says keep 1-2 feet of soil around the trunk undisturbed, apply fertilizers along the canopy line, and raise phosphorus by 20% in alkaline or acidic soils.	[V]	IMG 2026
No extra soil nitrogen in August: late N delays maturity, reduces colour and increases storage disorders.	[V]	DoE Aug 2026

FACT	TAG	SOURCE
SKUAST-Jammu HDP PoP words timing differently: basal dose about 3 weeks before bloom, second about 3 weeks after fruit set, third urea in June-July.	[V]	SKUAST-J HDP PoP

2.3 Foliar nutrition

NUTRIENT	IMG 2026 (USE THIS)	OTHER SKUAST SOURCES (CONFLICTING)	TAG	SOURCE
Calcium chloride	3-4 g/L at FD I, II and III, plus need-based pre-harvest	2023 guide: 4 g/L x3; REAC tentative: 3 g/L; Aug 2026 Soil Science: 0.4-0.5% (4-5 g/L) or Ca nitrate 0.5-0.75% every 10-15 days	[V]	IMG 2026, DoE Aug 2026
Boron	1-1.5 g/L need-based at pink bud, or petal fall if missed	REAC: boric acid 1.5 g/L at pink bud and petal fall; Aug 2026: boric acid 0.1%	[V]	same
Zinc sulphate	No routine spray in IMG 2026	2023 guide: 3-4 g/L at petal fall; REAC: 1 g/L at green tip/pink bud and after harvest; Aug 2026: zinc only after leaf analysis confirms need	[V]	same
Potassium sulphate (colour)	4-5 g/L need-based at FD III and pre-harvest	Aug 2026 Fruit Science: 5 g/L for late varieties; same message's Soil Science: 1% (10 g/L) for heavy crop or deficiency	[V]	same
Urea (post-harvest)	3-4% when 50% of leaves yellow naturally, in scab/Alternaria-infected orchards; 5% on orchard floor at dormancy if missed	IMG 2025: 2-3% two weeks after harvest plus 4% at 50% yellowing; SKUAST-J: 2-3%	[V]	IMG 2026
Urea (quick N recovery)	not in IMG 2026	J&K Horticulture FAQ: 0.5% (page not opened)	UNVERIFIED	FAQ
Calcium (HDP, SKUAST-Jammu)	n/a	0.3% Ca nitrate or CaCl2 fortnightly from mid-June to picking; CaCl2 can scorch leaves in heat. (The same PoP's "boron 1-1.5 per cent" is probably a typo; do not use.)	[V]	SKUAST-J HDP PoP

Background on post-harvest urea timing: a 2021 article says the best time is just before natural leaf fall in early November, and early or mid-October sprays delay hardening [V] (Kashmir Reader 2021). Prefer the IMG 2026 trigger (50% of leaves yellow).

2.4 Soil and leaf testing

FACT	TAG	SOURCE
Soil test after harvest; leaf tissue test at FD III. Leaf sampling window 15 July-15 August, 25 trees and 100-200 leaves per sample.	[V]	DoE Aug 2026, IMG 2026
Directorate of Horticulture Kashmir Soil and Leaf Analysis Laboratory, Rajbagh, Srinagar (est. March 1995).	UNVERIFIED	hortikashmir.gov.in
Soil testing lab at Shopian and mobile labs at Anantnag and Baramulla (undated page).	UNVERIFIED	horticulture.jk.gov.in

2.5 High-density orchards

FACT	TAG	SOURCE
HDP fertilizer, g/tree/yr (urea/DAP/MOP; vermicompost and FYM kg): yr 0 only VC 2, FYM 3; yr 1 108/54/100, VC 2.5, FYM 4; yr 2 206/108/200, 3, 5; yr 3 303/162/300, 3.5, 6; yr 4 401/217/400, 4, 7; yr 5 488/271/500, 5, 8. TENTATIVE (REAC). SKUAST-J has the same table, with year 5 as "5 and above" (N 225, P2O5 125, K2O 300 g).	[V]	REAC 2026, SKUAST-J HDP PoP
Fertigation principles only (no dose table): start N at rapid shoot growth, split over 4-6 weeks, finish by mid-July; fertigation can halve N doses.	[V]	SKUAST-J HDP PoP
Drip water need in July (Shalimar campus research): 2.1-2.5 L/plant/day in year 1 to 8.0-8.5 in year 6+.	[V]	SKUAST-J HDP PoP
Densities: medium 1,111/ha (3x3 m on MM-106, MM-111, M-7); high 2,222 or 3,333/ha (1.5x3 or 1x3 m on dwarfing stocks), bearing in year 2-3, commercial yield year 5-6; traditional 150-200/ha.	[V]	SKUAST-J HDP PoP
M-9: tree 25-30% of full size, shallow roots, needs irrigation and support, prone to suckering, susceptible to woolly aphid and fire blight. MM-111: well anchored, resists collar rot and woolly aphid, suits heavy poorly drained soils.	[V]	SKUAST-J HDP PoP
Department supplies M9, MM-106 and MM-111 rootstock to nurseries at 50% subsidy (2022 nursery scheme; current status not checked).	[V]	Rising Kashmir 2022
"3.0 x 1.5 m on M-9" (Horticulture FAQ) could not be opened; SKUAST-J gives 1x3 and 1.5x3 m.	UNVERIFIED	FAQ
Kashmir Division HDP: 1,103.79 ha, about 4,428 farmers; ~90% Gala and Red Delicious strains (Assembly reply, Sep 2026).	[V]	Greater Kashmir, 27 Sep 2026
Pollinizers by variety (no percentage given anywhere): Gala Redlum: Braeburn, Granny Smith, Cox, Fuji, Golden Delicious; Red Chief Camspur: Golden Delicious, Gala, Fuji, Golden Hornet, Elstar; Red Delicious Silver Spur: Elstar, Gala, Golden Delicious, Granny Smith; Granny Smith: Gala, Red Delicious, Golden Delicious, Idared.	[V]	SKUAST-J HDP PoP
Picking and cold-store life: Gala Redlum early-mid Aug (7-8 months); Gala Mitch mid-late Aug; Red Chief, Red Velox mid-Sep; Red Delicious Silver Spur, Super Chief late Sep (6-7 months); Golden Delicious Reinders early Oct; Granny Smith early-mid Oct; Fuji Zehn Aztec mid-late Oct (7-8 months).	[V]	SKUAST-J HDP PoP
Bee hives: IMG 2026 says 3-4 per hectare at 5-10% bloom; the Apr 2026 message says 2 per acre (about 4.9/ha) at 10-15% bloom or when king flowers open, and bouquets where pollinizers are lacking. Same message's apiculture section says "@3/ha".	[V]	IMG 2026, DoE Apr 2026
HDP establishment cost (3,333 trees/ha, 2021-era department scheme costing): about Rs 41 lakh/ha (Rs 22 lakh government, Rs 19 lakh beneficiary), plants at Rs 500.	[V]	SKUAST-J HDP PoP
Units: 1 acre = 8 kanal; 1 kanal ≈ 505.86 m^2; 1 ha ≈ 19.77 kanal.	[V]	Kashmir Reader 2021

3. Harvest, grading and storage

FACT	TAG	SOURCE
Maturity indices: starch index 2.5 (CA storage) or 3.5 (table) on a 0-6 scale; easy separation from spur; >75% red (variety-specific); TSS 12-14 °Brix; firmness 16-18 lb/sq inch. IMG 2026 changes seed colour to ">75% dark brown" and adds days after full bloom (variety-specific).	[V]	IMG 2026
Early varieties: harvest at starch-iodine 3-4/6, not for long storage. Harvest at proper size and at least 70% of varietal colour, upward twist, 2-3 pickings a week or fortnight apart.	[V]	DoE Aug 2026
Red Delicious optimal harvest (single 2023-24 study): 161 days from full bloom Srinagar, 163 Budgam, 162 Pattan, Anantnag and Shopian, 152 Kupwara.	[V]	Agricultural journals 2026
Handling: padded baskets, shade, move to godown quickly; pickers keep nails short.	[V]	DoE Aug 2026
Grades (SKUAST-K food scientist, opinion piece): about 8 lakh MT A-grade (needs CA), 7 lakh MT B-grade (straight to market), 6 lakh MT C-grade (processing only, should not leave the Valley as fresh fruit). No official J&K grading norm found.	[V]	Kashmir Life, 10 Feb 2026
1-MCP within 1-2 days of harvest at starch 2.5-3.0; CA fruit then keeps up to 25 days after removal against 8-10 untreated. Sachets Rs 14 per 10 kg, Rs 20 per 17-kg box (same opinion piece).	[V]	Kashmir Life, 10 Feb 2026
CA rent (one grower, Lassipora): Rs 35 per 17-kg crate per month, about Rs 180-220 per crate for 5-6 months; can rise at peak. Transport to the store about Rs 100-110 per 20-kg box over ~32 km (another grower).	[V]	101Reporters, 21 Oct 2026
2025: about Rs 180 per 20-kg box for three months (columnist). 2022: Rs 2/kg/month plus Rs 2/kg grading and Rs 2/kg packing (CA association president).	[V]	Kashmir Observer 2025, Kashmir Life 2022
Capacity: 76 CA stores, 2.92 lakh MT, against a 6 lakh MT need; mostly Lassipora and Aglar; Rs 600 crore budgeted for 40 new stores (2026-27); no scheme pays growers' storage rent. Production 21.09 lakh MT in 2024-25 (Economic Survey, as cited).	[V]	101Reporters
2026 season: cold stores booked months ahead, many full by late September; operators require grading first.	[V]	Kashmir Observer, 23 Sep 2026

No CA set-points (temperature, O2, CO2) from a Kashmir source were found.

4. Pests and diseases: corrections and status

4.1 Fire blight

FACT	TAG	SOURCE
EPPO: India "Absent, unreliable record" (evaluation based on 1992 information; one doubtful 1978 record on rose). No J&K record.	[V]	EPPO
No peer-reviewed or official report of fire blight in J&K was found. A 2024 SKUAST-K paper mentions it only as a pathogen spreading in China.	[V]	Sci Rep 2024
Symptoms: shoot tips wilt and blacken and bend into a hook; ooze in warm humid weather; dark sunken cankers where the bacterium overwinters.	[V]	netreefruit

Analysis: keep fire blight as "not known in J&K, report suspected cases to the Horticulture Department". Never give it as a diagnosis.

4.2 Codling moth

FACT	TAG	SOURCE
Corrected: codling moth (*Cydia pomonella*) **is present in the Kashmir valley**. SKUAST-K confirmed it by morphology and COI barcoding at Nadihal, Chak Gojri and Delina (Baramulla), after heavy premature fruit drop in 2019.	[V]	Sci Rep 14:11972, 2024
Before this it was known in India only from Ladakh; the Kashmir population most likely came from Leh (the paper says "possible route").	[V]	Sci Rep 2024
3 full generations plus a half overwintering generation; in 2022 the first ran standard weeks 16-25 (mid-April to late June), the third ended week 42 (mid-October). Baramulla karewa data.	[V]	Sci Rep 2024
April 2023 news: officials named Baramulla, Bandipora, Pulwama, Anantnag, Shopian and Budgam; present unidentified for about three years. (The "up to 80%" line is a generic statement, not a Kashmir measurement.)	[V]	FreshPlaza / Rising Kashmir
A 2023 pheromone-trap paper reports it in the Baramulla "incursion belt" and calls for containment elsewhere; whether it found it "only" there could not be checked.	UNVERIFIED	ResearchGate
SKUAST-K 2026 messages carry "fruit borer" (not named codling moth) management: survey, awareness, pheromone traps (8-10/ha April; 20/ha August), sprays if adults are trapped (rows in section 1.2).	[V]	DoE Apr 2026
Ladakh: declared a quarantine pest decades ago, with an embargo on moving apples out of Ladakh (2018 article).	[V]	Greater Kashmir 2018
India's domestic quarantine (DIP Act) provided for restricting inter-state movement of nine pests incl. codling moth, San José scale and apple scab (2010 policy paper; current DPPQS schedule not checked).	[V]	NAAS Policy Paper 44

4.3 Apple scab in 2026

FACT	TAG	SOURCE
Two weeks of rain in late July-early August 2026 triggered secondary scab in Shopian; J&K had 35% surplus rain in July. Necrotic leaf blotch, Gala cracking and waterlogging reported too.	[V]	Greater Kashmir, 9 Aug 2026
Secondary scab reported in Nagbal (Shopian), Dachnipora (Anantnag), Aharbal (Kulgam), Shadimarg (Pulwama).	[V]	Kashmir Thunder, 12 Aug 2026
By Diwali, dealers' association: about 50-60% of apples reaching mandis were Grade B/C (trade estimate, no official survey).	[V]	Kashmir Observer, 1 Nov 2026
Sopore, Nov 2026: B/C-grade fruit hit by scab, Alternaria, leaf miner and aphids fetched about Rs 200-250 a box.	[V]	Rising Kashmir, 3 Nov 2026

FACT	TAG	SOURCE
SKUAST-K scientist (interview): Delicious-type apples lose 30-40% a year to scab; about Rs 325 crore of fungicide used a year.	[V]	Kashmir Life, 8 Oct 2026
No source links a measured share of 2026 scab losses to fake fungicide; reported causes are prolonged rain and humidity.	[V]	Greater Kashmir, 24 Sep 2026
Biology: ascospores mature over 4-6 weeks from 1/4-inch green to 2-3 weeks after petal fall (primary window); rain-splashed conidia cause secondary cycles all season. Lesions show 9-17 days after infection; infection rare above 78°F.	[V]	Penn State
Kashmir isolates of *V. inaequalis* show a major shift to reduced sensitivity to difenoconazole and flusilazole, stable and with no fitness cost (Sci Rep). A 2018 SKUAST-K thesis found the biggest shift in myclobutanil, then flusilazole.	[V]	Springer Nature, Krishikosh thesis
5% urea on fallen leaves speeds breakdown; spore-reduction strategies cut spore load 50-80% (MSU; page blocked). Penn State encourages fall urea for leaf decomposition.	UNVERIFIED (figures); [V] (direction)	MSU, Penn State Marssonina

4.4 Mills (scab infection) table

Hours of continuous leaf wetness needed for infection at a given average temperature (average = (lowest + highest during the wet period) / 2; wetting ends when RH falls below 90%; add wet periods re-wetted within 8 hours). Celsius values are conversions.

AVG TEMP °F (°C)	PENN STATE REVISED MILLS, MIN H	NEWA (MACHARDY-GADOURY) MIN H	JONES "LOW" (LIGHT INFECTION) H	MODERATE H	HEAVY H
39 (3.9)	28				
41 (5.0)	21				
43 (6.1)	18	18	21	30	47
45 (7.2)	15				
46 (7.8)	13	14	16	24	37
48 (8.9)	12				
50 (10.0)	11	11	14	19	29
52 (11.1)	9				
54 (12.2)	8 (54-56)	8.5	11	16	24
57-59 (13.9-15.0)	7	7	10	14	22
60 (15.6)		6.5	9	13	20
61-75 (16.1-23.9)	6	6	9	12	18
77 (25.0)	8				
79 (26.1)	11				

Tag [V]. Sources: Penn State, netreefruit. Ascospores are not released in the dark, so night rain is counted from 8:00 a.m. [V] (netreefruit). The APS example (18°C, 9 h = light infection) is UNVERIFIED (APS), but matches the Jones "Low" column. Analysis: the app must name the table version driving its alerts; the Penn State minimum is the most sensitive, Jones "Low" fits "light infection" wording.

4.5 Other diseases

FACT	TAG	SOURCE
Alternaria leaf blotch in Kashmir HDP (2021-22): first spots 2nd week of May, blotches by 4th week of May, yellowing and leaf drop by late June.	[V]	IJAgS 95(8) 2025
Kashmir leaf blotch/leaf spot is associated with *A. alternata*, *A. tenuissima* and *A. arborescens*; scab is the top foliar disease, Alternaria second.	[V]	IJAgS 2025
"Frog-eye leaf spot" is the leaf phase of black rot (*Botryosphaeria obtusa*): purple-bordered spots with tan centres. Not a name for Alternaria.	[V]	Univ. of Kentucky
July 2025: Alternaria and necrotic leaf blotch (a physiological disorder) outbreaks; SKUAST-K pathologist estimated ~30% of farms affected, worse in dense unpruned canopies.	[V]	Greater Kashmir, 26 Jul 2025
SKUAST-K Aug 2026 gives separate product lists for "Alternaria leaf blotch and scab" and "Marssonina / sooty blotch / flyspeck".	[V]	DoE Aug 2026
Marssonina: spots from about mid-June on upper surface of mature leaves, 5-10 mm, grey-brown to black with purple edge, then yellowing and leaf drop. Ascospores released from late bloom to petal fall for 3-4 weeks, so protection starts then (Pennsylvania timing).	[V]	Penn State
Marssonina is established in Kashmir (listed among important diseases). The 2004-05 SKUAST-K survey incidence figures (17-29%) could not be found.	[V] (established); UNVERIFIED (figures)	IJAgS 2025, Krishikosh
Powdery mildew overwinters in terminal buds; scab fungicides from pink onwards should have mildew activity.	[V]	netreefruit
Sooty blotch and flyspeck are surface fungi, active from 1-2 weeks after petal fall to harvest, visible in late summer; pruning helps.	[V]	netreefruit
Collar rot: clean the collar, apply Chaubatia or Bordeaux paste, drench under canopy with Metalaxyl MZ 72 WP 0.5%. Root rot: drench with Carbendazim 50 WP 0.1% or Carbendazim + Mancozeb 0.5% in 15-20 cm holes 30 cm apart.	[V]	DoE Apr 2026
White root rot: 22 *Malus* species screened, none immune (2013); "no rootstock immune" is an inference.	[V]	Indian J. Hortic.

4.6 Insects and mites (including emerging pests)

FACT	TAG	SOURCE
Apple blotch leaf miner (*Leucoptera malifoliella*): invasive, confirmed in J&K (NCBI OR244421). 2023 NBAIR-SKUAST survey of Shopian, Anantnag, Pulwama, Kulgam: 13-100% defoliation, worst in Shopian, estimated 20-50% loss (to be ascertained).	[V]	ICAR-NBAIR alert
Growers mistake early leaf-miner blotches for Alternaria and overspray fungicide; miner feeding also opens the way to secondary Alternaria. Signs: mine/blotch with a larva inside; H- or boat-shaped white silk cocoons on bark and leaf undersides.	[V]	ICAR-NBAIR alert
Management: sanitation, pest-free plants, collect and burn leaves, scrape bark to expose pupae, gunny-bag trunk bands, yellow sticky traps (Aug 2026: about 10 m apart), watch April-May moth emergence; sprays work poorly as larvae sit inside the mine. SKUAST Aug 2026 now lists sprays (section 1.2).	[V]	ICAR-NBAIR, DoE Aug 2026
Leaf miner spread from Zainapora (Shopian) to Litter, Wachi, Yaripora, Lassipora, Tahab, Bijbehara (2024). Awareness programme for ~150 Kupwara growers, June 2026; no infestation level given.	[V]	Kashmir Life 2024, Daily Headlines Today
Oriental fruit fly (*Bactrocera dorsalis*) infests apple, apricot, nectarine, pear, peach and quince in the Kashmir valley (2019 paper).	UNVERIFIED	EPPO J&K
Fruit fly management (SKUAST 2026): bury dropped fruit, rake soil to expose pupae, methyl eugenol traps 25/ha from flowering, yellow sticky traps 25/ha to September.	[V]	DoE Aug 2026
Brown marmorated stink bug: only a 2017 species-list record, apparently from Jammu-province districts; no damage reports. Do not present as an emerging Kashmir pest.	UNVERIFIED	Rec. Zool. Surv. India
Mites: ERM overwinters as eggs on twigs and bark (dormant oil targets these; hatch tight cluster-petal fall); two-spotted mite overwinters as females in ground cover (oil does not reach it). Damage: bronzing, small fruit, drop, poor set next year. (Webbing as a two-spotted sign is general knowledge, not from this source.)	[V]	netreefruit
Kashmir *P. ulmi* acaricide resistance (PAU study): fenazaquin resistance factor 6.54 at Mattan, dimethoate 11.47 at Kunil (Kupwara); lab selection ~15-fold by generation 8. (Corrects the researcher's "no data found".)	UNVERIFIED	CORE
San Jose scale: grey rough bark; reddish-purple ring around each scale on fruit. SKUAST threshold >13 crawlers per cm^2 of twig.	[V]	netreefruit, DoE Aug 2026
Woolly apple aphid: reddish-brown under white wax, in bark cracks, cuts and wounds; root galls; above-ground numbers do not show root load.	[V]	netreefruit
MM.106 and MM.111 resist woolly apple aphid, but MM.106 is highly susceptible to crown and collar rot (US source).	[V]	apples.extension.org
Apple aphid: spray only if 40% of succulent shoots or leaves are infested; no mixing; stop insecticides 14 days before harvest.	[V]	DoE Aug 2026
Hairy caterpillar / leaf roller: collect and destroy egg masses; Chlorpyriphos 20 EC 100 ml/100 L if foliage damage. The species (*Lymantria obfuscata*, one generation a year) is from search only.	[V] (advice); UNVERIFIED (species)	DoE Aug 2026
Corrected: the Kashmir apple trunk borer is *Aeolesthes sarta*. *A. holosericea* is a separate species, not a synonym. No primary source for its 2-year life cycle in Kashmir.	[V]	Mantoo et al. 2010

5. Input quality, banned products and online-sale rules

5.1 Spurious and substandard inputs in J&K

FACT	TAG	SOURCE
2025-26: 5,772 pesticide samples tested, 85 substandard; 93 show-cause notices, 52 prosecutions, 22 FIRs, Rs 70,000 fines (Assembly, 24 Sep 2026).	[V]	Kashmir Life
2026-27 to Sep: 1,658 tested, 66 substandard; 19 licences cancelled or suspended, 19 prosecutions, 3 FIRs, Rs 1.33 lakh fines.	[V]	Kashmir Life
32 highly toxic pesticide formulations banned or restricted in J&K "in line with central advisories". No list or order number published.	[V]	Kashmir Life
Two pesticide QC labs (Lal Mandi, Srinagar; Talab Tillo, Jammu) within four notified QC labs; a State Pesticide Residue Testing Lab near completion. **Corrected:** do not say "six pesticide testing labs" (one Kashmir Life report).	[V]	Kashmir Reader, 25 Sep 2026, Rising Kashmir, 21 Feb 2026
Two-year figures to Dec 2025: 12,549 inspections, 10,222 samples, 190 misbranded/non-standard, 149 prosecutions, Rs 3.50 lakh fines; SOPs issued Sept 2025.	[V]	Kashmir Life, 28 Mar 2026
Shopian 2026 "Super M-45" (mancozeb) case: growers reported fruit drop after spraying; 235 kg seized on 27 Jun 2026; later the government reported 238 kg of batch SP-1902 across 12 dealers, with mismatched QR code, registration and licence details, label, manufacturing date and certificate of analysis; FIR 0200, Shopian PS, 23 Sep 2026. Lab reports and the joint committee's final report were still pending in early October.	[V]	Kashmir Reader, 1 Oct 2026, Kashmir Life, 27 Jun 2026, Rising Kashmir, 19 Jun 2026
SKUAST-K saw fruit fall in all 61 orchards surveyed in 37 villages; Shopian preliminary loss 6,593 trees on 335.75 kanal, ~10.7 lakh kg, ~Rs 4.42 crore.	[V]	Kashmir Reader, 1 Oct 2026
Feb 2025: specific batches of Indofil M-45 (mancozeb 75 WP) and Rallis CAPTAF (captan 50 WP) declared misbranded (batch-specific, not a brand-wide finding). Growers say lab results can take almost a year.	[V]	Kashmir Observer, 8 Feb 2025
Entry-point control: fertilizer sampled at rail rakes (Bari Brahmana, Udhampur, Anantnag) and Lower Munda road entry; Sept 2026 announcement of pesticide checkpoints at entry points.	[V]	Rising Kashmir, Daily Excelsior
Pesticides may legally be sold only through licensed dealers; buy registered products from licensed dealers.	[V]	Kashmir Reader, 8 Mar 2026
Fertilizer SOPs (FCO 1985): retailers display licence, stock and price list; sales linked to soil health card and Aadhaar; PoS/IFMS stock monitoring; complaints verified in 7 days, action report in 15.	[V]	Kashmir Life
Opinion claim: spurious agrochemicals above one-third of the valley market (earlier field studies).	[V] (that the author says so)	Kashmir Observer opinion

5.2 National bans and restrictions relevant to apple

FACT	TAG	SOURCE
Insecticides (Prohibition) Order 2023, S.O. 4294(E), 29 Sep 2023: dicofol, dinocap and methomyl banned outright (3 of the 27 in the 2020 draft).	[V]	Gazette
Monocrotophos 36% SL discontinued; existing stock only until expiry.	[V]	Gazette
Malathion: apple removed from approved uses. Dimethoate: label claims deleted for fruit eaten raw (covers apple). Chlorpyrifos lost only ber, citrus, tobacco; mancozeb only guava, jowar, tapioca. Carbofuran limited to 3% CG. Captan not in the order. (Current CIB&RC major-uses list not checked.)	[V]	Gazette
2018 order S.O. 3951(E) banned carbaryl, diazinon, dichlorvos, fenthion, methyl parathion, phorate, phosphamidon, triazophos, trichlorfon, benomyl, tridemorph and others (some phased to 31 Dec 2020). Secondary source.	[V]	Wikipedia list
Carbosulfan: draft ban order Aug 2026 (30-day comments); no final ban as of 13 Nov 2026.	[V]	Business Standard
Glyphosate: Registration Committee (30 Jul 2026) recommended use only by trained personnel or PCOs; final decision with DA&FW; sub judice.	[V]	Global Agriculture

5.3 Online sale, licensing and traceability

FACT	TAG	SOURCE
Insecticides (Second Amendment) Rules 2022 (rule 10E): a licensed seller may sell through an e-commerce platform for doorstep delivery; the platform must verify the State licence before engaging the seller; both follow the Consumer Protection (E-commerce) Rules 2020.	[V]	SCC Online
Rule 10E does not relax licence, product endorsement, Principal Certificate, approved premises or area limits; platforms need no pesticide licence (CropLife calls it a gap).	[V]	Global Agriculture, 27 Jan 2026
May 2026: CCPA had Amazon, Flipkart, Meesho and JioMart delist an unregistered herbicide; listings lacked composition, licence number and validity, hazard warnings and Principal Certificate details.	[V]	Business Standard
Insecticides (Amendment) Rules 2026, G.S.R. 493(E), 17 Jun 2026, in force 90 days after publication: digital licence applications, electronic stock registers, monthly e-returns within 15 days. (Secondary source; gazette not read.)	[V]	Indian Chemical Regulation
QR code on insecticide labels (Insecticides (First Amendment) Rules 2025, 3 Jun 2025; non-compliant stock not saleable after 30 months, about Dec 2027). G.S.R. number (367 or 368) and start of the 30 months unclear; vendor source.	[S]	QR Tiger, legalitysimplified
Earlier QR rule for retail packs (G.S.R. 211(E), 23 Mar 2023).	[S]	QR Tiger
No J&K-specific QR or traceability system for agrochemicals was found.	UNVERIFIED	n/a (negative finding)
Draft Pesticides Management Bill (released 7 Jan 2026, comments to 4 Feb 2026) would replace the Insecticides Act 1968; not enacted as of Nov 2026 (negative not provable). The 2020 Bill (RS, 23 Mar 2020) never passed.	[V] (draft); UNVERIFIED (status)	Business Standard, PRS
Fertilizer sellers need an FCO clause 8 registration certificate (Form A + Form O), separate for wholesale and retail. No FCO rule specific to online sale found.	[V]	Indian Kanoon
Subsidised fertilizer sold only after Aadhaar authentication on retailer PoS devices (since March 2018).	UNVERIFIED	Rajya Sabha Q.191

5.4 Nurseries and planting material

FACT	TAG	SOURCE
J&K Fruit Nurseries (Licensing) Act, 1987 (Act XXII of 1987), enforced from 1 Oct 1988. The 1978 date and "1991 Rules (SRO 167)" were not confirmed.	UNVERIFIED	India Code
Oct 2025: registration certificates to 205 new private nursery growers; other departments require the certificate.	[V]	Kashmir Images
NHB nursery recognition is voluntary and star-rated, first granted for two years.	[V]	NHB guidelines
June 2025: department to import high-density plants itself after late/substandard private supply; show-cause notice to "FruitWall" for canker-affected saplings.	[V]	ETV Bharat

Analysis for the shop: list only CIB&RC-registered products with licence number and validity, composition, hazard warnings and batch/QR shown; sell only through licensed J&K dealers; flag products named in J&K misbranding orders by batch; never list malathion or dimethoate for apple.

6. Labour, subsidies and costs

6.1 Labour

FACT	TAG	SOURCE
2026 harvest, Shopian: Rs 1,200-1,500 a day (worker shortage, bigger crop). Labour Department: a minimum wage is fixed, no upper cap.	[V]	Greater Kashmir, 22 Oct 2026
2025 harvest, Shopian: Rs 800-1,100 a day for picking; workers from Chenab Valley, Rajouri, Poonch and Bihar.	[V]	Kashmir Life, 28 Sep 2025
General daily labour Rs 600-700 plus food, tea and tobacco ≈ Rs 1,000 (2025 op-ed; it does not mention apples).	[V]	Kashmir Observer, 19 Mar 2025
2022 task rates: master pruner up to Rs 1,200/day (3-4 trees a day); packer Rs 800-1,000/day, max 30 boxes.	[V]	Kashmir Life 2022
Minimum wage (S.O. 513, 12 Oct 2022, from 17 Oct 2022): unskilled Rs 311, semi-skilled 400, skilled 483, highly skilled 552, admin 449 a day. No later J&K UT revision found; a draft J&K Code on Wages Rules 2026 is pending. Secondary sources.	[V]	labourlawindia, TeamLease

FACT	TAG	SOURCE
Corrected: Labour Bureau's J&K series cited is the "Packaging and Agriculture" occupation for men, not an all-agriculture average: Rs 718.18 (Jul 2026), Rs 727.27 (Aug 2026).	[V]	CEIC
Aug 2026 J&K Police advisory: newly arriving non-local labourers must register at the nearest police station; those at isolated sites told to move. No employer or landlord duty stated.	[V]	WION
Jammu division only: Kathua DM (Oct 2026, s.163 BNSS) bars contractors from engaging labourers without prior police verification; Kishtwar order (30 Jun 2026, 2 months, likely lapsed) gave employers 7 days; Jammu "Kirayedar" tenant portal (Dec 2025).	[V]	Greater Kashmir, Gulistan News, Rising Kashmir

6.2 High-density plantation subsidy

FACT	TAG	SOURCE
Revised Modified HDP Scheme: 5,500 ha of apple, walnut, almond, cherry over 2022-23 to 2026-27, overall 50% subsidy. It runs to 31 Mar 2027 (not March 2026); no successor announced.	[V]	Greater Kashmir, 29 Aug 2026
Up to Dec 2025: 1,119.21 ha, Rs 138.57 crore paid; 2026-27: 1,455 ha high/medium density with Rs 70 crore.	[V]	Kashmir Life, Feb 2026
Cost sharing: 50% subsidy, up to 40% optional bank loan (interest deferred 3 years, repayment from year 4), 10% farmer. Extras: micro-irrigation, virus-free material, anti-hail nets.	[V]	Kashmir Life, 1 Aug 2025
Minimum area now 1 kanal; J&K Bank loans with 3-4 year moratorium; five-feather plants about Rs 500 each (Mar 2026).	[V]	KNS, 27 Mar 2026
"Rs 27,000 per kanal" is the **anti-hail net** subsidy in HDP (Rs 27.50/m^2 in traditional orchards), not the plantation subsidy.	[V]	KNS
Corrected: the 40% (general) / 50% (Himalayan) assistance up to 2 ha per beneficiary is under **MIDH**, not HADP/JKCIP (Lok Sabha, Aug 2026).	[V]	KNO
Apply on the HADP Kisan Sathi portal (Oct 2026: 6,97,728 applications, 1,15,295 pending; horticulture 66,818).	[V]	Rising Kashmir, 27 Oct 2026
Credit-linked HADP cases go to banks only through Kisan Sathi; no offline cases (J&K Bank annexure, not opened).	UNVERIFIED	J&K Bank
2022: 50% capital support "up to Rs 22.15 lakh" (unit not stated).	[V]	Kashmir Reader 2022
Cost estimates: Rs 1-2 lakh per kanal (2024 op-ed, 120-150 plants at Rs 250-300); Rs 3,51,759 per kanal over first 5 years (unnamed 2026 study of 200 growers).	[V]	Daily Excelsior 2024, Kashmir Vision 2026
Traditional 10-12 trees/kanal against 160-170 under HDP; ~30% vs ~90% Grade-A (Chief Horticulture Officer).	[V]	Kashmir Life 2025

6.3 Market, freight and packing costs

FACT	TAG	SOURCE
Mandi commission: official 6%; 12-15% plus bags, weighing, loading reported (Aug 2025). May 2026: Directorate of Horticulture P&M told AMO Sopore to stop unauthorised commission.	[V]	KNO 2025, KDC 2026
2022 Shopian: 6% (market up to 13%), Rs 4/box unloading, Rs 2/box association; Delhi agents 12-14% with advance, 6-8% without.	[V]	Kashmir Life 2022
2013: Delhi HC set aside the notification letting Azadpur charge sellers 6%; current enforcement unverified.	[V]	The News Himachal
Freight to Delhi: usual Rs 70-100/box; Transport Dept "fair" Rs 100-120 (2025); crisis Rs 200-350; RTC cap Rs 90-130; association fixed Rs 130 Delhi, Rs 120 Punjab (Sep 2025). No 2026 normal-season rate found.	[V]	Greater Kashmir, ETV Bharat, Kashmir Observer
Northern Railway proposal (Oct 2026): 1.5x JP Scale for a round trip on JPP-RCS parcel coaches, i.e. 50% off return freight.	[V]	Kashmir Observer, 30 Oct 2026
Boxes (2022, latest found): cardboard about Rs 50-53 (15 kg), wooden Rs 90-130 (up to 18 kg), plastic crates Rs 180-200; ~60% packed in cardboard.	[V]	Kashmir Life 2022, Kashmir Pulse 2022
Corrected attribution: break-even about Rs 450 per 16-kg box (all grades) was given by the head of Pulwama Fruit Mandi, not a Shopian grower (2022).	[V]	Kashmir Life 2022
Prices, Nov 2026, Shopian: premium Rs 1,300-1,500/box, medium five-layer Rs 1,000-1,200.	[V]	Greater Kashmir, 7 Nov 2026
Kisan Khidmat Ghars: 2,000 planned, 1,309 operational (about Feb 2026); Kisan Sathi access to 120+ schemes, seeds/fertilizer/pesticide, micro-ATM/POS and KCC. Operators are entrepreneurs ("Krishi Udyamis" per a district notice).	[V]; UNVERIFIED (operator term, nodal agency)	Kashmir Life, Tribune, Udhampur notice

7. Still open

1. **Official IMG 2026 PDF** from SKUAST-K or the Horticulture Department (all 2026 guide rows come from a Scribd copy). Check against the printed booklet.
2. **SKUAST-K ruling on the 2026 conflicts:** Zineb + Hexaconazole dose (125 g vs 100 g) and strength; Mancozeb + Carbendazim unit and WP/WG; pre-harvest timing (15-20 vs 25 days, Propineb PHI 29); stop-spray rule (3-4 weeks vs 14 days); trap densities; mite thresholds; hive rate.
3. **HDP spray schedule:** none exists from SKUAST-K; the HDP fertilizer table is tentative (REAC). Whether REAC adopted the tentative age-wise schedule.
4. **Leaf miner and fruit borer chemicals**: how the Aug 2026 message options fit the stage table.
5. **CIB&RC label status** of listed molecules on apple (chlorpyrifos, thiacloprid, carbendazim mixes, captan) after 2023; current consolidated banned list.
6. **Codling moth:** current spread in the valley (2024-26) and whether any domestic quarantine now covers Kashmir (DPPQS notification unread).
7. **Official 2026 scab loss survey** (only trade estimates exist); recent Marssonina incidence data.
8. **J&K list of 32 banned/restricted formulations** and order number; Super M-45 lab results and committee report.
9. **QR-code and Insecticides Amendment 2026 gazette texts**; any FCO rule on online fertilizer sale; status of the Pesticides Management Bill.
10. **Nursery Act and Rules** text (fees, validity).
11. **HDP scheme after 31 Mar 2027**, official unit cost per ha or kanal, and the unit of the Rs 22.15 lakh ceiling.

12. **Labour:** 2025-26 rates for pruning, spraying, packing and grading; J&K agriculture scheduled-employment rate; any Kashmir-valley order on employer/landlord duties for non-local workers.
13. **Costs:** official CA tariff, 2026 carton prices, 2026 normal-season freight, current mandi deductions per box.
14. **Crop insurance:** not researched in this round; whether apple insurance will relaunch, and in what form, is unknown.
15. **Soil-testing services:** labs, fees and turnaround (department pages unreachable).
16. CA set-points and an official J&K grading norm.